Exempt, Evade, Enforce: Unpacking Property Tax Dynamics in the Short-Term Rental Market

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Abstract

How does third-party reporting affect compliance with taxes it does not directly target? I study this question in local property taxation, where liability depends on self-reported property use that local governments struggle to verify. In Florida's short-term rental (STR) market, hosts who rent more than 30 days per year lose their owner-occupied exemption, creating a sharp notch in tax liability. I exploit Airbnb's Voluntary Collection Agreements as a shock to indirect third-party reporting: the agreements require the platform to remit sales and accommodation taxes but do not report rental days. Applying a bunching estimator to listing-level STR data from 2015–2024, I find negligible bunching at the 30 day rental limit before the agreements and approximately 4,600 additional listings bunching after, with the most responsive hosts cutting bookings by more than 16 nights. Because hosts comply by curtailing activity rather than paying the commercial rate, their compliance yields no additional property tax revenue relative to evasion, leaving roughly $13.8 million unrealized over the sample period. Overall, I find that even indirect reporting generates meaningful compliance gains, and property tax rules curb STR activity.

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1 Introduction

How does third-party reporting affect compliance with taxes it does not directly target? This question speaks to a central concern in public finance: how governments can raise revenue while minimizing distortions to behavior and disruptions to markets. Third-party reporting is among the most effective tools for striking this balance. When an outside party reports a taxpayer's activity directly to the tax authority, underreporting becomes difficult and compliance rises (Kleven et al. 2011, Persson 2019, Pomeranz 2015, Naritomi 2019). However, this evidence comes largely from settings where the reported information maps directly to the collected tax (e.g., business, income, value-added, consumption taxes). Much less is known about whether reporting *indirectly* related to a tax also improves compliance.

In this paper, I investigate how indirect third-party reporting influences property tax compliance at the local level. Property taxes are the primary funding source for local governments, making their vulnerability to evasion especially problematic for local public finance.[1] This vulnerability stems from two intertwined challenges. First, property tax liability depends on self-reported use of the property by agents with a strong incentive to evade. Second, many local jurisdictions lack the capacity to verify this information because enforcement is costly and they have fewer resources than state or national governments. For these reasons, studying property taxes in the context of local government is important for protecting the fiscal base of public services, such as public schools and fire and police departments.

The short-term rental (STR) market in Florida provides a clean setting to study indirect reporting and property tax compliance. Under state law, a home rented for more than 30 days per year for two consecutive years forfeits its owner-occupied exemption and is taxed at the higher commercial rate. This rule creates a notch: a discrete jump in tax liability at the 30-day threshold. A STR host who would otherwise rent beyond the threshold can either reduce rental activity to remain below it, or rent more while underreporting days to retain residential treatment. I explore this revenue-tax tradeoff more in the conceptual framework.

I apply a bunching estimator to listing-level STR data from AirDNA to measure how hosts' rental behavior, and thus tax evasion, responds to a policy change that established Airbnb as a third-party tax reporter. The policy change I exploit is the introduction of Airbnb's Voluntary Collection Agreements (VCAs), which require Airbnb to remit sales and accommodation taxes to the appropriate tax jurisdictions on behalf of STR hosts. VCAs do not report host-level rental days, so they cannot mechanically trigger reclassification under

[1] Property taxes are a fundamental funding source for local governments in the U.S., particularly counties (Yushkov 2024, Urban-Brookings Tax Policy Center 2024). Many local governments rely on property taxes to fund public services and public schools. On average, the property tax provides 30% of revenue for local governments and raised $581 billion in 2021 (Lincoln Institute 2024).

the property tax rule. Any effect on property tax compliance from the VCAs is, therefore, a spillover from reporting targeted at a different tax.

I find no bunching at the rental-day threshold pre-VCAs and approximately 4,600 additional STRs bunching post-VCA, indicating that hosts adjust rental activity to retain residential treatment rather than continuing to evade. These findings show that even indirect reporting can generate sizable compliance gains, and that property tax rules meaningfully shape STR supply. The fiscal consequences, however, are less straightforward. Relative to when they evaded, hosts who bunch continue paying the residential rate, so their compliance generates no additional property tax revenue. Had these hosts reported rental activity over the threshold and paid the commercial rate, the government would have collected approximately $13.8 million in additional property tax revenue over the post-VCA sample period.

This paper contributes to the literature on tax compliance and information reporting by providing new evidence from local property tax systems. A large body of work shows that third-party reporting substantially reduces evasion in income and business taxation by limiting opportunities for misreporting (Kleven et al. 2011, Persson 2019). In contrast, less is known about the role of reporting in property taxation. Existing causal evidence on property taxes focuses primarily on salience and enforcement intensity—such as reminders, lien threats, and social norms (Perez-Truglia & Troiano 2018, Hallsworth et al. 2017)—rather than on information reporting itself. This paper shows that reporting arrangements designed to collect one tax can spill over to affect compliance with another.

This paper also relates to the literature on STR regulation and taxation. Prior work studies how STR bans and taxes affect housing markets and platform activity (Koster et al. 2021, Bei & Celata 2023, Garz & Schneider 2023, Li et al. 2022), while a broader literature documents the externalities and regulatory challenges associated with STR growth. The study most closely related to this paper is Bibler et al. (2021), which shows that Airbnb's Voluntary Collection Agreements (VCAs) increase STR prices, reduce bookings, and affect entry and exit through improved tax enforcement. Nevertheless, this study and other literature in the STR space do not address property taxes, a key source of local public funding. I fill this gap by providing causal evidence on how third-party reporting shapes property-tax compliance and evasion in the STR market. By linking rental-day thresholds to enforcement, I identify a new margin of evasion in the STR market and provide causal evidence on how indirect information reporting shapes property-tax compliance.

The next section provides background on the STR market and Florida property taxes. Sections 3 and 4 present the theoretical framework and the empirical methodology used for the analysis, respectively. After, I discuss the data in Section 5 and the results in Section 6. Section 7 positions the findings in the policy landscape, and Section 8 concludes.

2 Background

2.1 Third-party Reporting on STRs: Airbnb Voluntary Collection Agreements

Short-term rentals (STRs) occupy a unique and often ambiguous regulatory space since they combine residential and commercial characteristics. Generally, STRs refer to residential properties rented to guests for stays of less than a month. These short stays are typically booked and rented through a third-party platform, such as Airbnb, Vacation Rental by Owner (VRBO), and Booking.com.[2] Due to their short-term stays, STRs are governed by local lodging or occupancy tax laws rather than long-term lease regulations.

Local governments have instituted various regulations on STRs to measure rental activity and dampen its negative effects. Some of these regulations include rental permitting requirements, additional taxes, zoning restrictions, and annual rental day limits.[3] In terms of community impacts, informal tourist accommodations like STRs impact the operation of zoning, traffic flow, parking, waste services, noise, and residential privacy. Furthermore, cities must consider how to provide sufficient water, safety, fire, and emergency services with fluctuating residency numbers and in times of overcrowding (Palombo 2015). Other concerns for STRs entry in a city include effects on the existing hotel market and a reduction in the long-term rental supply (Li et al. 2022, Gurran & Phibbs 2017*a*).[4]

In response to the new STR regulations, Airbnb has since enacted voluntary regulations on their platform that increase reporting to local governments. One of these regulations is the Voluntary Collection Agreements (VCAs). VCAs are agreements between Airbnb and

[2]Airbnb held roughly 43% of the U.S. short-term rental market in 2025, followed by VRBO at 21% and Booking.com at 8%. Together Airbnb and VRBO represent about two-thirds of U.S. listings (Rental Scale Up 2025). Globally, these three platforms control more than 70% of short-term rental revenue (Yahoo Finance 2025).

[3]For example, to protect its long-term rental supply, New York City has banned rentals of entire homes for less than 20 days. San Francisco mandates that hosts reside in their rental home for a minimum of 275 days annually. Similarly, Portland restricts the percentage of apartments in a building that can have STR permits and requires hosts to occupy their homes at least 270 days per year (Gurran & Phibbs 2017*a*).

[4]Previous research on STRs has focused on their disruption of existing firms, such as hotels and resorts (Dogru et al. 2019, Dogru, Hanks, Mody, Suess & Sirakaya-Turk 2020, Zervas et al. 2015, Yeung & Coyle 2016, Dogru, Hanks, Ozdemir, Kizildag, Ampountolas & Demirer 2020) and the housing and rental market (Barron et al. 2021, Koster et al. 2021, Liang et al. 2022, Franco & Santos 2021, Todd et al. 2022, Valentin 2021). This literature documents both benefits—such as enabling homeowners to remain in high-cost areas (Guttentag 2015) and increasing visitor spending (Oskam & Boswijk 2016)—and substantial costs, including neighborhood disruption (Gurran & Phibbs 2017*b*) and rising rents (Lee 2016). STR growth also strains public services (Ke et al. 2021, van Holm & Monaghan 2021, Reinhard & Roth 2024) and raises administrative costs for local governments (Office of the City Treasurer 2021, Arora et al. 2025). These fiscal pressures underscore why understanding enforcement and compliance rates for the property-tax base is essential for local public finance.

the local tax authority for Airbnb to collect sales and accommodation taxes directly on the platform and remit them to the proper authority. Prior to the VCAs, the STR hosts were solely responsible for collecting and remitting all taxes, such as accommodation, sales, and tourism taxes, to the appropriate levels of government (city, county, and/or state). Between 2014 and 2021, Airbnb established over 275 VCAs with state, county, and city governments (Bibler et al. 2021).[5]

To detect property tax evasion in the STR market, this paper exploits the implementation of the Airbnb VCA with Florida on December 1, 2015.[6] I use the VCAs as a natural enforcement shock as well as a proxy for increased STR regulation and oversight by local governments. Although property taxes were not collected through Airbnb VCAs, entering into a VCA with Airbnb signals that the local government is increasing scrutiny and regulatory activity around STRs. This perceived increase in enforcement and visibility into rental activity extends the enforcement shock of VCAs to property tax compliance.

2.2 Property Tax Evasion through Fraudulent Owner-Occupied Exemptions

In detecting property tax evasion in the STR market, I focus on the evasion from fraudulent homestead exemption claims in Florida. The homestead exemption lowers a homeowner's annual property tax bill if their house is owner-occupied and the owner's principal residence. Homestead exemption is limited to one home per owner, so secondary homes and vacation houses do not qualify for homestead exemption.[7] These eligibility requirements also motivate the sample restrictions used throughout the paper. Specifically, the empirical analysis focuses on entire-home rentals managed by hosts with a single listing, who are the STR operators most likely to qualify for owner-occupied or homestead exemptions.

2.2.1 Property Taxes in Florida

Local governments in Florida levy residential property taxes based on a home's assessed value, the applicable district millage rates, and certain assessment limitations required by the Florida constitution. Due to limitations on the growth of assessed values, a home's

[5]Even with the implementation of VCAs, Airbnb does alert hosts that they may be responsible for remitting additional taxes based on their locality and tax jurisdiction. See https://www.airbnb.com/help/article/1036.

[6]According to Bibler et al. (2021), VCAs were generally announced only a few weeks prior to the effective date, limiting the scope for anticipatory behavioral responses.

[7]States usually consider a principal residence as the address listed on an owner's tax return, driver's license, and vehicle registration.

assessed value often does not equal the market value, except in a year of sale.[8] The total property tax liability equals the assessed value multiplied by the millage rate. More details on the calculation of Florida property taxes are available in Appendix F.

The Homestead Exemption provides substantial tax relief to Florida homeowners who claim a residential property as their permanent and primary residence. This exemption provides three distinct layers of property tax relief. First, $25,000 of property value is exempt from all property taxes, including school and non-school mills. Second, the homestead status exempts the portion of the property's value between $50,000 to $75,000 from non-school taxes.[9] These two tax provisions fall under Amendment 1 of the Florida Constitution, which passed in 2008.

The third tax savings that the homestead exemption provides is from the *Save Our Homes* (SOH) constitutional amendment from 1992. The SOH amendment restricts the annual growth in assessed value for a homestead at 3% or the rate of inflation, whichever is lower. Amendment 1 from 2008 expanded the SOH benefit to also allow homeowners to transfer up to $500,000 of accrued SOH savings to a new homestead within the state (Lincoln Institute of Land Policy and George Washington Institute of Public Policy 2024). Collectively, these provisions significantly reduce property taxes for eligible homeowners and accumulate those savings over time. The average savings from the homestead exemption in Florida on a $400,000 home can exceed $3,400 annually depending on the property's accrued SOH savings.[10]

Under Florida law, a property owner who rents their home for more than 30 days per year for two consecutive years is considered to have abandoned their homestead, thereby forfeiting the exemption.[11] A Florida Supreme Court case in April 2023 affirmed that even a homeowner renting part of their homestead is not entitled to any of the homestead exemption benefits (Lincoln Institute of Land Policy and George Washington Institute of Public Policy 2024). If the homestead exemption is lost, then the property is reassessed at full market value for the following tax year before any assessment caps are applied (Campo et al. 2024).

Florida law imposes substantial penalties for improper receipt of the homestead exemp-

[8] Amendment 2 of the Florida Constitution in 2018 made permanent a temporary provision that restricted growth in the assessed value of residential property to 10% for non-school taxes. Therefore, since the temporary provision began in 2008, assessed values have been capped each year until the assessed value equals market value or the home is sold.

[9] This $25,000 of the homestead's value is subject to property taxes that fund public schools.

[10] See Appendix G for a calculation of the average property tax savings from the homestead exemption in Florida.

[11] See Fla. Stat. §196.061(1). The 30-day threshold must be exceeded in two consecutive years; exceeding it in only one year does not result in the loss of homestead status. Reinstating the exemption is subject to additional filing and residency requirements, increasing the cost of losing homestead status.

tion.[12] Knowingly providing false information is a first-degree misdemeanor punishable by up to one year in jail or a $5,000 fine. Homeowners found to have improperly claimed the exemption within the prior ten years are liable for the unpaid taxes, a 50% penalty, 15% annual interest, and a recorded tax lien (Saint Lucie County Property Appraiser 2025, Sarasota County Property Appraiser 2025).

2.2.2 Opportunities for Fraud

Since homestead exemptions are granted and maintained until the home changes ownership, homeowners have an opportunity to engage in exemption fraud and tax evasion. A homeowner could claim the homestead exemption when they first move into the home and provide the proper documentation. As long as the assessor is not notified of a change in status or ownership, no annual verification or further documentation is needed. Later, the homeowner could list their home on a peer-to-peer accommodation platform, such as Airbnb and VRBO. Then, the homeowner could continue claiming the homestead exemption and receive the property tax benefits of a full-time, owner-occupied property regardless of how many days their home is rented.

3 Conceptual Framework

This section presents a conceptual framework for how a homeowner decides how many nights to rent their property as a STR in the presence of a property tax discontinuity. Property tax exemption thresholds, like the 30-day limit in Florida, introduce a notch in the property tax schedule. Crossing the threshold leads to a discrete jump in the homeowner's tax liability.

The following framework focuses on how a salient and enforceable threshold alters the intensive margin of STR behavior. Let homeowner i own a property with market value h_i. The homeowner rents out the property for N_i nights per year at a nightly price p and generates gross rental revenue $p \cdot N_i$. The homeowner faces operational costs $c_i(h_i, N_i)$, which depend on both the home's value and the number of rental nights. I assume costs increase with the home's value and number of rental nights. Accordingly, the first derivative of the cost function with respect to h_i and N_i are $\frac{\partial c}{\partial h_i} > 0$ and $\frac{\partial c}{\partial N_i} > 0$, respectively. The homeowner also faces a tax penalty $\Delta\tau(h_i, m_n)$ if they exceed the rental threshold N^*. This tax penalty is a function of the home's value and the applicable millage rate and has positive first derivatives as well ($\frac{\partial \Delta\tau c}{\partial h_i} > 0$ and $\frac{\partial \Delta\tau}{\partial m} > 0$).

[12] See Florida Statute 196.131(2) and 196.161(1).

Given this setup, homeowners will choose the N_i that maximizes their profit function. The profit function is:

$$\pi_i = p \cdot N_i - c_i(h_i, N_i) - \mathbb{1}_{N_i > N^*}[\Delta\tau(h_i, m_n)] \tag{1}$$

The indicator function $\mathbb{1}_{N_i > N^*}$ captures the presence of the tax notch. If the homeowner rents more than the threshold N^*, they face an immediate increase in property tax liability. Assuming the homeowner chooses N_i, their annual number of rental days, to maximize profits, the first-order condition (FOC) for an interior solution is:

$$\frac{\partial \pi_i}{\partial N_i} = p - \frac{\partial c_i(h_i, N_i)}{\partial N_i} \tag{2}$$

This implies that, in the absence of the notch, the homeowner rents until the marginal revenue equals marginal cost: $p = MC$.

A graphical representation of this profit-maximizing decision is depicted in Figure 1. The solid line up to N^* represents the payoff from renting when the homeowner retains the property tax exemption. The dashed line extending beyond N^* represents the counterfactual payoff schedule in the absence of the tax penalty. When the homeowner crosses the threshold, the tax penalty shifts the payoff schedule downward by Δtax. This creates a dominated region to the right of N^*: within this range, the homeowner receives a lower payoff from renting above the threshold than from locating at the threshold.

The indifference curves in Figure 1 illustrate how homeowners with different preferred levels of rental activity respond to the notch. A homeowner with relatively low demand for rental days, represented by IC_A, locates at $N_A = N^*$ and does not cross the threshold. A homeowner with a higher preferred number of rental days, represented by IC_C, chooses N_C even after accounting for the tax penalty because the additional rental revenue is large enough to offset the discrete tax cost. In contrast, a homeowner whose preferred rental activity would place them in the dominated region has an incentive to reduce rental days and bunch at N^* rather than pay the tax penalty. IC'_B represents this homeowner's counterfactual choice in the absence of the notch, where the homeowner would choose to rent N'_B nights, a rental level above N^*. Once the tax penalty is imposed, the feasible payoff schedule shifts downward above the threshold, and the homeowner instead locates at the kink, represented by IC_B. Homeowner B will choose to rent N^* nights, where the indifference curve IC_B intersects with the budget constraint at the notch. Thus, the notch creates a range of homeowners who would otherwise rent more than N^* and less than N_D but instead choose to locate at or just below the threshold.

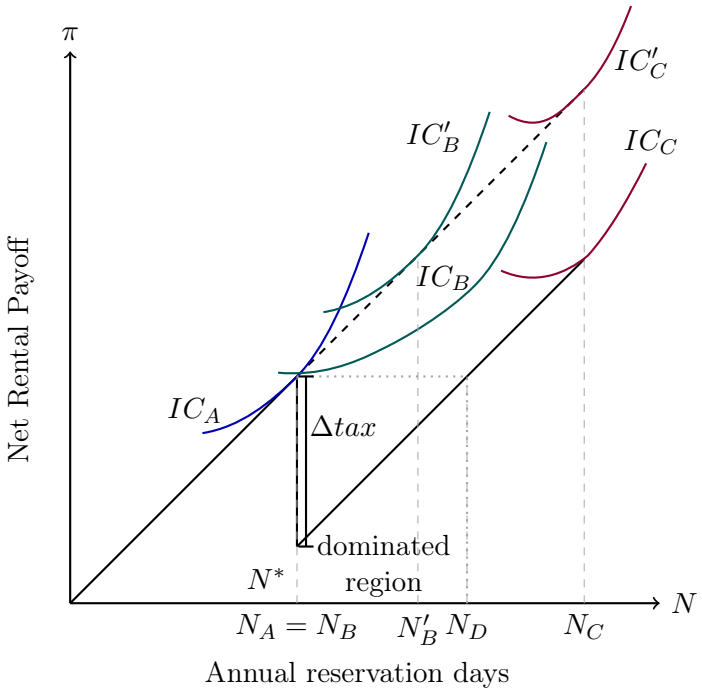


Figure 1: Bunching Incentives at Annual Rental Day Threshold

Notes: This figure illustrates the homeowner's choice of annual reservation days, N, in the presence of a property tax notch at N^*. The vertical axis measures net rental payoff, π. The solid line shows the feasible payoff schedule, while the dashed line shows the counterfactual payoff schedule absent the tax penalty. Crossing N^* triggers a discrete increase in property tax liability, Δtax, shifting the payoff schedule downward above the threshold. The dominated region is the range of rental days above N^* where the homeowner is better off bunching at the threshold than renting additional days and paying the tax penalty. The indifference curves IC_A, IC_B, IC'_B, IC_C, and IC'_C illustrate heterogeneity in homeowners' preferred rental intensity. The primed curves show counterfactual choices absent the tax penalty, while the unprimed curves show choices under the post-tax feasible payoff schedule. Homeowners whose unconstrained choices fall in the dominated region bunch at N^*, while homeowners with sufficiently high desired rental activity continue renting beyond the threshold despite the tax cost.

This optimization framework implies that observed bunching around the threshold reflects rational decision-making in environments with credible enforcement and high tax salience. For some homeowners, the discrete increase in taxes will dominate the marginal profit from additional rental nights. They will optimally choose to stop renting just below the threshold N^*. Others, whose rental revenue or cost structure makes it worthwhile, will rent beyond the threshold and forgo the exemption. Without credible enforcement, homeowners maximize profits by surpassing the rental day limit and underreporting rental activity.

If hosts behave according to this optimization logic, I should observe an excess mass of STR listings clustered just below the exemption limit. Hosts who prefer to retain the exemption will manage their annual rental days to remain below the cutoff, producing visible clustering as in Figure 2. A stronger enforcement regime and third-party reporting should strengthen incentives to remain below the rental day limit. Therefore, I anticipate a sharper

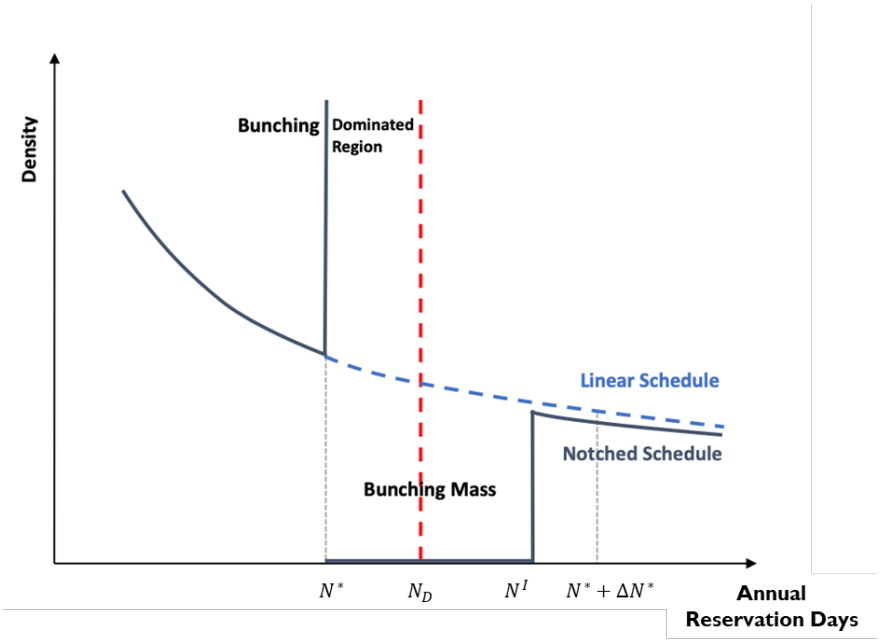


Figure 2: Bunching Conceptual Figure

Notes: This figure depicts perfect bunching behavior near a notched property tax threshold. STR hosts who would otherwise rent beyond N^* instead bunch just below the threshold due to a discrete jump in property tax liability. The shaded region represents the dominated region where no optimizing host chooses to locate. This figure is adapted from Figure 3b in Mavrokonstantis (2022).

bunching response and more pronounced clustering from STR hosts in the post-VCA period.

The degree of bunching provides a direct measure of hosts' responsiveness to the property tax notch. Using the excess mass at the threshold, I recover a reduced form elasticity of rental supply with respect to the property tax, as in the bunching literature. This elasticity captures how strongly hosts adjust their rental behavior in the presence of a property tax notch. Comparing elasticities before and after the VCAs reveals the sensitivity of hosts to third-party reporting in the property-tax system.

Although the above framework focuses on intensive-margin decisions, the rental-day threshold also has implications for the extensive margin decision of whether to participate in the market. Some homeowners may opt out of the STR market entirely due to the complexity of the thresholds or the potential jump in tax liability. Others may find that the tax-induced shift in their cost curve makes even short-term renting unprofitable. While this paper does not measure extensive-margin responses, these mechanisms provide a conceptual basis for understanding why such thresholds deter entry or encourage exit from the market.

4 Empirical Methodology

I employ a bunching estimator to measure the behavioral response of STR hosts to changing property tax rates in Florida. Similar to the approaches of Kleven & Waseem (2013) and Kopczuk & Munroe (2015), I exploit notches in the homestead exemption annual rental day limit to detect behavioral responses to property tax rates. I compare the actual distribution of annual rental nights (with the notch) to a counterfactual distribution (without the notch). I apply this bunching estimator to two periods: prior to the Airbnb VCA and after the VCA. By comparing the bunching results of the two periods which had different third-party reporting, I provide evidence of tax evasion in the STR market prior to the reporting increase.

Given the goals of this paper, a bunching estimator has several advantages over other causal inference methods to measure excess bunching and tax avoidance. The conceptual framework predicts that hosts facing a discrete increase in tax liability at the rental-day threshold will optimally locate just below the cutoff, generating excess mass in the distribution of rental days. The bunching approach directly tests this prediction and recovers both the number of responding agents and the magnitude of their behavioral adjustment. Another advantage of the bunching approach is that it exploits cross-sectional variation around the threshold to estimate how hosts respond to discrete changes in tax liability. Alternative approaches, such as difference-in-differences, would capture average changes in rental activity but would not identify the intensive-margin responses around the threshold that are central to this analysis.

4.1 Main Bunching Estimation

The bunching estimator measures the excess mass of STR properties that bunch at the rental day limit. At this 30-day notch, the change in property tax rates occur. Because of the large jump in property taxes owed from crossing the rental day threshold, the tax schedule has a notch, as opposed to a kink.

First, I plot the distribution of annual STR days based on the actual data with the following equation:

$$c_j = \sum_{i=0}^{p} \beta_i (N_j)^i + \sum_{i=N_L}^{N_U} \gamma_i \mathbb{1}[N_j = i] + \sum_{r \in R} \rho_r \mathbb{1}[\frac{N_j}{r} \in \mathbb{N}] + \sum_{k \in K} \theta_k \mathbb{1}[N_j \in K \wedge N_j \notin [N_L, N_U]] + \epsilon_j \tag{3}$$

The STRs are put into j bins based on their annual rental days N, and each bin is a range of annual rental days. The number of STR observations in each bin is c_j. The first term in the equation, $\sum_{i=0}^{p} \beta_i (N_j)^i$, is the polynomial of order p with coefficients β_i that fits the

binned data. The binned data are the counts of STR properties that fall into each bin. N_j is the maximum N in bin j of width δ.

The second term in the equation contains the bunching dummies that capture excess mass at the threshold. These dummy variables, γ_i, measure how much of the density is concentrated at the exemption limit relative to the rest of the distribution. N_L and N_U are the lower and upper bounds of the bunching region, respectively. They represent the area of the distribution that is impacted by the rental day limit. I estimate N_L and N_U by looking for visual changes in the rental day distribution.

The third and fourth terms in the estimation equation account for natural bunching patterns that are unrelated to the property tax threshold. The third summation term controls for round-number bunching through the coefficients ρ_r, where the set R includes round-number bunch points. The fourth summation term adjusts for any remaining fixed bunching points in the distribution through the set K, which contains prominent spikes outside the bunching region that are unrelated to the rental-day threshold. Including these terms improves the accuracy of the bunching estimator by isolating behavioral responses to the property tax threshold from other predictable, non-policy-related patterns in STR booking behavior. Based on the raw distribution of annual rental days in Florida, I include a fixed point control at 14 days and no round-number bunch points. The final term, ϵ_j, is the error term.[13]

Next, I estimate $\hat{c_j}$, the counterfactual binned count of STR properties. The counterfactual bin count is estimated as follows:

$$\hat{c_j} = \sum_{i=0}^{p} \beta_i (N_j)^i + \sum_{r \in R} \rho_r \mathbb{1}[\frac{N_j}{r} \in \mathbb{N}] + \sum_{k \in K} \theta_k \mathbb{1}[N_j \in K \wedge N_j \notin [N_L, N_U]] + \epsilon_j \tag{4}$$

The variables in Equation 4 are defined as above in Equation 3. However, the counterfactual estimation omits the bunching-region dummies γ_i to recover how the distribution would appear without the notch.

With the actual and counterfactual binned data, I then measure the excess number of STR hosts bunching at 30-days due to the notch. This excess bunching mass is estimated as:

$$\hat{B_0} = \sum_{j=N_L}^{N^*} (c_j - \hat{c_j}) \tag{5}$$

c_j remains the observed count of STR properties between the bunching region $[N_L, N^*]$. $\hat{c_j}$ is the counterfactual count of STR properties and excludes the bunching region. Thus, $\hat{B_0}$ sums the difference between the actual distribution c_j and the counterfactual distribution $\hat{c_j}$

[13]More details about the empirical equations can be found in Appendix C.

in the region affected by the rental day limit $[N_L, N^*]$. I refer to $\hat{B}_0$ as the bunching mass. Standard errors for $\hat{B}_0$ are bootstrapped 1,000 times.

In this context, $\hat{B}_0$ should be considered a lower bound for the number of bunchers for two reasons. First, some hosts have more than one property, in which case the total number of properties in my sample likely exceed the number of hosts. However, hosts can only have the homestead exemption on one residence. Thus, some STR properties in my sample would not be eligible for the homestead exemption regardless of annual rental days, which would attenuate my results.

Additionally, some STR hosts may choose not to rent and exit the market entirely because of the threshold. This market exit would impact the extensive margin of STR activity, which is not captured by these bunching estimates. So although these individuals are responding to the tax threshold, they would not be included in the estimate of the number of bunchers because they have exited the STR market.[14]

4.2 Estimation of Changes in Rental Behavior by STR Hosts

Next, I estimate how STR hosts change their rental behavior to stay under the rental day limit and avoid higher property taxes. $\hat{B}_0$ is the excess bunching mass calculated from Equation 5 and also estimates the number of extra STR properties clustering at the rental day limit. I examine the behavior of the average buncher by estimating b, the normalized excess mass. b is how many nights per year on average each STR host chooses not to rent to keep the homestead exemption. I estimate the average reduction in rental nights as:

$$\hat{b} = \delta \frac{\hat{B}_0}{\hat{c}_0} \tag{6}$$

The numerator $\hat{B}_0$ is the number of excess STRs at the notch in the bunching region. The denominator $\hat{c}_0$ is the counterfactual estimate of how many STRs would have located at the 30-day bin without the notch. Because $\hat{c}_0$ is the count in a single bin of $\delta = 2$ nights, the ratio $\hat{B}_0/\hat{c}_0$ measures the response in bins, and multiplying by the bin width δ converts it to nights. The tables report the normalized excess mass in both bins and nights.

As not all STR hosts respond optimally to the tax threshold, I adjust b to account for these differing responses. I define "bunchers" as those who manipulate their rental days to remain under the homestead exemption rental day limit and "non-compliers" as STR hosts who don't adjust their behavior. I estimate bunchers and non-compliers according to the theoretical framework of Mavrokonstantis & Seibold (2022) and related bunching models.

[14] See the last paragraph of Section 3 for more detail about how the property tax threshold could impact the extensive margins of STRs.

First, I recover the share of "non-compliers," α, which captures the share of STRs that fail to optimize due to friction in the market, ignorance of the threshold, or other adjustment costs that make it difficult for STR hosts to rent 30 days exactly. I estimate α as the ratio of the observed STRs to the counterfactual number of STRs in the dominated region just above the threshold. One minus this ratio provides an estimate of the proportion of hosts in the affected region who actively adjust their behavior in response to the property tax rule, as opposed to those who remain unaware of, constrained by, or indifferent to the threshold. Accordingly, the share of bunchers, denoted $\widehat{sharecompliers}$, corresponds to $1-\alpha$, the share of STRs that respond to the threshold as "compliers."

To identify the region in which this adjustment should occur, I define a dominated region $[N^*, N_D]$. The dominated region is the area where it is not optimal to locate due to the changing tax liabilities at the threshold. I estimate the dominated region by dividing the average change in property taxes ($\Delta\lambda$) from crossing the threshold by the average nightly revenue. The result, $N_D - N^* = \frac{\Delta\lambda}{avg.rate}$, is the average number of nights a property must rent to fully recoup the added tax burden. With $\Delta\lambda$ = \$3,000 and average nightly revenue of \$240, $N_D - N^* = 12.5$ nights, so the dominated region ends at 42.5 nights. In the main results, the estimator uses 2-night bins, thus the share of bunchers is computed over the bins from 30 to 44.

Accounting for the different behaviors from STRs in the dominated region, I adjust the estimation of b as:

$$\hat{b}_{adj} = \frac{\hat{b}}{\widehat{sharecompliers}} \tag{7}$$

where $\widehat{sharecompliers}$, is the proportion of STRs in the affected region who respond to the rental day threshold. I divide $\hat{b}$, in nights, by the share of bunchers to recover the Local Average Treatment Effect (LATE). The LATE $\hat{b}_{adj}$ estimates how bunchers specifically reduce their rental nights in response to the homestead exemption rental day limit. In other words, $\hat{b}_{adj}$ is the reduction in rental nights of the marginal buncher.

4.3 Tax Evasion: Rental Activity Before and After the VCA

To measure potential tax evasion, I employ the main bunching estimator with two additional subsamples: STRs operating before the Airbnb VCAs and those operating after. The pre-VCA period includes 2015 only, and the post-VCA period includes 2017–2024, as Airbnb's agreement with Florida became effective December 1, 2015. Since Florida's homestead exemption is dependent on two consecutive years of rental activity, I exclude 2016 since it could be used to qualify exemption status for 2015 (before the VCA) and 2017 (after the VCA).

As in Bibler et al. (2021) and Marion & Muehlegger (2008), I rely on the enforcement change in tax remittance to detect tax evasion. If the bunching estimators differ between the pre- and post-VCA periods, this suggests that STR hosts previously evaded higher property taxes by misreporting or underreporting their rental activity. A stronger bunching response after the VCAs would indicate that increased reporting and enforcement made it more difficult for STR hosts to claim the homestead exemption fraudulently and avoid property taxes. Visually, the difference would manifest as a more pronounced spike to the left of the 30-day cutoff in the post-VCA period relative to the pre-VCA period, signaling that more hosts are deliberately staying below the limit to retain the property tax exemption.

4.4 Identifying Assumptions and Validation

There are three identifying assumptions necessary for using a bunching estimator to calculate excess mass, dominated regions, and potential evasion. First, the counterfactual distribution must be smooth so that any bunching can be attributed to a behavioral response. Second, bunchers originate from above the threshold and thus, a marginal buncher can be defined. Third, the market must have constant friction for the behavioral response of the marginal buncher to be estimated (Kleven & Waseem 2013). Constant friction allows a person or entity to move to any point along the distribution with the same amount of effort. In this context, this assumption translates to STR hosts being able to increase or decrease their bookings with the same amount of ease regardless of their current rental days.

I assess the first assumption by plotting the distribution of annual rental days for the restricted sample of hosts managing a single property.[15] The histogram in Figure 3 shows that the distribution of rental days is smooth outside the bunching region. Since the counterfactual in the bunching estimation is constructed by fitting a smooth function to observations away from the threshold, this smoothness supports the assumption that the counterfactual distribution is well-approximated by the observed data outside the cutoff. The second assumption is satisfied in the conceptual framework but cannot be tested empirically. By definition, a "buncher" reduces their property's STR days in order to remain eligible for the homestead exemption.

For the third assumption to be met, a buncher must be able to reduce their number of rented nights similarly at different points of the distribution. Since Airbnb gives hosts immense flexibility in booking, bunchers can reduce nights booked by declining further bookings, canceling existing bookings, and removing their STR listing altogether. These activities

[15]Restricting the sample to STR hosts managing a single property focuses the analysis on those most likely to be claiming the homestead exemption on their rental property. Further details on this sample restriction are provided in Section 5.

give them ease in moving above and below the rental day limit threshold. This ease in changing annual bookings applies no matter where the STR is in the annual rental day distribution. Thus, the third assumption also holds.

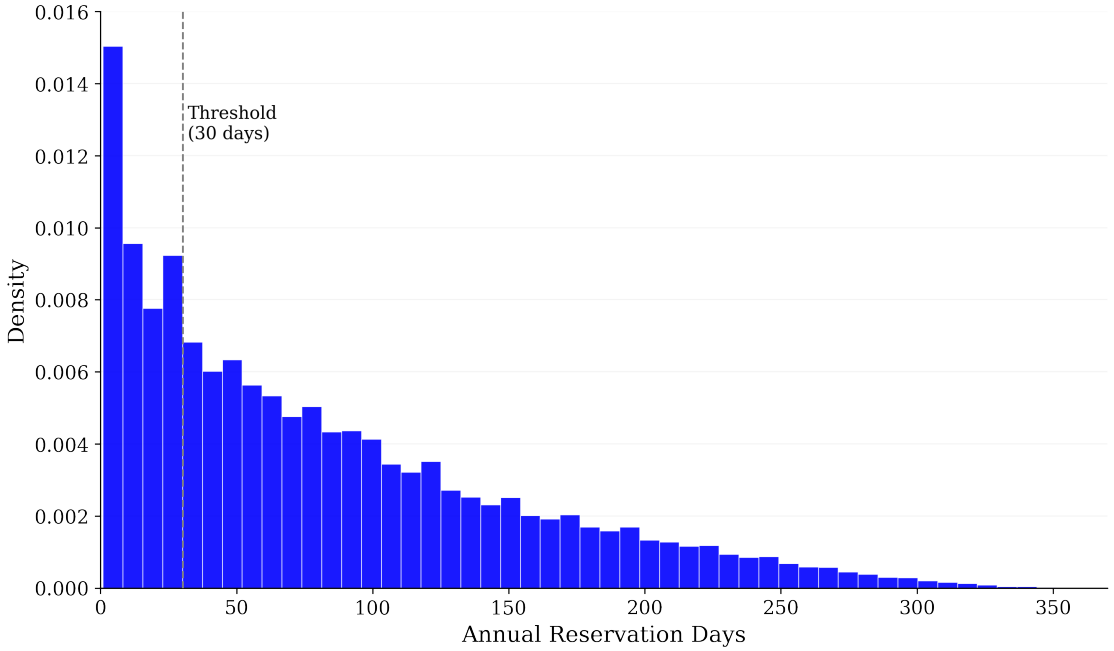


Figure 3: Distribution of Annual Reservation Days, 2015–2024

Notes: This figure displays a histogram of the distribution of annual reservation days for Florida STRs in the subsample. The subsample includes STRs that are a townhouse, condo, house, or villa and that have a host managing 1 unit. Dashed vertical lines indicate the short-term rental tax exemption threshold, 30 days. Observations with zero annual reservation days are excluded from the figure for clarity. Excluding zero, the mode for the distribution is 7 days.

5 Data

Data for this project come primarily from AirDNA, a third-party provider for data on STRs globally. AirDNA supplies data on rental properties listed on Airbnb, VRBO, or both by scraping the platforms and gathering data from data partners. Data on Airbnb listings are provided monthly from 2015 until 2024, while VRBO listings are monthly from 2017-2024. The variables from the AirDNA most pertinent for this paper are the number of bookings, number of reservations, reservation days, available days, and total revenue. I aggregate the monthly data to the annual level to create a dataset of 4,176,235 annual-property observations from 2015 to 2024.

I restrict the sample for my analysis to STRs most likely to respond to the homestead

exemption rental day limit using the property characteristics in the AirDNA data.[16] First, I limit the sample by property type, retaining properties that are condos, houses, townhouses, and villas and excluding the other types.[17] Next, I limit the sample to entire-home rentals, as partial-home rentals are treated differently under homestead exemption laws.[18] Lastly, I keep hosts that manage only one rental unit. I assume these hosts are more likely to be renting out their primary residence, which would qualify for the homestead exemption.[19] With these restrictions, the final sample has 1,168,880 property-year observations in Florida. The one-unit STR hosts account for approximately 30% of all STR reservations in Florida.

I supplement the STR data with data on Airbnb VCAs, typical home values, the Consumer Price Index (CPI), and property tax law. The implementation dates for the VCAs come from Bibler et al. (2020). The authors primarily gathered implementation dates from the Airbnb website or local news sources.

To quantify the average tax savings from the homestead exemption, I use typical home values from Zillow Research (2025) and county and school millage rates from Florida Department of Revenue, Property Tax Oversight (2025). I adjust all revenue variables in U.S. dollars using the CPI with 2024 as the base year. Consequently, the revenue can be compared across time and accounts for inflation. The CPI values come from the CPI-U survey of Urban consumers from the Bureau of Labor Statistics' January 2025 estimations (U.S. Bureau of Labor Statistics 2025).

Additionally, I collect data on FL property tax laws, owner-occupied and homestead exemptions, and property tax rental day limits. I use a variety of government websites, such as the Florida Department of Revenue, and third-party sources like The Lincoln Institute of Land Policy to obtain this information. Additional sources are listed in the references.

Summary statistics for STRs in Florida are presented in Table 3 in Appendix E. The average number of annual reservation days is 147 days per STR. STRs make on average $33,300 of revenue annually with an average nightly revenue of about $240.[20]

[16]The individual property characteristics include number of bedrooms and bathrooms, rental type (private room, shared room, or entire house), location (city, metropolitan statistical area), property type (house, townhouse, apartment, villa, etc.), host type (how many units the host rents), and the first listing date.

[17]Some other property types include hotel, cottage, guest house, cabin, RV, loft, and tiny house. As a robustness check, I include a wider variety of property types, adding cottage, guest house, loft, and tiny house. With this wider group of properties, the main estimates remain unchanged.

[18]Prior to the 2023 court case, Florida county appraisers divided a home's use between owner-occupied and rented areas, granting the homestead exemption only to the portion serving as the owner's primary residence. Since most of the analysis period is pre-2023, I drop partial-home rentals from the sample.

[19]See Appendix Table 2 for the number of observations remaining at each step of the sample restriction process.

[20]For the estimates of nightly price, I divide a STR's annual revenue by its annual reservation days. The result estimates the average revenue the STR host receives per night of renting the STR.

6 Results

6.1 Behavioral Responses to Notches and Third-Party Reporting

I compare the bunching behavior of STR hosts pre- and post-VCA to detect property tax evasion and changes in tax compliance. Bunching behavior in Florida differs starkly between the pre- and post-VCA periods as shown in Figure 4a and Figure 4b. Table 1 reports the estimates for Florida's pre- and post-VCA periods.

Bunching is minimal in the pre-VCA period (2015), with no statistically significant clustering of STRs at the 30-day rental threshold. However, after the VCA goes into effect, bunching behavior becomes much more pronounced. In the post-VCA period from 2017–2024, approximately 4,600 extra STRs cluster just below the 30-day threshold, and marginal bunchers reduce their bookings by 16.91 nights. These results are statistically significant at the 1% and 5% level and suggest that STR hosts in Florida adjusted their rental behavior after increased reporting and tax remittance by Airbnb. This behavior change is also consistent with the predictions of the conceptual framework.

Two robustness checks support the main estimates. First, because AirDNA began collecting VRBO listings in 2017, the start of the post-VCA period, I repeat the analysis using Airbnb listings only (Appendix H). Second, I replace the polynomial counterfactual with the rescaled distribution of hosts with 21 or more units, who are ineligible for the homestead exemption (Appendix I).[21] Both estimates of the normalized excess mass are smaller than the main post-VCA estimate of 0.450 in Table 1 but remain positive and statistically significant at the conventional levels.

To assess whether the results depend on the functional form of the counterfactual, Appendix J re-estimates the main 2017–2024 specification for one-unit hosts with every combination of polynomial order from 1 to 9 and bin width from 1 to 4 nights, and with alternative controls for round numbers. Across these specifications, excess mass at the 30-night threshold remains positive and statistically significant, normalized excess mass ranges from roughly 0.8 to 1.7 nights, and the share of bunchers stays between 0.035 and 0.075. The main estimate of b, the normalized excess mass, 0.900 nights is near the bottom of this range. The specifications that fit the data best after accounting for degrees of freedom imply larger responses, so the main estimates remain a conservative lower bound.

While the bunching estimates do not identify each host's precise tax liability, they provide a basis for approximating unrealized property tax revenue. If each bunching host would

[21] The hosts with 21+ units could be claiming the homestead exemption on their current residence and renting it out. If all hosts did this, this eligibility would be at most 5% of the sample of STRs with hosts managing 21 or more units.

State	Florida (30-day limit)	
Year(s)	2015	2017–2024
Estimated Excess Mass	-1.171	4,589.321***
	(7.957)	(1,416.941)
Normalized Excess Mass (Bins)	-0.019	0.450**
	(0.126)	(0.184)
Normalized Excess Mass (Nights)	-0.037	0.900
Share of Bunchers	0.009	0.053
LATE (Nights)	-4.25	16.91
N	1,283	193,148

Table 1: Bunching Estimation Results: Likely Homeowners with 1 Unit

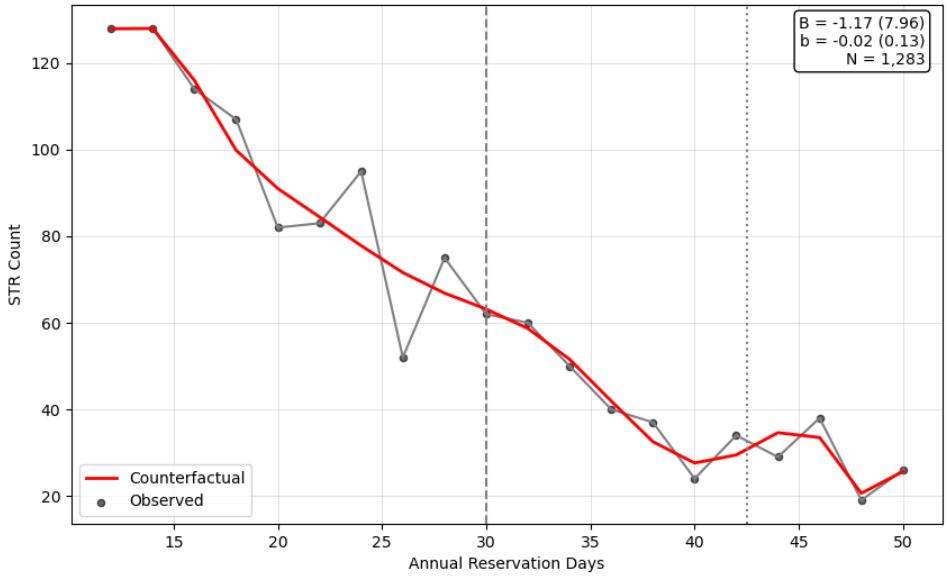
Notes: This table reports bunching estimator results for the "Likely Homeowners with 1 Unit" subsample—STRs that are a townhouse, condo, house, or villa managed by a host with 1 unit. Data are from AirDNA. The columns estimate bunching at the 30-day rental limit (bin width: 2 days; 10 bins on each side; fixed point control for 14). N is the number of observations in the bin range. Estimated excess mass is the number of additional STRs located at the threshold. Normalized Excess Mass (Bins) is the estimated excess mass divided by the counterfactual count in the 2-day threshold bin. Normalized Excess Mass (Nights) multiplies the bins measure by the bin width of 2. Share of bunchers is one minus the ratio of observed to counterfactual counts in the dominated region, capturing the fraction of STRs that respond to the cutoff (compliers); the dominated region is the 12.5 nights above the threshold ($3,000 exemption divided by $240 per night), estimated over the bins from 30 to 44. LATE (Nights) is the normalized excess mass in nights divided by the share of bunchers, the reduction in rental nights of the marginal buncher.
$p < 0.10$*, $p < 0.05$**, $p < 0.01$***.

face an average additional annual liability of $3,000 upon losing residential treatment, the estimated bunching mass implies approximately $13.8 million in unrealized property tax revenue in Florida over 2017–2024 (4,600 bunching hosts × $3,000 = $13.8 million.) This is the revenue local governments would have collected had these hosts reported rental activity above the threshold and paid the commercial rate, rather than retaining the homestead exemption.[22] The calculation is illustrative, but it captures the fiscal consequence of bunching. Hosts can continue earning STR income while local governments forgo the additional property tax revenue associated with commercial classification.

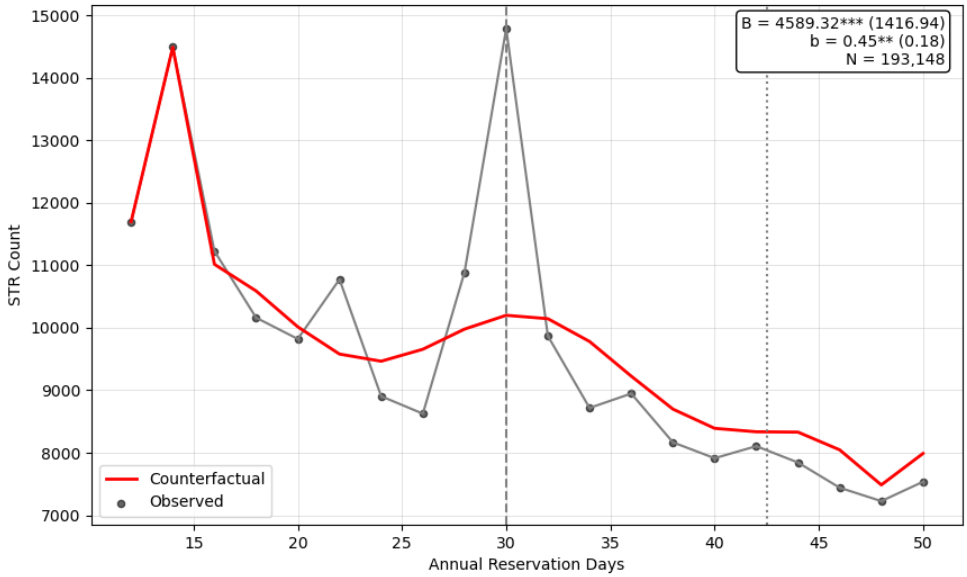
6.2 The Dominated Region and Optimizing Frictions

Next, I assess whether the behavior of STR hosts is consistent with the profit-maximizing behavior depicted in the Conceptual Framework of Section 3. According to the Conceptual Framework, STRs that locate in the dominated region are not profit-maximizing and thus, could improve their profits by locating outside the region. Given the average annual tax

[22]See Appendix F for details on estimated savings from the homestead exemption.



(a) Pre-VCA (2015)



(b) Post-VCA (2017–2024)

Figure 4: Bunching at the 30-Day Property Tax Threshold Before and After Airbnb VCAs

Notes: The figures plot the distribution of annual reservation days for likely homeowner-hosts with one unit. The dashed gray line marks the 30-day homestead exemption threshold, and the dotted gray line at 42.5 days marks the end of the dominated region. Black dots represent actual binned counts; the red line shows the counterfactual distribution from the bunching estimator. Bin width is 2 days. I include a fixed point control at 14 days. Ten bins are used on each side of the threshold. B is estimated excess mass at the threshold and b is normalized excess mass. Standard errors from a residual bootstrap with 1,000 replications are in parentheses. p<0.10*, p<0.05**, p<0.01***.

savings of $3,000 from the homestead exemption and average nightly revenue of $240, the dominated region is [30, 42.5] days for Florida. Therefore, an average property must rent 12.5 days in Florida to recoup the cost of the additional property tax.

By comparing the dominated region with the marginal buncher's response, I find that bunchers in Florida respond rationally to the change in tax rate at the rental day threshold. The marginal buncher reduces rental nights by 16.91, from about 46.9 nights to 30, about 4.4 nights more than the 12.5-night width of the dominated region. Accounting for operational costs would explain why Florida bunchers would bunch more than 12.5 nights, since the 12.5 night estimate is based on revenue only, not profits.

The presence of any STRs in the dominated region, while surprising, could be due to several factors. First, because of data limitations, I cannot identify which STRs are truly owner-occupied or actually claiming the exemptions, so the estimated responses may include ineligible properties. I do not expect a behavioral or bunching response from properties that are ineligible for the homestead exemption at the rental-day limit. Second, since the homestead exemption in Florida is forfeited after exceeding the 30-day threshold in two consecutive years, some hosts may rationally locate in the dominated region in a given year if they plan to remain below the threshold the following year.[23] For these two reasons, the estimates in this study should be interpreted as lower bounds on bunching and tax evasion.

Second, optimizing frictions likely explain part of the observed mass in the dominated region. Optimizing frictions occur when hosts face informational or logistical constraints that prevent precise optimization around the cutoff. For example, a host may intend to exceed the rental threshold, and does, but does not receive enough bookings to pass the dominated region. In which case, the host remains stuck in the dominated region although it would be optimal to avoid that region. In that case, some STRs in the dominated region may reflect unrealized rental demand or imperfect adjustment rather than deliberate noncompliance or ignorance of the threshold.

6.3 Reduced Form Elasticity Estimates: Rental Nights to Tax Rate

While the previous results described rental behavior of STR hosts, the reduced-form elasticity captures how sensitive STR rental activity is to changes in property taxes. I calculate the elasticity as follows:

$$\epsilon_{N,t} = \frac{\%\Delta N}{\%\Delta T} = \frac{\hat{b}/N^*}{\Delta\lambda/\lambda} \tag{8}$$

[23]To the extent this behavior occurs in Florida, some observations in the dominated region classified as non-compliers may instead reflect hosts strategically planning their rental activity given the two-year rule. This would attenuate the estimated share of bunchers and bias the corresponding LATE estimates upward.

The percentage change in rental nights $\%\Delta N$ is the average reduction in nights $\hat{b}$ in response to the cutoff. I also estimate this with $\hat{b}_{adj}$, which is the reduction in nights by STR hosts who are bunchers and more responsive to the rental day limit. I find the percentage change in property taxes $\%\Delta T$ by dividing the change in taxes from losing the homestead exemption, $\Delta\lambda$, by the total tax liability with the homestead exemption, λ. I interpret the elasticities as the percentage change in rental nights for the STR host from a 1% increase in the property tax rate. The calculations for the elasticity estimates are included in Appendix G.

For STRs just below the 30-day threshold in Florida, the average STR reduces rental activity by approximately 0.90 nights. Using the illustrative examples in Appendix G, this corresponds to an average elasticity of 0.020–0.056. The range of the estimate depends on the amount of the SOH protection that the homestead has accumulated over time.[24]

For marginal bunchers in Florida, the effect is substantially larger. These hosts reduce their bookings by 16.91 nights at the threshold. Depending on the amount of SOH protection, a 1% increase in property taxes leads to a 0.38–1.05 percent decrease in rental nights for marginal bunchers.[25] This range is broadly consistent with Bibler et al. (2021), who find that a 10% increase in the tax rate corresponds to a 3.6% reduction in nights booked. At the lower bound of my estimates, I estimate a comparable 10% increase in property taxes is associated with a 3.8% decrease in rental nights. Overall, the elasticity estimates show that the threshold affects a small, highly responsive group of marginal hosts, even though the average STR host is relatively insensitive to the associated property-tax increase.

7 Discussion

The bunching evidence speaks directly to the question of whether third-party reporting affects compliance with taxes it does not target. Airbnb's VCAs report nothing about host-level rental days, yet bunching below the 30-day threshold intensifies after their introduction. Even indirect reporting, then, appears sufficient to change how hosts manage rental activity around the property tax exemption.

Reporting on sales and accommodation taxes likely affects property tax behavior due

[24]The FL elasticity estimates should be interpreted as illustrative because the tax-savings calculations likely understate the true value of the homestead exemption for many Florida homeowners. The examples in Appendix G include only county and school district millage rates and therefore provide a lower-bound estimate of the tax savings associated with homestead status. In practice, municipal and special district taxes may further increase the cost of losing the exemption. Moreover, the value of the exemption depends critically on the amount of accumulated SOH protection, which varies substantially across homeowners and can generate much larger tax increases than those reflected in the illustrative examples.

[25]The lower value uses Example B (large SOH gap; a tax increase of about $3,400) and the upper value uses Example A (small SOH gap; a tax increase of about $2,000). See Appendix G for the calculations.

primarily to perceived enforcement. The VCAs bring Airbnb into the formal tax collection process and signal that the platform cooperates with tax authorities, raising hosts' perceived likelihood that their rental activity could be detected. Before the VCAs, hosts could exceed the threshold with little risk of detection. Afterward, STR activity became more visible to authorities, even though rental days themselves were not reported. Notably, Airbnb's communications to hosts about the VCAs do not mention the homestead exemption. Hosts therefore appear to respond to a general increase in the perceived likelihood of detection rather than to new information about the property tax rule itself.

Broader tax salience may play a secondary, complementary role. Homestead exemption eligibility is not verified annually, and many hosts may be unaware that the exemption is conditional on staying below the rental-day limit. In one Florida county, for example, a tax appraiser reported that 80% of exemption fraud cases in 2017 stemmed from ignorance rather than deliberate evasion (Thomson Reuters Institute 2018), although self-reports from detected cases likely understate intentional evasion. Because the VCAs do not mention the exemption, any salience effect is likely indirect and less prominent than perceived enforcement.[26]

The size of the financial incentive helps explain why hosts respond so strongly once they adjust their behavior. Because the homestead exemption and its rental-day limit predate the VCAs, the incentive itself cannot account for the change in bunching after the agreements; rather, it amplifies the response. The homestead exemption in Florida is particularly valuable because of the SOH cap. Over several years, SOH protection can save a homeowner tens of thousands of dollars, particularly in high-value markets. Upon losing the homestead exemption, the property is immediately reassessed at market value for all future tax years. The homeowner therefore faces not only a higher current property tax bill but also the loss of accumulated assessment limitation benefits that constrain future tax growth. Homeowners who have held their properties for many years may have assessed values substantially below market values, making the loss of homestead status especially costly. These future tax consequences likely amplify the response and elasticity at the threshold, especially by marginal bunchers.

The fiscal consequences of the VCAs depend on how hosts who previously evaded respond once reporting increases. Before the agreements, a host could rent beyond the 30-day

[26]Evidence from other settings suggests that information alone can move compliance. In North Alabama, the Alabama Mountain Lakes Tourist Association partnered with Granicus to send notification letters to over 2,500 STR owners reminding them of their lodging tax obligations. Within two months, the area saw a 21% increase in billed revenues through the compliance system and an 8% increase in statewide STR tax collection among the 16 counties covered (Granicus 2023). When noncompliance stems from a lack of information rather than intentional evasion, relatively low-cost interventions such as awareness campaigns may improve compliance without resorting to costly audits.

threshold, underreport rental days, and continue paying the lower residential rate. After the agreements, that host faces three options, each with different implications for property tax revenue relative to evasion. First, the host can continue renting above the threshold and accept commercial treatment, in which case the government collects the incremental tax revenue from the host shifting from the residential rate to the commercial rate. Second, the host can reduce rental days to remain below the threshold. This host becomes compliant but continues paying the residential rate, so property tax revenue is unchanged relative to evasion. In this case, the government forgoes the commercial tax revenue it would have collected under the first response. Third, the host can end up in the dominated region just above the threshold, losing residential treatment without earning enough additional rental income to justify the higher tax. The government collects the incremental liability from these hosts, but their position does not reflect an optimizing choice and more likely stems from booking uncertainty, scheduling frictions, or incomplete knowledge of the rule.

The bunching estimator captures the second response. Had these hosts complied at their prior level of activity, reporting more than 30 rental days and paying the commercial rate, the government would have collected approximately $13.8 million in additional property tax revenue over the study period in Florida.[27] Because they instead reduce rental activity to remain below the threshold, they continue paying the residential rate, and the realized property tax gain from these hosts relative to evasion is zero. Any property tax gains from improved compliance therefore come from hosts in the first and third groups rather than from bunchers. The $13.8 million figure should not be interpreted as the net fiscal impact of STRs or of the VCAs, since STRs also generate sales, accommodation, and tourism-related tax revenue. Rather, it illustrates that the fiscal consequences of enforcement depend not only on whether it improves compliance, but also on how taxpayers adjust their underlying activity.

These findings should be interpreted with caution given certain limitations of the data. Most notably, the data do not identify STRs with the homestead exemption, so estimates in this study represent lower bounds on bunching and tax evasion. Future research with property-level tax records or verified exemption claims could address this limitation. Additionally, the bunching estimates capture only the intensive-margin effect of property tax

[27] I calculate unrealized property tax revenue as $\widehat{L} = \widehat{B} \times \Delta\tau$, where $\widehat{B}$ is the estimated excess bunching mass and $\Delta\tau = T^{\text{commercial}} - T^{\text{residential}}$ is the incremental annual property tax liability from losing residential treatment. Using $\widehat{B} = 4{,}600$ and $\Delta\tau = \$3{,}000$ yields $\widehat{L} = \$13.8$ million. Allowing $\Delta\tau$ to range from $2,000 to $3,500 yields $\widehat{L} \in$ [$9.2 million, $16.1 million].

policy on STR supply and therefore likely understate the overall effect.[28] More research is needed to understand the extensive-margin effects of property tax thresholds.

8 Conclusion

This paper asks whether reporting designed for one tax can improve compliance with another, using Florida's short-term rental (STR) market as a test case. Using listing-level data and a bunching estimator, I examine how hosts respond to rental-day thresholds tied to large property tax exemptions. Furthermore, I exploit the introduction of Airbnb's Voluntary Collection Agreements (VCAs) as an enforcement shock to measure how hosts alter rental activity along both compliance and evasion margins. I contribute both to the literature on STR regulation and tax compliance by studying the impact of property tax design and indirect third party reporting on STR activity.

The bunching patterns before and after the VCAs tell a clear story of reducing tax evasion and STR activity. When Airbnb began collecting and remitting sales and accommodation taxes on hosts' behalf through Voluntary Collection Agreements (VCAs), the platform became part of the tax system without ever sharing the rental-day counts that determine exemption status. Nevertheless, hosts responded. Excess mass just below the 30-day homestead exemption threshold grew to roughly 4,600 listings after the agreements took effect when excess mass had been essentially absent beforehand. The hosts most sensitive to the cutoff reduced their rentals by over 16 nights to retain the property tax exemption.

Whereas prior work documents enforcement spillovers across taxpayers within the same tax (Pomeranz 2015), this paper shows that reporting can spillover across taxes for the same taxpayer. The evidence points primarily to perceived enforcement as the channel driving the increase in compliance. By bringing Airbnb into the formal tax collection process, the VCAs signal that the platform cooperates with tax authorities and raise hosts' perceived likelihood of detection. Notably, Airbnb's communications about the agreements do not mention the homestead exemption, so hosts appear to respond to greater visibility rather than to new information about the rule itself. Broader tax salience may play a secondary role. Collecting taxes on hosts' behalf may prompt some hosts to examine their other obligations and discover a threshold they did not know existed. The size of the response reflects the value of what is at stake. In Florida, losing the owner-occupied homestead exemption triggers

[28]The intensive margin captures how existing STR hosts adjust the number of nights they rent their properties in response to property tax policy. However, property tax thresholds may also affect the extensive margin by discouraging entry into the STR market or encouraging existing hosts to exit altogether. Because the bunching estimator relies on observed rental activity, it cannot capture homeowners who choose not to enter the STR market because of the risk of losing owner-occupied tax benefits.

reassessment at market value and forfeits accumulated Save Our Homes protection, so crossing the threshold can be costly for long-tenured owners. Because this incentive predates the VCAs, it amplifies the response rather than explaining it. Compliance responses to indirect reporting are therefore likely largest where exemptions are most valuable.

The fiscal and policy implications of these findings depend on the government's objective. If the goal is to limit the commercialization of owner-occupied housing, rental-day thresholds paired with improved reporting are effective. If the goal is to raise property tax revenue with minimal distortion, the outcome is less favorable. Relative to a world in which hosts evade, the government collects more revenue only from hosts who now accept commercial treatment; hosts who bunch remain at the residential rate. Compared with renting beyond the threshold and paying the commercial rate, however, hosts' decision to stay below the cutoff creates an efficiency loss in the rental market since hosts distort their rental behavior. Moreover, it costs the government approximately $13.8 million in unrealized property tax revenue. These fiscal stakes are especially consequential in tourism-heavy areas, where STR activity increases demand for police, EMS, and other public safety services (Ke et al. 2021, van Holm & Monaghan 2021, Reinhard & Roth 2024) and raises pressure on emergency departments, waste systems, and other municipal operations (Arora et al. 2025, Office of the City Treasurer 2021).

For policymakers seeking to preserve the intent of owner-occupied exemptions, the results suggest three pathways. First, third-party reporting can be paired with tax rules that do not create such a sharp incentive to bunch, such as a gradual phaseout of the exemption or a smoother increase in tax liability as rental activity rises. This would preserve the ability to tax commercial use while allowing rental decisions to reflect underlying economic opportunities rather than an arbitrary cutoff. Second, if much noncompliance stems from a lack of salience, low-cost interventions such as notification letters and clear communication of exemption rules can improve compliance without costly audits. Third, integrating platform data with property tax records would allow assessors to verify exemption claims directly rather than relying on indirect spillovers. In an era where platform economies blur the line between private and commercial use, the interaction between tax design and third-party reporting will determine whether property tax exemptions remain instruments of fairness or become loopholes for evasion.

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A Appendix: Additional Short Term Rental Industry Background

This appendix provides additional background on the short-term rental (STR) industry and enforcement practices that inform the empirical analysis. It summarizes the structure and market concentration of major STR platforms and discusses variation in local enforcement capacity relevant to property tax compliance.

The modern STR industry emerged with the founding of Airbnb in 2007, when Brian Chesky and Joe Gebbia began renting out an air mattress in their living room as a bed and breakfast. The following year, they launched the Airbnb platform (Lagorio-Chafkin 2010). On the Airbnb platform, people seeking temporary accommodations can connect with hosts offering their spare rooms or entire homes for lodging. Hosts choose the price per night for their accommodations, and Airbnb collects fees from both hosts and guests (Koster et al. 2021).

Other platforms, such as Vacation Rental by Owner (VRBO) and Booking.com, offer similar services. Because of the technological innovations underlying these platforms, STR hosts can operate with relatively low overhead costs and compete directly with traditional accommodation providers, including hotels and bed and breakfasts (Guttentag 2015, Morgan & Kuch 2015, Orsi 2013).

As a result, the STR market has become highly concentrated among a small number of platforms. In 2025, Airbnb held roughly 43% of the U.S. short-term rental market, followed by VRBO at 21% and Booking.com at 8%. Together, Airbnb and VRBO account for about two-thirds of U.S. listings (Rental Scale Up 2025). Globally, these three platforms control more than 70% of short-term rental revenue (Yahoo Finance 2025). This concentration underscores the importance of platform-level reporting and data-sharing for understanding STR activity and tax compliance.

Against this backdrop, enforcement of homestead exemption rules varies substantially across counties, and publicly available data on audit rates or investigative capacity remain limited. However, emerging evidence suggests that enforcement has intensified in some jurisdictions as advances in data analytics have made it easier for property appraisers to cross-reference tax rolls, booking activity, and other administrative records to identify suspicious properties (Thomson Reuters Institute 2018). Some counties have created dedicated fraud-investigation units or citizen hotlines, such as the Orange County Property Appraiser's office, which reports recovering more than $550 million in improper exemptions after hiring former law-enforcement professionals and encouraging anonymous tips from residents (Erb 2016). Similarly, the Duval County, Florida property appraiser identified approximately 1,900 homes

erroneously claiming the homestead exemption in 2017, resulting in $11 million in recovered taxes (Thomson Reuters Institute 2018).

B Appendix: STR Market in Florida

This appendix provides additional background on the short-term rental (STR) market in Florida that informs the empirical analysis. The figures below document the growth and geographic distribution of STRs in Florida over the sample period. The following discussion describes the regulatory environment surrounding Airbnb's Voluntary Collection Agreements and their relationship to property tax enforcement.

Figure 5 presents the share of STRs in the major FL metro areas. Generally, STRs are concentrated in coastal cities in Florida as well as in Orlando. Figure 6 shows the growth of STRs in Florida by host type from 2015-2024. Most of the STR growth comes from hosts managing one property or more than 20 properties.

The homestead exemption and its associated 30-day rental day limits both predate Airbnb, which launched in 2008, and the widespread adoption of short-term rentals (STRs). In the mid-2010s, Airbnb began collecting and remitting accommodation and sales taxes on behalf of STR hosts to local governments through VCAs. Consequently, these governments received more information about STR activity within their jurisdictions.[29] Even though VCAs do not remit property taxes, I use the VCAs as a proxy for greater oversight, regulation, and enforcement in the STR market, which should lead to more enforcement in property tax collection as well.

[29]The extent to which local governments in Florida use this data for enforcement is not fully documented. In some cases, counties have leveraged this information to identify exemption fraud. For example, task forces such as the one in Orange County, Florida cross-referenced property tax records with Airbnb listings to uncover fraudulent homestead exemptions. See Kelly Philips Erb, "Investigations, Tips Lead To More Than $550 Million In Property Tax Fraud Recovery," *Forbes*, 2016. https://www.forbes.com/sites/kellyphillipserb/2016/06/02/investigations-tips-lead-to-more-than-550-million-in-property-tax-fraud-recovery/.

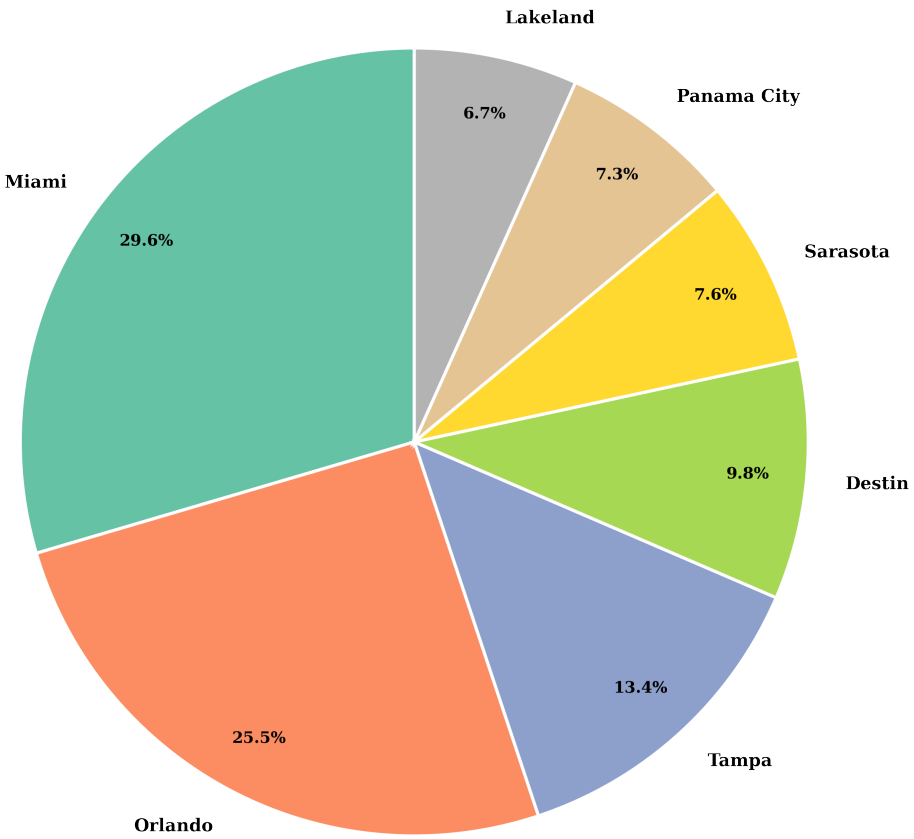


Figure 5: Metro Areas for STRs in FL in 2024

Notes: This figure shows the percentage of STRs in Florida in the major metropolitan areas across the state. Data is from AirDNA for the year 2024. The STRs included in the figure represent only two-thirds of the total STRs in Florida.

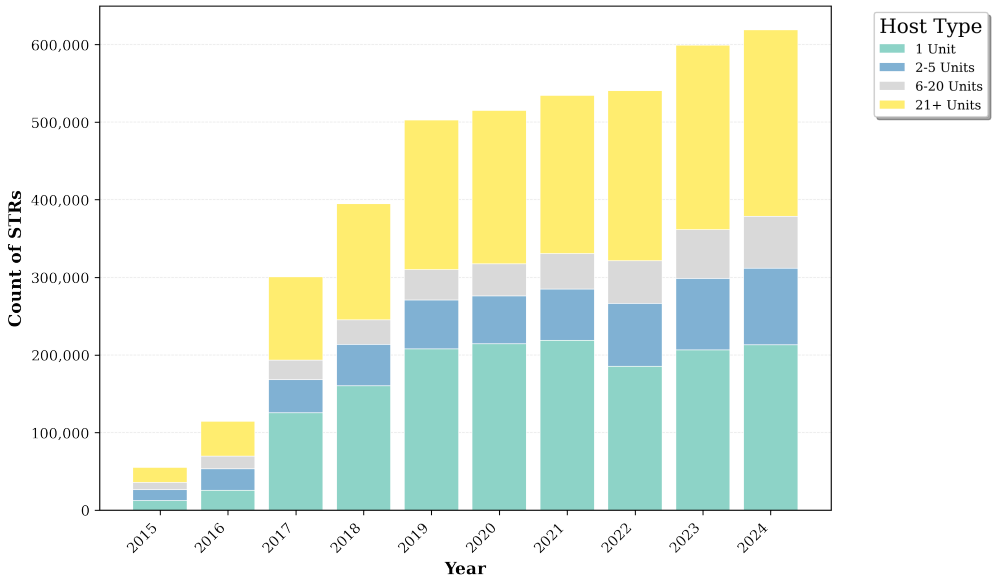


Figure 6: Number of STRs in FL

Notes: This figure displays the number of STRs in Florida per year from 2015-2024 as reported by AirDNA. The bars in the chart are divided based on the host type, which is how many STR units the host is managing.

C Appendix: Bunching Estimator Details

This appendix presents the technical details of the bunching estimator underlying the empirical results. I outline how the distribution of annual STR rental days is binned and modeled, how excess mass at the exemption threshold is identified, and how the counterfactual distribution is constructed and adjusted to satisfy the integration constraint.

I plot the distribution of annual STR days based on the actual data with the following equation:

$$c_j = \sum_{i=0}^{p} \beta_i (N_j)^i + \sum_{i=N_L}^{N_U} \gamma_i \mathbb{1}[N_j = i] + \sum_{r \in R} \rho_r \mathbb{1}[\frac{N_j}{r} \in \mathbb{N}] + \sum_{k \in K} \theta_k \mathbb{1}[N_j \in K \wedge N_j \notin [N_L, N_U]] + \epsilon_j \tag{9}$$

The STRs are put into j bins based on their annual rental days N, and each bin is a range of annual rental days. The number of STR observations in each bin is c_j. The first term after the equal sign is the polynomial of order p with coefficients β_i that fits the binned data. The binned data are the counts of STR properties that fall into each bin. N_j is the maximum N in bin j of width δ.

The second term in the equation are the bunching dummies that capture excess mass at the threshold. These dummy variables, γ_i, measure how much of the density is concentrated at the exemption limit relative to the rest of the distribution. N_L and N_U are the lower

and upper bounds of the bunching region, respectively. They represent the area of the distribution that is impacted by the rental day limit. I estimate N_L and N_U by looking for visual changes in the rental day distribution.

The last two terms in the estimation equation account for natural or mechanical bunching that is unrelated to the property tax threshold. The third summation term controls for round-number bunching through the coefficients ρ_r. The set R includes round values where STR hosts may cluster bookings. For instance, STRs frequently offer discounts for week-long stays, leading to visible spikes in the distribution of annual rental days at multiples of seven. The coefficients ρ_r capture the average excess mass at these round-number points. Including this control allows the estimator to separate tax-induced bunching at the 30-day limit from predictable, calendar-driven clustering in the data.

The fourth summation term adjusts for fixed bunching points elsewhere in the distribution that are not related to the rental day threshold or a natural bunching point. The set K includes prominent spikes in the rental day distribution that fall outside the bunching region $[N_L, N_U]$ but could otherwise distort the estimated distribution. The coefficients θ_k measure and absorb these unrelated concentrations of mass, ensuring that the fitted values are not biased by excess clustering at other points in the distribution. Similar to round number bunching, these fixed bunching points are visually detectable upon examining the rental day distribution.

Next, I estimate $\hat{c_j}$, the counterfactual binned count of STR properties and adjust the counterfactual to meet the integration constraint correction. The counterfactual bin count is estimated as follows:

$$\hat{c_j} = \sum_{i=0}^{p} \beta_i (N_j)^i + \sum_{r \in R} \rho_r \mathbb{1}[\frac{N_j}{r} \in \mathbb{N}] + \sum_{k \in K} \theta_k \mathbb{1}[N_j \in K \wedge N_j \notin [N_L, N_U]] + \epsilon_j \tag{10}$$

The variables in Equation 10 are defined as above in Equation 9. However, the counterfactual estimation omits the bunching-region dummies γ_i to recover how the distribution would appear without the notch. The integration correction constraint accounts for bunchers' response to the notch. When bunchers come from above the threshold and locate at the notch, then the proceeding bins have less STRs than what would have existed without the notch. This correction shifts $\hat{c_j}$ upward to compensate, so that the missing mass under the counterfactual equals the excess bunching at the notch (Chetty et al. 2011, Mavrokonstantis & Seibold 2022).

D Appendix: Sample Construction

Table 2 details the number of observations remaining at each step of the sample restriction process for Florida. Beginning with the full AirDNA dataset, I sequentially restrict to listings likely owned by homeowners, listings managed by hosts operating five or fewer units, entire-home listings, and finally listings managed by hosts operating a single unit. This last restriction constitutes the main estimation sample used throughout the analysis.

Table 2: Sample Restrictions: Florida

Sample Restriction	Observations
Full sample	4,176,235
Likely homeowner or property type missing	3,148,586
Host manages 1–5 units	1,634,717
Entire home listing	1,461,251
Host manages 1 unit (main sample)	1,168,880

Notes: Each row reports the number of listing-year observations remaining after applying the indicated restriction. Restrictions are applied sequentially.

E Appendix: Descriptive Statistics

Table 3 presents the descriptive statistics for Florida STRs in the restricted sample. The average Florida rental has 3 bedrooms and 2.5 baths and has been operating for almost 5 years. Average annual reservation days is 147, and the occupancy rate is approximately 50%. Florida STRs on average generate $33,300 of revenue annually. Accordingly, the average nightly rate in Florida is about $240 with higher average rates in Miami and lower ones in Orlando and Tampa. The average length of a reservation in Florida is 6.2 days.

Table 3: FL Descriptive Statistics: Likely Homeowners with 1 Unit, 2015-2024

Variable	All of FL	Orlando	Miami	Tampa
A. Property Characteristics				
Num. of Bedrooms	3.09	4.49	2.41	2.32
Bathrooms	2.54	3.51	2.07	1.87
Share of House Listings	0.49	0.48	0.54	0.51
Share of Condo Listings	0.34	0.14	0.36	0.41
Share of Properties on Airbnb and Vrbo	0.08	0.05	0.09	0.12
Years as a STR	5.11	4.95	4.77	4.93
B. Other Characteristics				
Annual Number of Reservations	31.44	34.60	35.78	33.20
Annual Reservation Days	147.01	150.89	140.49	158.27
Average Stay in Days	6.17	5.06	5.26	6.69
Average Occupancy Rate	0.53	0.55	0.52	0.56
Annual Revenue in Dollars	33,311.62	30,557.57	36,297.71	32,508.11
Revenue / Revenue Potential	0.81	0.82	0.81	0.80
Average Nightly Rate	236.16	213.37	283.48	207.39
Observations	1,168,880	239,183	136,272	89,119

*Notes: This table presents summary statistics for a subsample of Florida STRs from 2015-2024. The "Likely Homeowners with 1 unit" subsample consists of STRs that are a townhouse, condo, house, or villa and that have a host managing 1 unit. The reported statistics are means. The variables in Panel B are weighted by the number of annual reservation days for each property. Data comes from AirDNA.

F Appendix: Property Tax Calculations

Today, the property tax remains a fundamental funding source for local governments in the U.S., particularly for counties (Yushkov 2024). Many local governments rely on property taxes to fund public services and public schools. On average, the property tax provides 30% of revenue for local governments and raised $581 billion in 2021 (Lincoln Institute 2024). The different assessment ratios and millage rates from the homestead exemption are meant to redistribute the property tax burden more equitably. However, levying an equitable property tax requires understanding the extent and potential for tax evasion.

Thirty-three states offer a version of the homestead exemption for owner-occupied residential property. For most of the 33 states, these exemptions are granted to all homeowners. However, select states restrict the program to certain groups, such as veterans, people over 65, first responders, and low-income earners (Lincoln Institute 2024).

Florida Property Taxes

Florida's constitution limits school mills and non-school mills (e.g., county and city) to 10 each unless voters have approved a higher rate. Additionally, the new millage rates must be less than 110% of the rollback rate, which is the millage rate that would raise the same revenue as the previous year. For education funding, the Florida legislature sets a uniform school millage rate for the state, effectively mandating the amount of funds that each school district must raise for school operations.

Below, I present a mathematical representation of the tax liability for a non-homestead property and a homestead property in Florida. The total property tax liability for an non-homestead home in Florida in year t can be expressed as:

$$A_t = min(h_t, A_{t-1} \cdot 1.10) \tag{11}$$

$$\lambda_t = A_t \cdot (\frac{m_n + m_s}{1000}) \tag{12}$$

where the first equation defines the assessed or taxable value A_t for home h_t in year t. For non-homestead, the assessed value is the lesser of the home's market value h and the 110% of the previous year's assessed value A_{t-1}. The millage rates are split into two parts: m_n, the non-school millage rate applied to county and municipal services, and m_s, the millage rate specifically levied for school funding.

The corresponding tax liability for an owner-occupied homestead property is:

$$A_t = min(h_t, A_{t-1} \cdot min(1.03, 1 + CPI_t)) \tag{13}$$

$$A_{t,F} = A_t - 25,000 \tag{14}$$

$$\lambda_t = A_{t,F} \cdot (\frac{m_n + m_s}{1000}) - E \cdot \frac{m_n}{1000} \tag{15}$$

$$E = \begin{cases} 25{,}000, & \text{if } A_t > 75{,}000, \\ A_t - 50{,}000, & \text{if } 50{,}000 < A_t \leq 75{,}000, \\ 0, & \text{if } A_t \leq 50{,}000. \end{cases} \tag{16}$$

The assessed value for the homestead A_t in year t is the minimum of the market value of the home h and the capped assessed value based on the SOH protection. Under SOH, the capped assessed value for year t is the previous year's assessed value A_{t-1} multiplied

by 1.03 or 1+CPI, whichever is lower. Since the Consumer Price Index (CPI) determines the inflation rate for year t, I use CPI_t as shorthand for the inflation rate. The homestead exemption exempts the first $25,000 of assessed value to be exempt from all mills. Therefore, the taxable value used to determine tax liability for year t is $A_{t,F} = A_t - 25,000$. In total property taxes, a homeowner pays the non-school m_n and school millage m_s rates on the $A_{t,F}$ minus the last term in the equation, $E \cdot m_n$. The last term $E \cdot \frac{m_n}{1000}$ exempts up to $25,000 from the non-school millage m_n for homestead properties over $50,000.

G Appendix: Average Property Tax Savings

G.1 Homestead Exemption and Save Our Homes (SOH) Protections

Under Florida law, when a homestead exemption and its associated Save Our Homes (SOH) protections are forfeited (for example, by renting the property for more than 30 days per year for two consecutive years under Fla. Stat. §196.061(1)), the property is immediately reassessed at full market value and loses the homestead exemptions.

Let h_t denote the market value of the home in year t, A_t the SOH-capped assessed value, m_n the non-school (local) millage rate, and m_s the school millage rate.

The annual tax liability **with** the homestead exemption and SOH protections is:

$$\lambda_{t,\text{HS}} = (A_t - 25{,}000) \cdot \frac{(m_n + m_s)}{1000} - 25{,}000 \cdot \frac{m_n}{1000} \tag{17}$$

and **without** the exemption (after reassessment to market value):

$$\lambda_{t,\text{noHS}} = h_t \cdot \frac{(m_n + m_s)}{1000} \tag{18}$$

The difference between the two represents the annual property tax increase from losing the exemption:

$$\Delta\lambda_t = (h_t - A_t + 25{,}000) \cdot \frac{(m_n + m_s)}{1000} + 25{,}000 \cdot \frac{m_n}{1000} \tag{19}$$

Numerical Examples The illustration below uses an average house value of $h_t = \$400{,}000$ and the average population-weighted, county-only millage rates from the Florida Department

of Revenue, Property Tax Oversight (2025): $m_n = 8.349$, and $m_s = 5.946$.[30]

$$\frac{m_n + m_s}{1000} = 0.014295, \quad \frac{m_n}{1000} = 0.008349$$

Example A (moderate SOH gap): $A_t = \$300{,}000$

$$\Delta\lambda_t = (400{,}000 - 300{,}000 + 25{,}000) \times 0.014295 + 25{,}000 \times 0.008349 = 1{,}995.60$$

The annual property tax **with the homestead exemption** is:

$$\lambda_{t,\text{HS}} = (300{,}000{-}25{,}000){\times}0.014295{-}25{,}000{\times}0.008349 = 275{,}000{\times}0.014295{-}208.725 = 3{,}722.40$$

Thus, the homeowner pays roughly \$3,720 annually with the exemption and would pay \$5,720 without it, yielding an annual savings of approximately \$2,000.

Example B (large SOH gap): $A_t = \$200{,}000$

$$\Delta\lambda_t = (400{,}000 - 200{,}000 + 25{,}000) \times 0.014295 + 25{,}000 \times 0.008349 = 3{,}425.10$$

The annual property tax **with the homestead exemption** is:

$$\lambda_{t,\text{HS}} = (200{,}000{-}25{,}000){\times}0.014295{-}25{,}000{\times}0.008349 = 175{,}000{\times}0.014295{-}208.725 = 2{,}292.90$$

So the homeowner pays roughly \$2,290 annually with the exemption and would pay \$5,720 without it, resulting in annual savings of about \$3,430.

These values correspond directly to the λ (denominator) and $\Delta\lambda$ (numerator) terms used in the elasticity calculation for Florida. The tax with the exemption ($\lambda_{t,\text{HS}}$) serves as the reference base when expressing the property tax increase as a percentage change.

The examples demonstrate how the SOH cap compounds over time: longer ownership periods and greater appreciation produce larger differences between assessed and market

[30] m_s is the population-weighted average school millage rate, equal to the sum of School Board Operating and School Board Debt Service levies. m_n is the population-weighted average non-school, non-municipal millage rate, which includes county government operating, debt service, dependent special districts, independent special districts, and municipal service taxing units (MSTUs). Both rates are computed from the Florida Department of Revenue *Millage and Taxes Levied Report* (Florida Department of Revenue, Property Tax Oversight 2025) and exclude municipal levies. Homeowners residing within city limits face a higher total millage burden. Population weights are county-level estimates from the Bureau of Economic and Business Research (U.S. Census Bureau 2023).

values, thereby increasing the effective tax savings from the homestead exemption. The cumulative savings over multiple years can easily reach tens of thousands of dollars for long-tenured homeowners in high-growth markets.

Remarks.

- The algebra in (19) explicitly reflects the *immediate* reset of assessed value to the market value h_t when homestead/SOH protections are lost; the reassessment is the reason the leading term contains $h_t - A_t$.
- The examples above are annual increases (not cumulative present values). If one reports the cumulative additional tax burden over T future years after reassessment, one should sum the annual $\Delta\lambda_{t+\tau}$ for $\tau = 0, \ldots, T-1$ while accounting for subsequent growth in millage rates and assessed values (because, after the reset, the reassessed basis is h_t and the former SOH cap no longer applies). This cumulative amount can reach several tens of thousands of dollars for long-tenured owners with large SOH gaps — but it is *not* the annual figure.

Elasticity Calculations for Florida I compute the elasticity of rental nights with respect to the property tax as

$$\epsilon_{N,t} = \frac{\%\Delta N}{\%\Delta T} = \frac{\hat{b}/N^*}{\Delta\lambda/\lambda}, \tag{20}$$

where $N^* = 30$ is the rental-night threshold and $\Delta\lambda/\lambda$ is the percentage increase in the property tax when the homestead exemption is lost. Because the estimates use bins of 2 nights and the normalized excess mass is measured relative to the counterfactual count in one bin, the response in nights is twice the normalized excess mass: the average response is $\hat{b} = 2 \times 0.450 = 0.90$ nights, and the response of the marginal buncher is $\hat{b}_{adj} = 16.91$ nights.

For the **STR from Example A**, $\hat{b} = 0.90$ nights and $N^* = 30$:

$$\epsilon_{N,t} = \frac{0.90/30}{1,995/3,722} = \frac{0.030}{0.536} = 0.056$$

A 1 percent increase in the property tax reduces rental nights by about 0.056 percent.

For the **STR from Example B**, $\hat{b} = 0.90$ nights and $N^* = 30$:

$$\epsilon_{N,t} = \frac{0.90/30}{3,425/2,293} = \frac{0.030}{1.494} = 0.020$$

A 1 percent increase in the property tax reduces rental nights by about 0.020 percent. For

the average STR, the response is highly inelastic in both examples, because most hosts near the threshold do not respond.

I calculate the same elasticity for the **marginal buncher**, using the local average treatment effect (LATE): the normalized excess mass divided by the share of bunchers, in nights. The marginal buncher would have rented about 46.9 nights without the threshold and reduces their rental nights by about 16.91 nights to stay at 30, which is about 4.4 nights beyond the 12.5-night dominated region.

For the **STR from Example A**, $\hat{b}_{adj} = 16.91$ nights and $N^* = 30$:

$$\epsilon_{N,t} = \frac{16.91/30}{1,995/3,722} = \frac{0.564}{0.536} = 1.05$$

A 1 percent increase in the property tax reduces the marginal buncher's rental nights by about 1.05 percent, an approximately unit elastic response.

For the **STR from Example B**, $\hat{b}_{adj} = 16.91$ nights and $N^* = 30$:

$$\epsilon_{N,t} = \frac{16.91/30}{3,425/2,293} = \frac{0.564}{1.494} = 0.377$$

A 1 percent increase in the property tax reduces the marginal buncher's rental nights by about 0.38 percent. As expected, the marginal buncher responds much more to the threshold than the average STR. Because many hosts in the dominated region do not respond, and because the sample cannot separate hosts who claim the homestead exemption from those who do not, these elasticities should be read as lower bounds.

G.2 Summary of Estimated Property Tax Savings

Table 4 summarizes the key statutory parameters and the estimated average annual property tax savings from the homestead exemption in Florida. The estimates incorporate both the standard homestead exemption and the SOH assessment cap, use illustrative examples for typical assessed-value gaps, and account for the immediate reassessment of the property to full market value when the exemption is lost.

Table 4: Comparison of Average Property Tax Savings for Owner-Occupied Homes

Parameter / Measure	**Florida (Homestead + SOH)**
Median Home Value ($)	400,000
Assessment Ratio (Owner-Occupied)	— (uses SOH-capped A_t)
Assessment Ratio (Non-Owner)	— (reset to full market value h_t)
Assessment Growth Cap	3% or CPI (SOH)
School Millage Exemption	None beyond first $25,000
Typical Non-School Millage (m_n)	8.349 mills
Typical School Millage (m_s)	5.946 mills
Homestead Exemption Amount	Up to $50,000 (two tiers)
SOH Benefit Included	Yes (explicitly modeled $h_t - A_t$ gap)
Estimated Annual Tax Savings	$2,000–$3,500
Representative Monthly Savings	$160–$280
Key Legal Threshold	>30 days rental for 2 consecutive years
Primary Legal Reference	Fla. Stat. §196.061; Furst v. Rebholz (2023)

Notes: Florida calculations use illustrative millage rates of $m_n = 8.349$ and $m_s = 5.946$ mills, and market-to-assessed-value gaps ($h_t - A_t$) of $100,000–$200,000. Florida values reflect the immediate annual increase in property tax liability after reassessment to full market value when homestead and SOH protections are lost (see Eq. 19). Cumulative losses over multiple years can be substantially larger as the reassessed property continues to appreciate without the SOH cap.

H Appendix: Airbnb-Only Robustness Check

A potential concern with the main analysis is that AirDNA began scraping VRBO listings in 2017, coinciding with the start of the post-VCA period in Florida. If VRBO listings systematically differ from Airbnb listings in their bunching behavior, the introduction of VRBO data in 2017 could confound the pre/post comparison rather than reflecting a true behavioral response to the VCA. To assess this, we replicate the main analysis restricting the sample to Airbnb listings only.

Figure 7 displays the bunching estimates of annual reservation days for the Airbnb-only sample from 2017–2024. The pre-period figure is not shown here as it is identical to the pre-period figure in the main analysis. Because AirDNA did not scrape VRBO listings prior to 2017, the pre-period samples are Airbnb-only by construction. The results are qualitatively consistent with the main analysis, though the estimated excess mass is attenuated relative to the full sample.

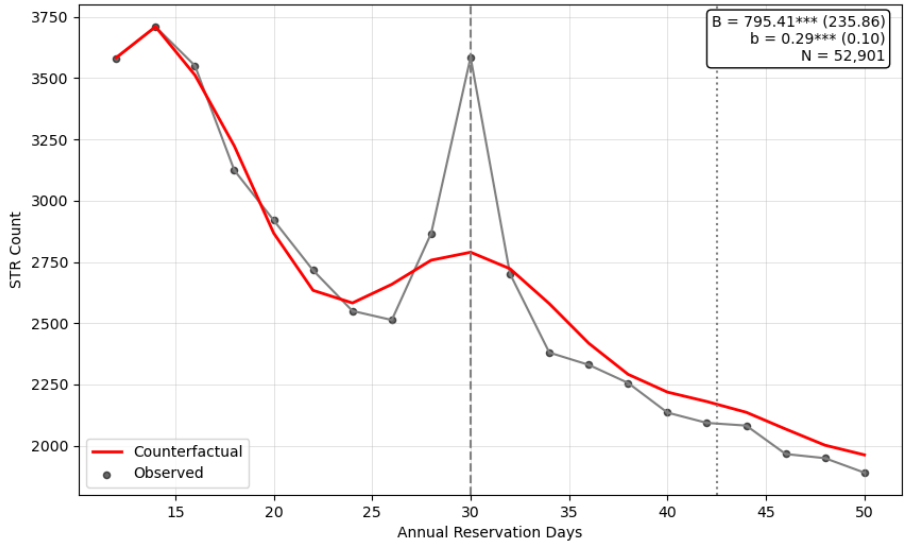


Figure 7: Distribution of Annual Reservation Days, Airbnb Only, 2017 -2024

Notes: The figure plots the distribution of annual reservation days for likely homeowner-hosts with one unit. The dashed gray line marks the 30-day homestead exemption threshold, and the dotted gray line at 42.5 days marks the end of the dominated region. Black dots represent actual binned counts; the red line shows the counterfactual distribution from the bunching estimator. Bin width is 2 days. I include a fixed point control at 14 days. Ten bins are used on each side of the threshold. B is estimated excess mass at the threshold and b is normalized excess mass. Standard errors from a residual bootstrap with 1,000 replications are in parentheses. Years of analysis are 2017-2024.
p<0.10*, p<0.05**, p<0.01***.

Table 5: Bunching Estimates, Airbnb-Only Sample

State	Florida (30-day limit)	
Year(s)	2015	2017–2024
Estimated Excess Mass	-1.171	795.408***
	(7.957)	(235.862)
Normalized Excess Mass (Bins)	-0.019	0.285***
	(0.126)	(0.098)
Normalized Excess Mass (Nights)	-0.037	0.570
Share of Bunchers	0.009	0.034
LATE (Nights)	-4.25	16.61
N	1,283	52,901

Notes: This table replicates Table 1 restricting to Airbnb listings only. The pre-period for Florida is 2015 and the post-period is 2017–2024. Standard errors from a residual bootstrap with 1,000 replications are in parentheses. N is the number of observations in the bin range. Estimated excess mass is the number of additional STRs located at the threshold. Normalized Excess Mass (Bins) is the estimated excess mass divided by the counterfactual count in the 2-day threshold bin. Normalized Excess Mass (Nights) multiplies the bins measure by the bin width of 2. Share of bunchers is one minus the ratio of observed to counterfactual counts in the dominated region, capturing the fraction of STRs that respond to the cutoff (compliers); the dominated region is the 12.5 nights above the threshold ($3,000 exemption divided by $240 per night), estimated over the bins from 30 to 44. LATE (Nights) is the normalized excess mass in nights divided by the share of bunchers, the reduction in rental nights of the marginal buncher. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

I Appendix: Hosts with 21 or More Units as the Counterfactual Robustness Check

As an alternative to the main estimation, I use the distribution of annual reservation days among hosts with 21 or more units as the counterfactual for one-unit hosts. Because the homestead exemption applies only to a permanent residence, these hosts can claim it on at most one of their many properties, so their distribution near the threshold should reflect factors unrelated to the exemption, such as demand for monthly stays or platform features. Consistent with this, estimating the main specification for hosts with 21 or more units gives a normalized excess mass of 0.146 (0.065) in 2015 and 0.197 (0.078) in 2017–2024. They bunch at 30 days both before and after the VCA, so their bunching is not a response to it. The estimation uses the same periods as the main analysis (2015 before the VCA and 2017–2024 after), the same bin width of 2 days, and the same 10 bins on each side of the threshold, and the bunching window is the bin that ends at the 30-day threshold.

Because hosts with 21 or more units list many more properties than one-unit hosts, the two distributions must be put on the same scale. Let m_j be the number of listings

of hosts with 21 or more units in bin j, n_j the number of one-unit listings, and W the bunching window. I first convert the comparison counts to shares, $s_j = m_j / \sum_k m_k$, and then rescale them by a single constant so that, across the bins outside the bunching window, the counterfactual sums to the one-unit count:

$$\hat{c}_j = s_j \cdot \frac{\sum_{k \notin W} n_k}{\sum_{k \notin W} s_k}. \tag{21}$$

The counterfactual therefore takes its shape from hosts with 21 or more units and its level from one-unit hosts, and the level is determined only by bins that the threshold should not affect. The excess mass is $B = \sum_{j \in W}(n_j - \hat{c}_j)$, and the normalized excess mass is $b = B/\hat{c}_{z^*}$, where $\hat{c}_{z^*}$ is the counterfactual count in the threshold bin. This is the same normalization used for the main estimates. Standard errors come from the same residual bootstrap as the main estimates, with 1,000 replications. The bootstrap holds the shape of the comparison distribution fixed, so the standard errors do not reflect sampling variation among hosts with 21 or more units.

Table 6 compares these estimates with the main estimates, and Figure 8 plots the one-unit distribution against the rescaled comparison distribution. Before the VCA, one-unit hosts show no significant bunching relative to hosts with 21 or more units, with a normalized excess mass of 0.134 (0.385), consistent with the main pre-VCA estimate of −0.019 (0.126). After the VCA, the estimated excess mass is 2,424.8 (473.9) listings and the normalized excess mass is 0.196 (0.039), significant at the 1 percent level. This estimate is smaller than the main post-VCA estimate of 0.450 (0.184), but it remains positive, statistically significant, and more precisely estimated. Relative to this counterfactual, the share of bunchers in the dominated region is small and imprecisely estimated (0.012, with a standard error of 0.015, compared with 5.3 percent in the main estimates), so I do not interpret the implied LATE of 32.6 nights.

The smaller estimate is expected. Because hosts with 21 or more units also bunch at the threshold, their bunching is built into this counterfactual, so the estimate measures bunching by one-unit hosts over and above the bunching of hosts with 21 or more units. This removes any clustering near 30 days that is common to both groups, and the gap between the two estimates (0.254) is comparable to the post-VCA normalized excess mass of 0.197 for hosts with 21 or more units, although the two are not directly additive because each is normalized by a different counterfactual count.

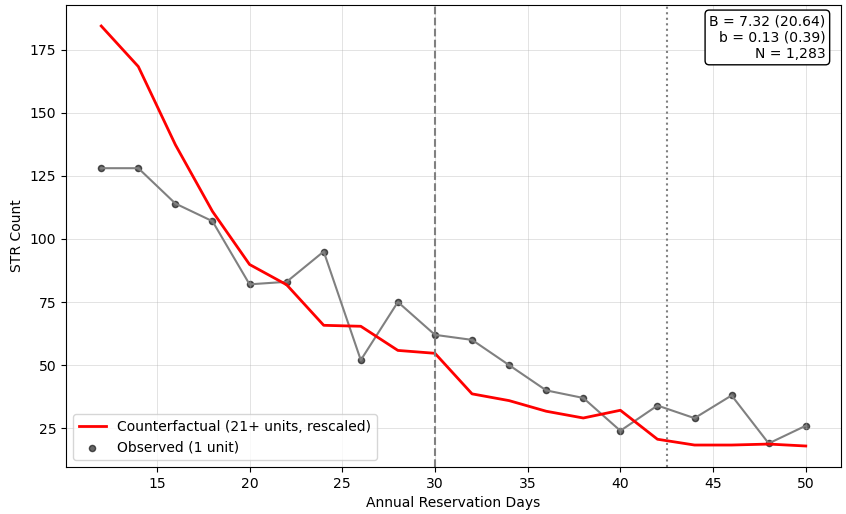
This comparison distribution is also an imperfect counterfactual away from the threshold. Above about 40 days, the number of one-unit listings falls off somewhat faster than the number of listings of hosts with 21 or more units. For these reasons, I treat this design

as a robustness check rather than the main specification. Because it nets out all bunching shared with hosts who do not respond to the VCA, and because the main sample cannot separate eligible from ineligible properties, the estimate of 0.196 should be interpreted as a conservative lower bound on the response of one-unit hosts who claim the homestead exemption.

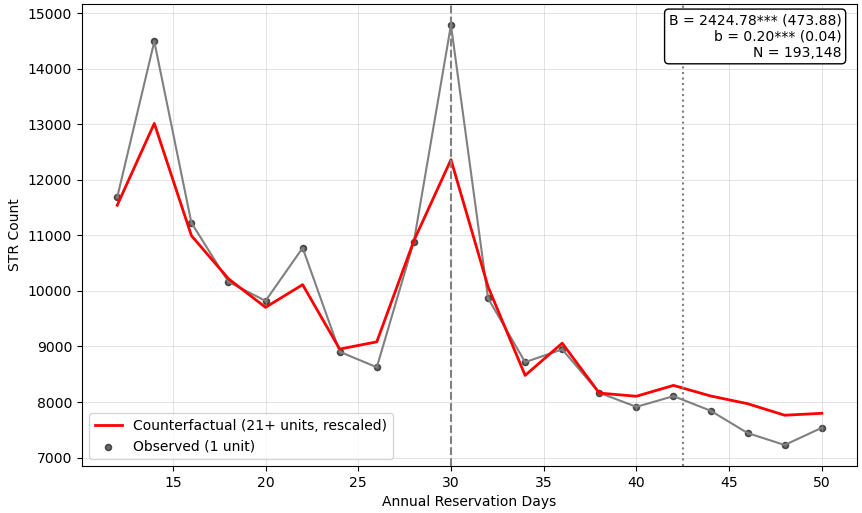
Table 6: Bunching Estimates for One-Unit Hosts Using Hosts with 21 or More Units as the Counterfactual

Counterfactual	Polynomial (Main Estimates)		Hosts with 21+ Units	
Year(s)	2015	2017–2024	2015	2017–2024
Estimated Excess Mass	-1.171	4,589.321***	7.317	2,424.777***
	(7.957)	(1,416.941)	(20.645)	(473.884)
Normalized Excess Mass (Bins)	-0.019	0.450**	0.134	0.196***
	(0.126)	(0.184)	(0.385)	(0.039)
Normalized Excess Mass (Nights)	-0.037	0.900	0.268	0.392
Share of Bunchers	0.009	0.053	-0.327	0.012
LATE (Nights)	-4.25	16.91	-0.82	32.62
Marginal Buncher (Nights)	25.7	46.9	29.2	62.6
N (1 Unit)	1,283	193,148	1,283	193,148
N (21+ Units)			3,336	220,473

Notes: This table reports bunching at the 30-day rental limit for the main sample of likely homeowner-hosts with 1 unit under two counterfactuals. The first two columns repeat the main estimates, where the counterfactual is a polynomial fit to the one-unit distribution (Table 1). The last two columns use the distribution of likely homeowner-hosts with 21 or more units and entire-home listings, converted to shares and rescaled so that it matches the one-unit count outside the bunching window. Data are from AirDNA. Bin width is 2 days, with 10 bins on each side of the threshold. N (1 Unit) and N (21+ Units) are the numbers of observations in the bin range for each group. Estimated excess mass is the number of additional STRs located at the threshold. Normalized Excess Mass (Bins) is the estimated excess mass divided by the counterfactual count in the 2-day threshold bin. Normalized Excess Mass (Nights) multiplies the bins measure by the bin width of 2. Share of bunchers is one minus the ratio of observed to counterfactual counts in the dominated region, capturing the fraction of STRs that respond to the cutoff (compliers); the dominated region is the 12.5 nights above the threshold ($3,000 exemption divided by $240 per night), estimated over the bins from 30 to 44. LATE (Nights) is the normalized excess mass in nights divided by the share of bunchers, the reduction in rental nights of the marginal buncher. Standard errors from a residual bootstrap with 1,000 replications are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.



(a) Pre-VCA (2015)



(b) Post-VCA (2017–2024)

Figure 8: Bunching Among One-Unit Hosts Using Hosts with 21 or More Units as the Counterfactual

Notes: The figures plot the distribution of annual reservation days for likely homeowner-hosts with 1 unit and entire-home listings. Black dots represent actual binned counts. The red line shows the counterfactual: the distribution of likely homeowner-hosts with 21 or more units in the same period, converted to shares and rescaled to match the one-unit count outside the bunching window. The dashed gray line marks the 30-day homestead exemption threshold, and the dotted gray line at 42.5 days marks the end of the dominated region. Bin width is 2 days, with 10 bins on each side of the threshold. B is estimated excess mass at the threshold and b is normalized excess mass. Standard errors from a residual bootstrap with 1,000 replications are in parentheses. p<0.10*, p<0.05**, p<0.01***.

J Appendix: Robustness to the Form of the Counterfactual

The bunching estimates depend on the counterfactual distribution of annual reservation days, which I recover by fitting a polynomial to the binned distribution of one-unit hosts around the 30-night threshold. The main specification uses 2-night bins, a ninth-order polynomial, and a fixed effect for the bin containing 14 nights. To show that the results do not hinge on these choices, I re-estimate the main 2017–2024 specification for one-unit hosts with every combination of polynomial order from 1 to 9 and bin width from 1 to 4 nights and different fixed point bunching controls. Everything else I hold fixed: the threshold at 30 nights, the dominated region from 30 to 42.5 nights, the fixed effect for the bin containing 14 nights, and 1,000 bootstrap draws. The estimation window covers 20 nights on each side of the threshold, rounded out to whole bins, except with 4-night bins, where the estimator requires at least six bins on each side and the window therefore runs from 6 to 54 nights. I do not report bin widths of 5 nights or more, since the window would then have to extend to stays of 1 or 2 nights, where the distribution is dominated by very short stays, and the estimates become unstable. Because normalized excess mass is measured relative to the counterfactual count in the threshold bin, it scales with the bin width, so I compare specifications using normalized excess mass in nights rather than in bins.

Table 7 varies the controls for round numbers in the main specification. The main specification includes a fixed effect for the 14-night bin. Including only this fixed effect reproduces the main estimates exactly. Replacing the 14-night fixed effect with an indicator for bins at multiples of 7 nights, to absorb weekly stays at 14, 28, and 42 nights, raises normalized excess mass to 1.12 nights, the share of bunchers to 0.057, and the LATE to 19.6 nights, with a slightly lower RMSE. Including both controls gives similar results: normalized excess mass of 1.08 nights and a LATE of 19.1 nights. In every case the estimates are at least as large as in the main specification.

Table 8 reports the results of the different bin widths and polynomial orders. With bin widths of 1 to 3 nights, the estimated excess mass is positive and statistically significant in every specification, normalized excess mass ranges from 0.81 to 1.53 nights, and the share of bunchers ranges from 0.035 to 0.074. Normalized excess mass in the main specification, 0.900 nights, is near the bottom of this range: 1-night bins give similar values (0.81 to 0.96 nights), and with 2- or 3-night bins every specification except the eighth-order polynomial with 2-night bins (0.894 nights) gives a larger value, so the main estimate remains a lower bound on the response of one-unit hosts. The LATE is between 16 and 31 nights in all but one of these specifications; the main specification's 16.9 nights is again at the low end,

and low-order polynomials, which imply a smaller share of bunchers, give the largest values. Normalized excess mass is significant at the 5 percent level in every specification except those with 3-night bins and polynomials of order 7 to 9, where few degrees of freedom remain. With 4-night bins, normalized excess mass ranges from 1.0 to 1.7 nights and the share of bunchers from 0.04 to 0.07, but these estimates are less precise: the quadratic gives a negative share of bunchers, and the ninth-order polynomial uses every available degree of freedom.

Table 8 also reports two measures of fit. The root mean squared error (RMSE) of the counterfactual falls as the polynomial order rises, which is mechanical, since adding terms cannot worsen the in-sample fit. The residual standard error, which accounts for the degrees of freedom used, is instead lowest for a linear counterfactual with 1- to 3-night bins and for a fourth-order polynomial with 4-night bins. The specifications that fit best by this measure therefore imply larger bunching responses than the main specification, which reinforces the interpretation of the main estimates as a lower bound.

Table 7: Round-Number and Fixed-Effect Controls in the Main Specification

	Main (FE 14)	Round Number 7	Round Number 7 and FE 14
Estimated Excess Mass	4,589.3***	5,322.3***	5,178.1***
	(1,416.9)	(1,602.6)	(1,662.9)
Normalized Excess Mass (Bins)	0.450**	0.562**	0.539*
	(0.184)	(0.238)	(0.285)
Normalized Excess Mass (Nights)	0.900	1.124	1.077
Share of Bunchers	0.053	0.057	0.056
	(0.019)	(0.022)	(0.022)
LATE (Bins)	8.46	9.79	9.54
LATE (Nights)	16.91	19.58	19.08
RMSE (per Night)	295.3	278.3	273.1
Residual SE (per Night)	455.1	428.8	449.9
Residual df	8	8	7
N	193,148	193,148	193,148

Notes: All columns use the main specification (one-unit hosts, 2017–2024, bin width 2, polynomial of order 9). The main specification includes a fixed effect for the 14-night bin and a round-number control at 100 nights, which never enters because no bin in the 10–50 night window is a multiple of 100; the second column drops it. Round Number 7 replaces the 14-night fixed effect with one indicator for the bins at multiples of 7 nights (14, 28, and 42); the last column includes both. RMSE and residual SE are defined as in Table 8. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: Bunching Estimates by Polynomial Order and Bin Width, One-Unit Hosts, Florida 2017–2024

	Polynomial Order								
	1	2	3	4	5	6	7	8	9
Bin width: 1 Night (window 10–50, dominated region to 43)									
Estimated Excess Mass	4,575.2***	4,581.1***	4,590.1***	4,448.3***	4,456.4***	4,308.6***	4,308.3***	4,178.9***	4,175.3***
	(963.7)	(972.0)	(972.7)	(949.8)	(953.2)	(926.0)	(926.8)	(917.6)	(916.8)
Normalized Excess Mass (Bins)	0.958***	0.961***	0.964***	0.908***	0.911***	0.855***	0.855***	0.808***	0.807***
	(0.207)	(0.216)	(0.216)	(0.213)	(0.214)	(0.207)	(0.208)	(0.207)	(0.206)
Normalized Excess Mass (Nights)	0.958	0.961	0.964	0.908	0.911	0.855	0.855	0.808	0.807
Share of Bunchers	0.038	0.037	0.052	0.052	0.057	0.051	0.051	0.049	0.049
	(0.043)	(0.036)	(0.025)	(0.024)	(0.019)	(0.018)	(0.017)	(0.017)	(0.017)
LATE (Bins)	25.41	26.00	18.52	17.40	16.06	16.90	16.91	16.47	16.48
LATE (Nights)	25.41	26.00	18.52	17.40	16.06	16.90	16.91	16.47	16.48
RMSE (per Night)	466.4	466.3	466.1	450.5	451.1	434.0	433.9	420.6	420.0
Residual SE (per Night)	485.5	492.2	499.2	489.7	498.0	486.8	494.8	487.8	495.7
Residual df	36	35	34	33	32	31	30	29	28
Flag	W	W							
Bin width: 2 Nights (window 10–50, dominated region to 44)									
Estimated Excess Mass	5,220.7***	5,234.4***	5,274.1***	5,009.8***	5,058.5***	4,839.5***	4,776.6***	4,568.5***	4,589.3***
	(1,307.6)	(1,338.6)	(1,351.6)	(1,339.5)	(1,357.8)	(1,351.5)	(1,377.8)	(1,354.4)	(1,416.9)
Normalized Excess Mass (Bins)	0.546***	0.548***	0.554***	0.512***	0.520***	0.486***	0.477***	0.447***	0.450**
	(0.141)	(0.152)	(0.154)	(0.156)	(0.160)	(0.163)	(0.167)	(0.167)	(0.184)
Normalized Excess Mass (Nights)	1.091	1.096	1.108	1.024	1.040	0.973	0.954	0.894	0.900
Share of Bunchers	0.040	0.039	0.054	0.051	0.057	0.052	0.052	0.053	0.053
	(0.043)	(0.036)	(0.025)	(0.024)	(0.020)	(0.020)	(0.020)	(0.019)	(0.019)
LATE (Bins)	13.63	13.97	10.26	10.01	9.18	9.31	9.24	8.46	8.46
LATE (Nights)	27.27	27.94	20.51	20.01	18.37	18.61	18.47	16.92	16.91
RMSE (per Night)	341.7	341.4	342.9	324.1	325.1	312.5	306.9	294.7	295.3
Residual SE (per Night)	372.4	384.3	399.4	391.8	409.1	410.7	423.1	428.1	455.1
Residual df	16	15	14	13	12	11	10	9	8
Flag	W	W	W	W					
Bin width: 3 Nights (window 9–51, dominated region to 45)									
Estimated Excess Mass	6,615.3***	6,877.4***	7,013.1***	6,727.6***	6,992.3***	6,747.3***	7,092.0***	6,744.2***	6,243.8**
	(1,887.1)	(1,989.8)	(2,010.7)	(2,023.1)	(2,160.8)	(2,159.0)	(2,456.6)	(2,348.2)	(2,581.4)
Normalized Excess Mass (Bins)	0.458***	0.485***	0.500***	0.470***	0.498***	0.472**	0.508*	0.472*	0.422
	(0.136)	(0.156)	(0.161)	(0.168)	(0.191)	(0.199)	(0.278)	(0.255)	(0.312)
Normalized Excess Mass (Nights)	1.375	1.456	1.499	1.410	1.493	1.416	1.525	1.415	1.265
Share of Bunchers	0.045	0.035	0.061	0.059	0.068	0.065	0.067	0.074	0.068
	(0.049)	(0.044)	(0.030)	(0.030)	(0.029)	(0.028)	(0.031)	(0.026)	(0.028)
LATE (Bins)	10.23	13.71	8.15	8.03	7.27	7.25	7.57	6.38	6.24
LATE (Nights)	30.70	41.14	24.44	24.08	21.82	21.74	22.72	19.14	18.72
RMSE (per Night)	317.8	305.4	301.5	290.0	289.8	280.3	285.1	242.2	217.0
Residual SE (per Night)	362.3	367.0	384.3	395.3	426.6	451.9	514.0	504.1	553.2
Residual df	10	9	8	7	6	5	4	3	2
Flag	W	W	W	W	W	W	W		
Bin width: 4 Nights (window 6–54, dominated region to 46)									
Estimated Excess Mass	5,113.9*	7,082.5***	7,730.3***	6,790.8***	6,839.4***	6,603.5***	7,348.2**	7,385.0**	6,555.7*
	(2,767.2)	(2,636.4)	(2,505.1)	(2,272.1)	(2,463.2)	(2,435.6)	(3,154.8)	(3,119.8)	(3,828.3)
Normalized Excess Mass (Bins)	0.249*	0.381**	0.431***	0.360**	0.363**	0.346**	0.401	0.404	0.343
	(0.138)	(0.154)	(0.158)	(0.144)	(0.165)	(0.166)	(0.294)	(0.292)	(1.823)
Normalized Excess Mass (Nights)	0.995	1.523	1.723	1.438	1.452	1.385	1.603	1.615	1.371
Share of Bunchers	0.043	-0.020	0.064	0.061	0.063	0.059	0.063	0.071	0.063
	(0.064)	(0.052)	(0.032)	(0.029)	(0.028)	(0.027)	(0.033)	(0.029)	(0.037)
LATE (Bins)	5.76	-19.05	6.73	5.90	5.80	5.85	6.34	5.69	5.43
LATE (Nights)	23.04	-76.21	26.92	23.59	23.21	23.39	25.37	22.77	21.72
RMSE (per Night)	665.7	476.8	309.7	244.3	247.1	239.9	248.5	228.9	199.1
Residual SE (per Night)	780.6	597.7	419.4	362.4	409.7	459.3	582.7	759.1	–
Residual df	8	7	6	5	4	3	2	1	0
Flag		S	W				W		R
N	193,148								

Notes: Each column re-estimates the main 2017–2024 one-unit specification with the counterfactual polynomial order in the column header and the bin width in the panel heading. Everything else is as in Table 1: the threshold is at 30 nights, the dominated region runs from 30 to 42.5 nights, the specification includes a fixed effect for the bin containing 14 nights, and standard errors come from 1,000 bootstrap draws. The main specification uses a bin width of 2 and a polynomial of order 9. The estimation window covers 20 nights on each side of the threshold, rounded out to whole bins; with 4-night bins it runs from 6 to 54 nights because the estimator requires at least six bins on each side. The dominated region is covered by whole bins, so it ends at the night shown in each panel heading. Normalized excess mass and the LATE are reported in bins and in nights (bin width × bins). RMSE is the root mean squared error of the fitted counterfactual over the bins used in the fit. Residual SE is $\sqrt{SSR/\text{residual df}}$. Both are divided by the bin width so they are comparable across panels. Flags: R, the regression has no residual degrees of freedom; S, share of bunchers outside $(0, 1)$; W, implied marginal buncher beyond the upper end of the estimation window. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

K Appendix: Bunching by Year

The main estimates pool all years from 2017 to 2024. If hosts learned about the 30-day threshold gradually, or if enforcement or awareness grew over time, bunching would be larger in later years, and the pooled estimate would average over a growing response. To check this, I re-estimate the main specification separately for each year from 2015 to 2024. The specification is identical to Table 1: one-unit hosts of likely owner-occupied property types, 2-night bins, a ninth-order polynomial, a fixed effect for the bin containing 14 nights, and a dominated region from 30 to 42.5 nights. Because normalized excess mass is measured relative to the counterfactual count in the threshold bin, it is comparable across years even though the number of listings changes from year to year.

Table 9 reports the estimates and Figure 9 plots the observed and counterfactual distributions for each year. There is no bunching in 2015, before the VCA, and normalized excess mass is small and insignificant in 2016, which the main analysis omits. From 2017 onward, excess mass at 30 nights is positive and significant at the 1 percent level in every year, and normalized excess mass is significant at the 5 percent level in every year except 2018, where it is largest (0.758 bins) but imprecisely estimated. Normalized excess mass ranges from 0.301 bins in 2017 to 0.758 bins in 2018. Every yearly estimate after 2016 is therefore positive, of similar size to the main pooled estimate of 0.450, and a lower bound on the response of one-unit hosts.

Table 9: Bunching at 30 Nights by Year, Florida One-Unit Hosts

Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Estimated Excess Mass	-1.2	16.6	490.2***	952.3***	772.7***	706.4***	342.8***	460.9***	395.4***	468.8***
	(8.0)	(10.3)	(173.5)	(293.5)	(228.1)	(226.2)	(122.2)	(137.9)	(123.5)	(141.9)
Normalized Excess Mass (Bins)	-0.019	0.109	0.301**	0.758	0.561**	0.501**	0.374**	0.436**	0.311***	0.364***
	(0.126)	(0.071)	(0.126)	(0.507)	(0.243)	(0.223)	(0.164)	(0.170)	(0.115)	(0.135)
Normalized Excess Mass (Nights)	-0.037	0.219	0.603	1.515	1.122	1.003	0.748	0.872	0.622	0.729
Share of Bunchers	0.009	0.008	0.034	0.092	0.064	0.060	0.043	0.050	0.038	0.045
LATE (Nights)	-4.25	27.35	17.99	16.56	17.48	16.66	17.39	17.57	16.20	16.19
Marginal Buncher (Nights)	25.7	57.4	48.0	46.6	47.5	46.7	47.4	47.6	46.2	46.2
N	1,283	2,896	32,514	23,364	25,389	25,655	17,271	19,977	24,431	24,547

Notes: Each column estimates the main specification of Table 1 on one year of data: one-unit hosts of likely owner-occupied property types, 2-night bins from 10 to 50 nights, a ninth-order polynomial, a fixed effect for the bin containing 14 nights, and a dominated region from 30 to 42.5 nights (bins to 44). Normalized excess mass is reported in 2-night bins and in nights (twice the estimate in bins). The LATE is normalized excess mass in nights divided by the share of bunchers, and the marginal buncher is 30 nights plus the LATE. Bootstrap standard errors (1,000 draws) in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

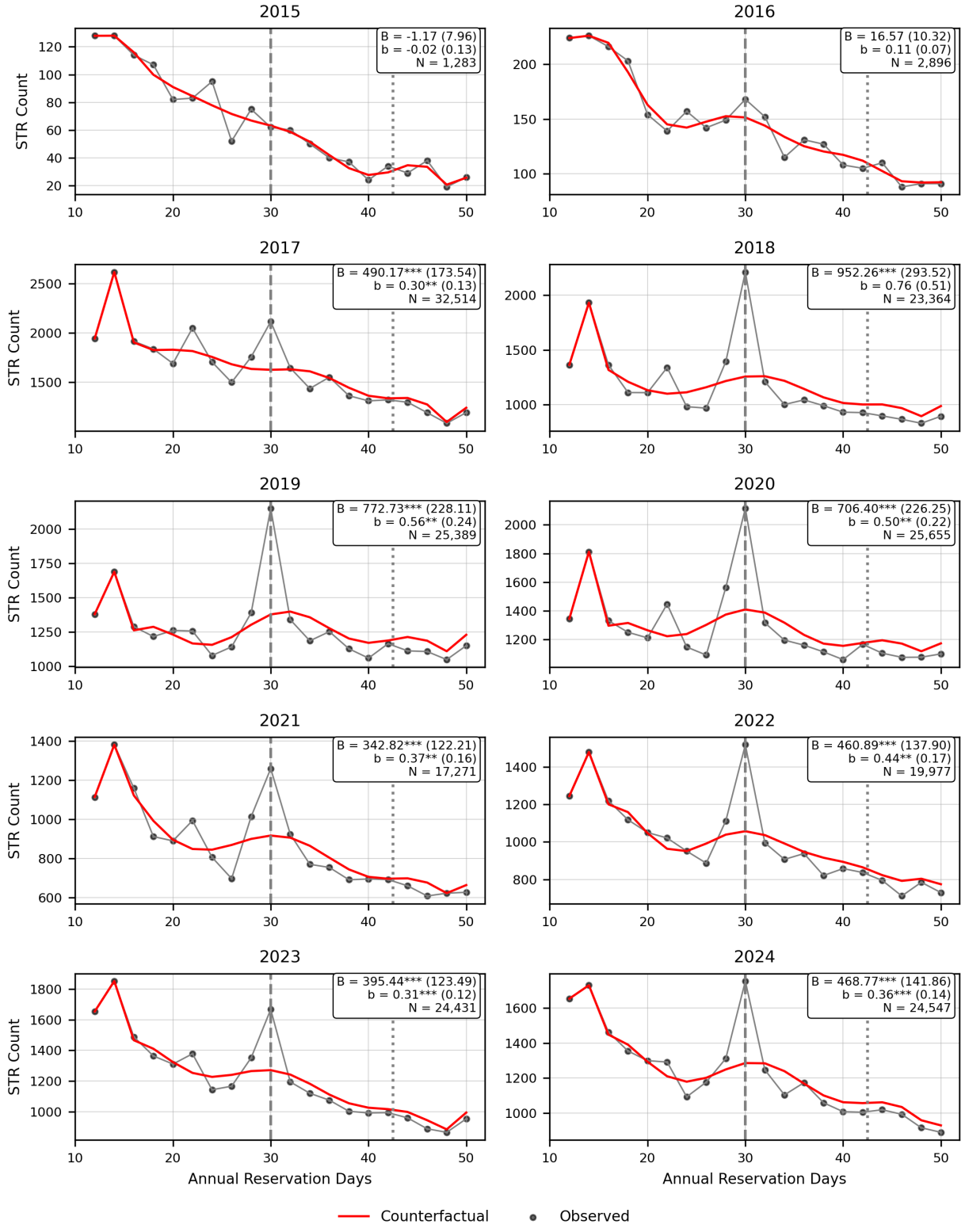


Figure 9: Bunching at 30 Nights by Year, Florida One-Unit Hosts

Notes: Each panel plots the number of one-unit listings in each 2-night bin of annual reservation days (black) and the estimated counterfactual (red) for one year, using the specification of Table 9. The dashed line marks the 30-night threshold and the dotted line marks the end of the dominated region at 42.5 nights. The vertical scale differs across panels.

L Appendix: Bunching Among Hosts Already Renting in 2015

The main comparison sets the 2015 distribution of reservation days against the 2017–2024 distribution, but the hosts in the two samples are not the same. Many listings enter after 2015 and many 2015 listings leave. If the hosts who entered after the VCA were different in ways that led them to rent exactly 30 nights, for reasons unrelated to the homestead exemption, the post-VCA bunching could reflect a change in who is in the sample rather than a change in how hosts behave. To check this, I fix the sample of properties in 2015 and follow only those properties after the VCA.

The 2015 cohort contains every property in the main 2015 sample (one-unit hosts, entire homes of likely owner-occupied property types) that was booked for at least one night in 2015, which gives 2,858 properties. I keep these properties' rows in the main post-VCA sample, so every later observation meets the same sample restrictions as Table 1, and no property that entered after 2015 is included. I then estimate the main specification on this cohort. Attrition is substantial: 1,734 of the 2,858 properties are booked for at least one night in some year between 2017 and 2024, and the share of the cohort that is still booked falls from 56 percent in 2017 to 14 percent in 2024. The cohort sample is therefore small, with 1,623 property-years within the estimation window in 2017–2024.

Table 10 reports the results and Figure 10 plots the observed and counterfactual distributions. In 2015, before the VCA, this cohort shows no bunching at 30 nights (column 1; normalized excess mass of -0.019, standard error 0.126). After the VCA, the same properties bunch. In 2017–2024 normalized excess mass is 0.369 bins (0.140), significant at the 1 percent level (column 3), and including 2016 gives 0.302 (0.126) (column 2). These estimates are close to the full-sample estimate of 0.450 (0.184) in column 4 and well within its confidence interval. The share of bunchers is 0.053, the same as in the full sample, and the LATE is 14.0 nights, compared with 16.9 in the full sample. Because the comparison holds the properties fixed, the appearance of bunching after the VCA is a change in the behavior of hosts who were already renting, not only a change in who rents. As in the main results, these estimates are a lower bound on the response of one-unit hosts.

Column 5 repeats the exercise for the properties of hosts with 21 or more units in 2015. These properties also show some bunching after the VCA (0.103 bins, standard error 0.045), but it is about a quarter of the one-unit cohort's estimate and smaller than the 0.177 estimated for all 21+ unit hosts in 2017–2024 (Appendix I). The difference between the one-unit and 21+ unit cohorts is therefore not driven by different entry into the two groups after 2015.

Table 10: Bunching at 30 Nights Among Properties Already Renting in 2015, Florida

	(1)	(2)	(3)	(4)	(5)
Hosts	2015 one-unit cohort			All one-unit	2015 21+ unit cohort
Years	2015	2016–2024	2017–2024	2017–2024	2017–2024
Estimated Excess Mass	-1.2	43.0***	35.6***	4,589.3***	23.8**
	(8.0)	(15.5)	(11.0)	(1,416.9)	(10.0)
Normalized Excess Mass (Bins)	-0.019	0.302**	0.369***	0.450**	0.103**
	(0.126)	(0.126)	(0.140)	(0.184)	(0.045)
Normalized Excess Mass (Nights)	-0.037	0.605	0.738	0.900	0.206
Share of Bunchers	0.009	0.046	0.053	0.053	0.014
LATE (Nights)	-4.25	13.26	14.02	16.91	14.63
Marginal Buncher (Nights)	25.7	43.3	44.0	46.9	44.6
N	1,283	2,356	1,623	193,148	3,777

Notes: The 2015 one-unit cohort is the 2,858 properties in the main 2015 sample (one-unit hosts, entire homes of likely owner-occupied property types) booked for at least one night in 2015. Columns (1) to (3) use only these properties, and column (5) builds the same cohort from hosts with 21 or more units in 2015 (7,907 properties). Column (4) is the main estimate from Table 1. Every column uses the main specification: 2-night bins from 10 to 50 nights, a ninth-order polynomial, a fixed effect for the bin containing 14 nights, and a dominated region from 30 to 42.5 nights (bins to 44). Normalized excess mass is reported in 2-night bins and in nights (twice the estimate in bins). The LATE is normalized excess mass in nights divided by the share of bunchers, and the marginal buncher is 30 nights plus the LATE. Bootstrap standard errors (1,000 draws) in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

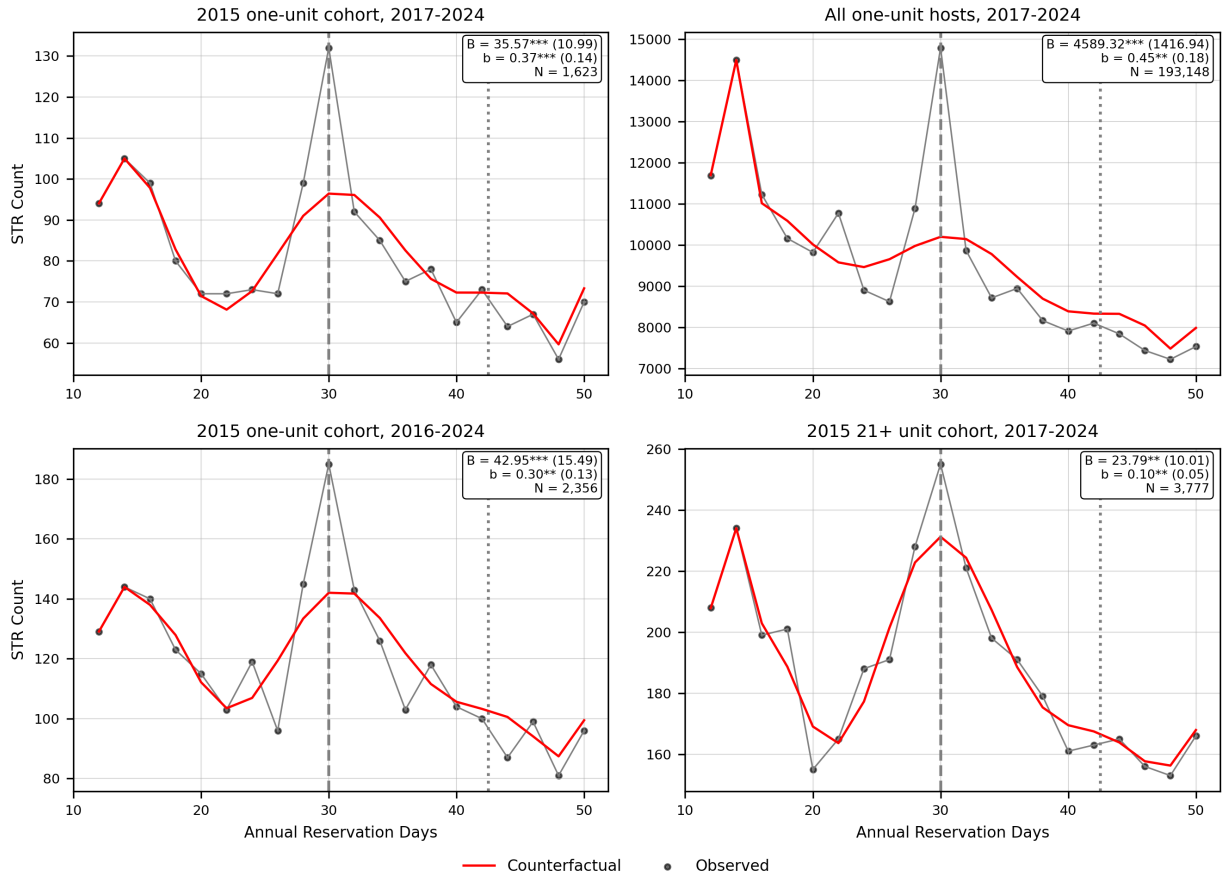


Figure 10: Bunching at 30 Nights Among Properties Already Renting in 2015, Florida

Notes: Each panel plots the number of listings in each 2-night bin of annual reservation days (black) and the estimated counterfactual (red), using the specification of Table 10. The dashed line marks the 30-night threshold and the dotted line marks the end of the dominated region at 42.5 nights. The vertical scale differs across panels.

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ORIGINAL ARTICLE

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Reindustrialization meets rural reality: The limits of foreign direct investment-led development

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Abstract

As industrial policy drives new waves of foreign direct investment (FDI) into the U.S., rural areas are becoming key destinations. Using synthetic control, difference-in-differences, and propensity score matching, I examine whether rural FDI translates job growth into higher incomes. Evidence from Kia Motors' $1.2 billion investment in Troup County, Georgia, and a national panel of rural counties (1995–2019) shows that FDI raises employment—by 13%–16% locally and 3% nationally—but has no average effect on income. Gains occur where housing and labor markets are strong, suggesting FDI must be paired with policies that improve housing supply, mobility, and absorptive capacity.

KEYWORDS
foreign direct investment, industrial policy, regional development, rural economies

JEL CLASSIFICATION
R58, R11, H25, H71

1 | INTRODUCTION

Federal trade and industrial policies have triggered a new wave of foreign direct investment (FDI) into the United States. In 2025, Hyundai Motor Group and Roche Pharmaceuticals committed $21 billion and $50 billion, respectively, to U.S.-based manufacturing, with Roche citing the need to insulate supply chains from future trade disruptions (Reuters Staff, 2025a, 2025b). These two companies join a flood of other foreign firms expanding U.S. operations to secure market access and reduce exposure to global risk. With the sudden increase in FDI, chronicling the impacts of past FDI projects remains vital to preparing for the incoming investment.

What distinguishes this wave of FDI from previous ones is the likely spillover into rural counties. At the federal level, new industrial policies emphasize the revitalization of struggling U.S. regions, many of which are rural areas. For

Abbreviations: BEA, Bureau of Economic Analysis; BLS, Bureau of Labor Statistics; BLS-LAUS, Bureau of Labor Statistics Local Area Unemployment Statistics; CSA, Callaway Sant'Anna (2021); CZ, commuting zone; DiD, difference-in-differences; FDI, foreign direct investment; LQ, location quotient; PCPI, per capita personal income; POW, place of work; PSM, propensity score matching; QCEW-BLS, Quarterly Census of Employment and Wages—Bureau of Labor Statistics; RMSPE, root mean squared prediction error; RUCC, Rural-Urban Continuum Code; TWFE, two-way fixed effects; USDA, U.S. Department of Agriculture.

An earlier version of this paper appeared in the dissertation of Alexandra Troidl.

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instance, the Inflation Reduction Act of 2022 directs substantial clean energy and infrastructure investments toward rural regions, including the largest investment in rural electrification since 1936 (Environmental Entrepreneurs (E2) 2024; U.S. Department of Agriculture, 2023). Site selection consultants and corporate executives consider available land a top factor in location decisions, impacting investment timelines and project viability (Greiner, 2025; Site Selectors Guild, 2024). Consequently, U.S. states have launched major initiatives to address the shortage of industrial land by preparing new development-ready sites (Evans, 2025). Meanwhile, manufacturers are increasingly considering rural communities and other areas outside major metros for their projects (Irwin-Hunt, 2024). Together, these trends highlight how urban areas, which have been saturated by FDI in the past, will not have the capacity to absorb the incoming investment. Therefore, rural counties will become more prominent players and new targets for FDI.

Previous studies show that FDI supports capital formation, knowledge transfers, and the adoption of foreign best practices through linkages between foreign and domestic firms (Alfaro et al., 2004). It also indirectly boosts domestic investment by improving profitability, infrastructure, and the local talent pool (Amighini et al., 2017). In poorer economies, FDI stimulates growth and introduces new technologies. Further evidence suggests that FDI has a crowd-in effect on domestic investment, stimulating domestic growth and development by expanding intermediate demand, reducing fixed costs for related firms, and deepening local supply chains (Farla et al., 2016; Guo et al., 2024). While prior research has examined regional variation in FDI effects in lagging European regions (e.g., Barbieri et al., 2018; Petrakos et al., 2003), little is known about how these dynamics play out in rural areas.

This paper studies whether these known benefits of FDI materialize in rural areas. This open question requires study for two main reasons. First, rural counties face persistent structural disadvantages that prevent them from experiencing agglomeration economies. These include limited labor pooling, thinner supplier networks, and lower innovation spillovers—all of which are central to the agglomeration benefits described by Greenstone et al. (2010) and Moretti (2010). This raises questions about the efficacy of FDI in these contexts. Secondly, compared with urban areas, rural counties tend to have lower educational attainment, weaker infrastructure, older populations, and more concentrated industrial bases. These constraints limit their ability to take advantage of new economic opportunities, even when large-scale investments occur. Given the challenges rural counties face, this paper investigates whether FDI improves employment and income outcomes in rural America, and if so, under what conditions.

The foreign origin of the capital is analytically relevant to these research questions, not merely incidental. Unlike domestic investors, foreign firms enter rural markets with higher informational and coordination costs, less familiarity with local institutions and regulations, and a greater initial reliance on non-local managerial talent and imported inputs rather than local sourcing. These features can dampen short-run income effects even when employment rises and shape the pace at which complementary domestic activity develops around the investment. This distinguishes the present study from the domestic plant-siting literature such as Greenstone and Moretti (2004) and Greenstone et al. (2010). The mechanisms linking large manufacturing investment to local income growth may operate differently when the investing firm must simultaneously build institutional knowledge, supply-chain relationships, and workforce integration from scratch as with FDI. Additionally, FDI is a policy-relevant category as state and local governments explicitly recruit foreign investors through dedicated incentive packages, trade missions, and site-readiness programs. All of these factors make it important to evaluate what these strategies targeting FDI deliver in rural communities.

I assess the extent to which FDI projects impact regional development in rural counties with two approaches. First, I employ the synthetic control method to study the impact of a $1.2-billion investment by Kia Motors in rural Troup County, Georgia. Second, I conduct an aggregate analysis of FDI's impacts on U.S. rural counties with propensity score matching (PSM) and difference-in-differences (DID). I use data from the Bureau of Economic Analysis (BEA) and the Bureau of Labor Statistics Local Area Unemployment Statistics (BLS-LAUS) for the years 1995–2019. Data on FDI projects (or lack thereof) come from fDi Markets, a company that tracks and reports data on FDI projects globally.

Both analyses draw similar conclusions: FDI is on average associated with an increase in employment and a null effect on per capita income in rural counties. The amount of employment growth following FDI depends on the county. For instance, Troup County sees employment increases approximately 4–5 times higher than the estimate from the aggregate propensity score analysis. Furthermore, while rural FDI reliably increases employment, its impact on local incomes is heterogeneous. Where incomes do rise, they are accompanied by higher housing growth, more establishments, and stronger wage gains. These findings suggest that attracting FDI projects is not by itself a guarantee of economic prosperity. FDI must be complemented by supporting policies that strengthen rural labor and housing markets.

This paper makes three main contributions to the study of regional development and investment policy. First, as the first U.S.-wide analysis of FDI in rural areas, this paper adds to the emerging policy discussion on U.S. regional development through FDI. Recent shifts in U.S. trade and industrial policy create new opportunities and challenges for

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rural regions that have seen little FDI to date. The Troup County case study offers a rare opportunity to examine the localized impacts of such investment in depth. It also serves as a valuable preview of how future industrial policy may play out in similar communities. In doing so, this paper extends the domestic plant-siting literature to the foreign investment context, where the mechanisms linking large manufacturing investment to local income growth may differ in important ways, as discussed previously. Therefore, this paper treats FDI as a distinct policy category while acknowledging that some mechanisms may generalize to large domestic investments.

Second, FDI is often framed as a remedy for sluggish regional development, and this paper highlights important limitations of using FDI to generate income gains. Beyond estimating average treatment effects, the analysis reveals the local conditions under which rural FDI translates into higher local incomes. By examining economic conditions such as housing supply, labor market spillovers, and commuting patterns, I shed light on why some communities benefit more from FDI than others. These insights speak directly to policymakers designing complementary policies to support lasting economic growth.

The third contribution of this paper comes from the county-level lens of the study. While labor market areas may offer a more complete picture of regional spillovers, economic development incentives are typically administered at the *county* or *state* level. Across the nation, state and county governments offer tax abatements, infrastructure spending, and workforce training programs to attract large foreign-owned facilities. Given the time and resources spent to recruit new businesses, policymakers in these jurisdictions must evaluate whether public investments produce sufficient returns within their own borders. If county-level impacts are attenuated by commuting and housing shortages, which is consistent with the evidence in this paper, then the use case for localized subsidies becomes more complex.

The remainder of the paper is organized as follows. Section 2 situates this work within the literature on regional development, FDI, and local multipliers. Section 3 presents the background, methodology, data, and results for the Troup County case study. Next, Section 4 describes the methodology, data, and results for the analysis of all U.S. rural counties. Section 5 discusses mechanisms and policy implications. Finally, Section 6 concludes.

2 | RELATED LITERATURE

The literature on regional development and FDI documents a complex and often conditional relationship. Cross-country analyses generally find that the benefits of FDI are more likely to materialize in economies with higher levels of human capital, better-developed financial markets, and adequate infrastructure. These impacts occur through channels such as technology transfer, capital formation, and productivity gains (Alfaro et al., 2004; Borensztein et al., 1998; Carkovic & Levine, 2005; Hermes & Lensink, 2003). Without such absorptive capacity, foreign investment may have muted or even negative effects (Alvarado et al., 2017; Baiashvili & Gattini, 2020; Bermejo Carbonell & Werner, 2018; De Mello, 1997). This combination of infrastructure, labor market depth, and institutional readiness is defined as absorptive capacity, which, as the broader local multiplier literature suggests, determines whether economic shocks translate into sustained income gains (Amiti et al., 2024; Greenstone et al., 2010; Moretti, 2010). Absorptive capacity explains how developed and middle-income countries often capture more of FDI's potential gains, while low-income economies benefit less.

Beyond aggregate growth outcomes, another strand of the literature examines how FDI affects local wages and finds mixed results. FDI often increases the presence of other foreign firms in the host county, which should raise labor demand and average wages. However, if the labor pool remains unchanged, FDI may have no effect on wages (Kato-Vidal, 2013). Foreign firms may also pay higher wages to reduce turnover, protect proprietary knowledge, attract skilled workers, or comply with host-country regulations (Breau & Brown, 2011; Kucharčíková et al., 2015). This ambiguity in wage effects helps explain why FDI can boost employment without necessarily raising local incomes, a dynamic central to this paper's findings.

Research on FDI in lagging regions further emphasizes the importance of local conditions for economic development. Studies of lagging economies in Europe highlight how infrastructure, education levels, and industrial linkages determine whether foreign investment translates into sustained development (Kucharčíková et al., 2015; Resmini, 2000). In the U.S., the plant siting literature has examined the impacts of large manufacturing facilities, many involving foreign capital. Greenstone and Moretti (2004) show that counties winning large plants experience significant manufacturing job growth and earnings gains but at high fiscal cost. Greenstone et al. (2010) find evidence of agglomeration spillovers to incumbent firms, but these benefits depend on the presence of complementary local assets. While valuable, these studies typically do not distinguish rural from urban contexts, and they rarely address long-run income effects.

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A parallel strand of work examines place-based rural development programs. Evaluations of the Appalachian Regional Commission (Isserman & Rephann, 1995), California's Enterprise Zone program (O'Keefe, 2004), and the Delta Regional Authority (Pender & Reeder, 2011) provide quasi-experimental evidence that targeted investments can raise incomes and employment. Similar to the FDI literature, results from these rural programs vary greatly with initial conditions (Feser, 2013). Even the impact of substantial investments depends on other enabling factors such as infrastructure, industry clusters, and housing availability.

The local multiplier literature offers a framework for understanding the patterns found in the other literatures. New tradable-sector jobs can generate additional non-tradable-sector employment through increased local demand (Amiti et al., 2024; Moretti, 2010). However, multiplier magnitudes depend on housing market elasticity, labor supply responsiveness, and commuting patterns. These factors impact whether benefits accrue in the host region or diffuse into neighboring jurisdictions (Monte et al., 2018; Moretti, 2010).

In summary, these literatures point to three conclusions. First, the economic returns to FDI are highly uneven and mediated by place-specific absorptive capacity. Second, large-scale investments can produce significant spillovers. However, these spillovers are often contingent on infrastructure, labor skills, and market integration. Third, there is a lack of systematic, county-level evidence on how rural U.S. economies respond to FDI, especially in terms of long-term income effects and the mechanisms that link job creation to income growth. This paper addresses these gaps directly and tests whether FDI delivers not only jobs but also the sustained income gains that regional development programs target.

3 | INFERENCE FROM THE KIA PLANT CASE STUDY

I begin with a case study to analyze the impact of FDI on the employment and income of rural counties. This case study focuses on the investment of Kia Motors into Troup County, Georgia in 2006. I employ the synthetic control method to construct a synthetic Troup County then use this counterfactual to measure the FDI's impact with a DID estimator. I estimate that the Kia Motors investment is associated with an annual 13%–16% increase in employment on average and a null effect on PCPI.

3.1 | Case study background

In 2006, Troup County, Georgia experienced one of the largest FDIs of any rural area when Kia Motors decided to establish its operations in the county. This investment included a $1.2-billion assembly plant which would provide approximately 2500 jobs. In addition, Kia also expected several suppliers to establish their operations in the area in the following years. These include Hyundai-Mobis, Dongwon Autopart Technology, Sejong Georgia LLC, Sewon America, Daehan Solution, Johnson Controls, Glovis, Powertech America, and Kumho Tire. These subsequent projects represented an additional 3600 jobs and $830 million investment in the region.

Before locating in Troup County, the automaker requested site preparation assistance, workforce training, tax incentives, and proof that the plant would be welcomed by the local community. Georgia spent $48.5 million on site preparation, $81 million on road improvements, and also built a $14.5 million training facility that would be managed by the QuickStart program. In addition to these incentives, Troup County also offered the automaker $130 million in property tax abatements that would gradually dissipate over a 16-year period and an additional $21 million toward infrastructure development.

From a site selection perspective, this move was unexpected. Troup County did not have the necessary infrastructure and supply chain linkages, as evidenced by the deal requirements, that other areas could offer. The only benefit of the area was that it was located near a major highway and in close proximity to the Hyundai plant in Montgomery, Alabama.

Locals were excited about the new plant and expected a housing boom and a revitalization of the area with new families locating near the auto plant. While a few new businesses have since emerged, the "boom" was smaller than expected and county unemployment still remained high in 2011. As with all large business deals, there are also concerns regarding the tax concessions and whether the economic benefits of the investment outweigh the costs (Maister, 2011).

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TABLE 1 Descriptive statistics for Troup County, 2000.

Variable	Troup County	Georgia
Population (in thousands)	58.78	8186.45
Poverty rate (%)	14.80	13.00
Unemployment rate (%)	3.60	3.70
Share manufacturing establishments	0.08	0.05
Per capita personal income ($)	24,859	28,851
Employment to population ratio	58.93	48.92

Note: Data is from U.S. Census Bureau, Bureau of Labor Statistics (BLS), United States Department of Agriculture (USDA), and Bureau of Economic Analysis (BEA).

Table 1 presents descriptive statistics for Troup County in 2000. Compared to the rest of Georgia, Troup County has a higher poverty rate and lower PCPI. Additionally, 8% of Troup County's business establishments belong to the manufacturing sector, compared to only 5% for the state of Georgia.

3.2 | Methodology: Difference-in-differences regression with synthetic control method

For the Kia Motors case study, I use the synthetic control method to provide a counterfactual for Troup County. The synthetic control method uses a donor pool and weights to create a synthetic county for comparison. The donor pool consists of rural counties that did not receive FDI and are thus untreated. Weights are chosen to minimize the difference between the outcome variables of the treated county and the synthetic county during the pre treatment period. In Supporting Information S2: Appendix, I provide more detail on how the synthetic county is constructed and what variables are used to determine the weights of the donor counties.

With the synthetic county, I then employ a DID econometric design to estimate the treatment effect of the Kia plant investment on Troup County's PCPI and total employment. The regression takes the following form:

$$lnY_{it} = \beta_0 + \beta_1 Post_t + \beta_2 Troup_i + \beta_3 Post_t \cdot Troup_i + \epsilon_t \quad (1)$$

lnY_{it} represents the log of the outcome variables: PCPI and total county employment by employee's residence and employee's place of work (POW). $Post_t$ is an indicator variable for whether year t is in the pre treatment period (0) or post treatment period (1). $Troup_i$ is an indicator variable for whether county i is the treated county, Troup County (1), or the synthetic county (0). Lastly, the coefficient of interest, β_3, captures the average treatment effect on the outcome variable in Troup County for the post-treatment period.

I elect two treatment years to be analyzed separately. The first analysis uses 2006, the year the FDI was announced, and the second uses 2009, the year the Kia plant began production. This second treatment year is used in case the effects of the investment are delayed until plant opening. Given the literature, I would expect the hiring of workers to both impact county incomes and total employment more than building the plant or adding supplemental infrastructure, both of which were done in the years after the investment but prior to plant opening. For the first study, the pretreatment period is 10 years prior (1995–2006) and the post-treatment period is 9 years after the investment (2007–2015). The second study has the same range (1995–2015) with a split in pre- and post-treatment periods at 2009.

3.3 | Case study data

3.3.1 | Rural classifications

I rely on The Rural-Urban Continuum Code (RUCC) from the USDA to classify rural and metro areas. The RUCC builds upon the official metro and nonmetro designations determined by the Office of Management and Budget. It further separates counties into three levels of metro development and six levels of nonmetro development. Counties are

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TABLE 2 FDI capital investment as a share of county GDP.

Sample	*N*	Mean (%)	Median (%)	75th pct. (%)	90th pct. (%)
All rural treated	597	25.42	5.24	16.12	42.74
RUCC 4–5	173	8.83	2.67	9.57	20.50
RUCC 6–7	353	17.91	5.68	16.98	36.64
RUCC 8–9	71	103.17	16.69	94.81	207.76
Troup County: FDI/GDP = 37.93%; approx. 89th percentile					

Note: FDI capital investment as a share of county GDP is computed using cumulative FDI capital investment from fDi Markets divided by Bureau of Economic Analysis county-level GDP in the year of the county's first FDI announcement. Sample includes the 597 treated rural counties in the propensity score matching analysis on common support. GDP is measured in current dollars. Troup County has an FDI capital share of 37.9%, placing it at approximately the 89th percentile of the full distribution.

Abbreviations: FDI, foreign direct investment; GDP, gross domestic product; RUCC, Rural-Urban Continuum Code.

ranked on a scale of one to nine, with scores of 1-3 representing urban counties and scores of 4–9 representing rural or nonmetro counties. These county codes are updated every 10 years.

The donor pool for the case study consists of 56 rural counties that did not receive any FDI between 2003 and 2018 according to fDi Markets data. Since Troup County's RUCC is four, the RUCC for the donor pool counties is either four or five for 1993 and 2003. A county with a code in the 4–5 range in 1 year but not in the other year was eliminated from the donor pool. I also exclude U.S. islands and territories from the donor pool. Additionally, I exclude Troup County's neighboring counties from the donor pool to avoid contamination from potential spatial spillovers of the Kia Motors investment.

3.3.2 | FDI data

The FDI data come from fDi Markets and provide the treatment status and timing for my sample. fDi Markets reports on any new foreign investment or expansion of a foreign investment that leads to new jobs and capital investment.[1] It gathers data on FDI from public sources, such as the Financial Times newswires, media sources, industry organizations, investment promotions agencies newswires, and purchases data from market research and publication companies. While priority is given to company sources, a FDI project announcement is verified with multiple sources. fDi Markets collects information on a FDI project's capital investment and jobs in real time as much as possible.[2] At the end of each month, they publish their findings from the previous month, update the data with actual investment and job numbers, and delete canceled projects.

To contextualize the scale of FDI relative to local economic activity, I compute each treated county's cumulative FDI capital investment as a share of county gross domestic product (GDP) in the year of its first FDI announcement, using BEA county-level GDP data from the CAGDP1 series. Table 2 reports the distribution of this measure for the 597 treated rural counties in the PSM sample, disaggregated by RUCC category. Across all treated counties, the median FDI capital share is 5.2% of county GDP and the 75th percentile is 16.1%, indicating that for most rural counties FDI represents a meaningful but not dominant share of local economic activity. The distribution is right-skewed: the mean of 25.4% is pulled upward by a small number of very large investments relative to very small local economies, particularly among the most remote counties.

The RUCC breakdown reveals important heterogeneity in FDI's relative economic importance across rural county types. In RUCC 4–5 counties, which are relatively close to metropolitan areas and have larger local economies, the median FDI share is just 2.7% of GDP. In RUCC 6–7 counties the median rises to 5.7%, reflecting both a greater relative investment scale and smaller county economies. Among RUCC 8–9 counties the mean FDI share reaches 103%, though the median of 16.7% is more representative of the typical county in this group. The elevated mean for RUCC 8–9 reflects the very small GDP base of the most remote rural counties rather than implausibly large FDI flows. A single modest investment can represent a substantial fraction of a very small local economy, consistent with the absorptive capacity constraints documented in Section 4.5.1. For reference, Troup County's FDI capital share of 37.9% places it at approximately the 89th percentile of the full distribution, confirming that the Kia Motors investment was large relative to the typical rural FDI project but not without precedent in the sample.

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Referencing a 2024 snapshot of the fDi Markets database, I create a donor pool of counties that did not receive FDI between 2003 and 2018. Then, I restrict the sample to counties whose rural continuum code in 1993 and 2003 was four or five. The resulting donor pool has 56 untreated rural counties. In a robustness check later on, I change the donor pool to counties that received FDI in 2016–2019 and did not receive FDI from 2003 to 2015. I refer to these 15 counties as later treated counties. In that specification, I compare Troup County to a synthetic county made of counties that eventually become treated after the analysis period.

3.3.3 | Outcome variables

The dependent variables for the case study are PCPI and total employment at the county level from 1995 to 2015. Data for PCPI comes from the BEA (Bureau of Economic Analysis, 2022a). PCPI is the sum of the personal income of residents in the county divided by the county population. Population figures come from the U.S. Census Bureau's mid-year estimates (July 1).[3] Interpreting PCPI becomes difficult when there is presence of a large institutional population, such as a college or prison, or a group atypical to the resident population which temporarily inflates or deflates the per capita income, such as a major construction project (Bureau of Economic Analysis, 2022b).[4] Since Troup County has neither a major university nor prison,[5] I do not find these interpretation difficulties a concern.

For measuring FDI's impact on income, I use PCPI for three distinct reasons. First, PCPI, as opposed to other income variables, measures approximate changes in county wages without firm-level employee wage data. Second, it allows the comparison rural counties with differing populations, unlike aggregate income. Lastly, PCPI provides a proxy for the average standard of living and quality of life for a typical county resident.

The data for total county employment by employee residence comes from the BLS-LAUS program (U.S. Bureau of Labor Statistics, 2023b). BLS-LAUS defines total county employment as the number of employed persons who reside in the county. To be counted as employed, an individual must have received some amount of pay or profit during the survey week, worked unpaid for 15 h at a family business, or been on a temporary absence from their regular job (U.S. Bureau of Labor Statistics, 2023a). I interpret the total employment variable from the BLS-LAUS as the number of county residents who received pay via employment or held a job during the week of the survey.

I also investigate the impact of FDI on jobs in Troup County using data from the BEA. While the BLS-LAUS data considers an individual's residency when tallying employment, the BEA counts employment by POW, effectively estimating the number of jobs in the county. The BEA's total county employment includes full-time and part-time jobs, wage and salary jobs, and self-employed jobs.[6] I interpret the total county employment from the BEA as the number of full-time, part-time, and self-employed jobs in the county. From looking at both of these employment measures, I distinguish between FDI's impacts on employment for Troup County residents and on total jobs held by residents and non-residents.

3.4 | Case study results

3.4.1 | Synthetic control method: Weight and fit

This section reviews how the outcomes variables for the synthetic county mimics that of Troup County in the pre-treatment period. I conduct the synthetic control method analysis as outlined in Section 3.2 for all outcome variables with the treatment year of 2006 and separately for 2009. I list the donor counties and weights for the synthetic county in Table 3 for PCPI, Table 4 for employment by residence, and Table 5 for employment by POW. Muskingum County, Ohio and Dodge County, Wisconsin appear consistently in all the synthetic units. Other states represented often are Mississippi, Missouri, Oklahoma, and Florida.[7]

Table 6 details how the outcome variable differs between the synthetic unit and Troup County on average for the pretreatment years. The synthetic control method specifications for PCPI produce on average a gap of 0.01 and 0.02 between the natural log of PCPI for Troup County and the synthetic unit during the pretreatment period for treatment years 2006 and 2009, respectively. These gaps are about 0.10% and 0.20% of the mean PCPI for Troup County in 2005, the year before the FDI. For employment, I detect essentially no gap between Troup and the synthetic unit during the pretreatment period. Together, these findings confirm that the synthetic unit has average outcomes almost identical to that of Troup County in the pretreatment period.

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TABLE 3 Synthetic control weights for Troup County for PCPI.

Treatment year = 2006		Treatment year = 2009	
Ross County, OH	0.3650	Baldwin County, GA	0.3150
Scotts Bluff County, NE	0.2280	Muskingum County, OH	0.2410
Baldwin County, GA	0.2000	Blue Earth County, MN	0.1860
Juneau Borough/city, AK	0.0940	Ross County, OH	0.1100
Dodge County, WI	0.0770	Humboldt County, CA	0.0590
Albany County, WY	0.0290	Juneau Borough/city, AK	0.0410
Muskingum County, OH	0.0060	Dodge County, WI	0.0360
		Defiance County, OH	0.0140

Note: This table lists all the donor counties and weights for the synthetic unit for Troup County for the PCPI analysis. The data for constructing the synthetic unit come from the Bureau of Economic Analysis, years 1995–2015.

Abbreviation: PCPI, per capita personal income.

TABLE 4 Synthetic control weights for Troup County for total employment by residence.

Treatment year = 2006		Treatment year = 2009	
Muskingum County, OH	0.3060	Muskingum County, OH	0.4320
Scott County, MO	0.1950	Jackson County, OK	0.1860
Jackson County, OK	0.1530	Citrus County, FL	0.1410
Citrus County, FL	0.1420	Scott County, MO	0.0940
Dodge County, WI	0.1140	Dodge County, WI	0.0790
Monroe County, FL	0.0450	Morgan County, IL	0.0370
Morgan County, IL	0.0450	Albany County, WY	0.0310

Note: This table shows the counties that make up the synthetic unit for Troup County for the analysis of total county employment by residence. Data come from Bureau of Labor Statistics Local Area Unemployment Statistics, years 1995–2015.

TABLE 5 Synthetic control weights for Troup County for total employment by POW.

Treatment year = 2006		Treatment year = 2009	
Muskingum County, OH	0.5640	Muskingum County, OH	0.4810
Citrus County, FL	0.1440	Stephens County, OK	0.2170
Gila County, AZ	0.1330	Citrus County, FL	0.1450
Stephens County, OK	0.0440	Dodge County, WI	0.1240
Dodge County, WI	0.0420	Marion County, MO	0.0220
Marion County, MO	0.0320	Dodge County, NE	0.0120
Jones County, MS	0.0220		
Bolivar County, MS	0.0190		

Note: This table shows the counties that make up the synthetic unit for Troup County for analysis of total county employment by POW. The data source is the Bureau of Economic Analysis, years 1995–2015.

Abbreviation: POW, place of work.

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TABLE 6 Comparison of outcome variables: Troup versus synthetic.

Outcome variable	Troup County	Synthetic unit	Gap	Gap/Troup's 2005 value	RMSPE	RMSPE/Troup's 2005 value (%)
Log of PCPI (1995–2005)	10.09	10.08	0.0100	0.0010	0.01	0.10
Log of PCPI (1995–2008)	10.13	10.15	−0.0200	−0.0020	0.02	0.20
Log of employment by residence (1995–2005)	10.24	10.24	0.0000	0.0000	0.02	0.20
Log of employment by residence (1995–2008)	10.24	10.24	0.0000	0.0000	0.01	0.10
Log of employment by POW (1995–2005)	10.54	10.54	0.0000	0.0000	0.01	0.09
Log of employment by POW (1995–2008)	10.55	10.57	−0.0200	−0.0019	0.01	0.09

Note: This table compares the natural log of PCPI, total employment by residence, and total employment by POW for Troup County and the synthetic unit for the 2006 and 2009 treatment year specifications. The reported values are unit averages over all the pretreatment years. The gap is the difference between Troup County's value and the synthetic unit's and is shown as a fraction of the 2005 Troup County value. The RMSPE is the sum of the squared difference between the outcomes for Troup County and the synthetic unit across the years specified. Data come from the Bureau of Labor Statistics Local Area Unemployment Statistics and the Bureau of Economic Analysis, years 1995–2008.

Abbreviations: PCPI, per capita personal income; POW, place of work; RMSPE, root mean squared prediction error.

I graph the time series of the outcomes variables from 1995 to 2015 in Figure 1. In the post-treatment period, Troup County's PCPI decreases compared to the synthetic unit. Troup County's total employment levels are higher compared to the synthetic unit in the post treatment period for both the 2006 and 2009 treatment years. These figures indicate that the Kia investment positively impacted employment by residency and by POW.

3.4.2 | Inference from synthetic control method placebo tests

Next, I assess the significance of my results with RMSPE ratios and placebo tests, as described by Cunningham (2021). The Root Mean Square Prediction Error (RMSPE) is the average gap between the treated county and the synthetic county before treatment or after treatment. For the placebo tests, I conduct the synthetic control analysis as though a donor unit is treated. I repeat this for the 56 donor counties and record the pre-treatment and post-treatment RMSPEs.

Then, I construct a distribution of placebo effects to evaluate the magnitude of the treatment effect in Troup County relative to that of the placebo tests. This exercise provides evidence that either the results stem from randomness, in which case Troup County's ratio is low, or from the treatment, in which case the ratio is high.[8] If Troup County's ratio is in the top 5% or at least the second highest, then I infer that the observed treatment effects were unlikely to have happened by random chance. Thus, the rank cutoff establishes an alpha level of 0.05.

Figure 2a for 2006 and Figure 2b for 2009 present the histograms from the placebo tests for PCPI. The figures have some outliers on the histograms. However, Troup County does not stand out from the others in terms of the post- to pre-treatment RMSPE ratio. Furthermore, with a 2006 treatment year, Troup County's RMSPE ratio is 6.472. The ratio ranks 12 out of 56 counties, giving it a *p*-value of 0.2105. In the 2009 treatment year specification, the ratio for Troup County is 3.0119 and ranked 28 out of 56 with a *p*-value of 0.491. With both *p*-values far above the 5% significance level, I cannot say with confidence that the negative effect on PCPI estimated from the synthetic control method is not from randomness.

The histograms for employment by residency and POW in Figure 2 provide evidence that the employment effects the synthetic control method measured do not stem from randomness. In the 2009 specification for employment by residency, Troup County has the second highest ratio. For employment by POW, Troup County has the first or second highest ratio for both treatment years. The placebo tests further support that the Kia Motors FDI substantially increased employment in Troup County, regardless of employee residence.

In addition to the histograms, I create gap graphs, which offer an alternate view of the placebo test results. I graph the difference between the outcome variables for each county and its synthetic unit from 1995 to 2015. These gap graphs are displayed in Figure 3. For clarity in the figures, I only include 30 counties who have the highest RMSPE ratios.

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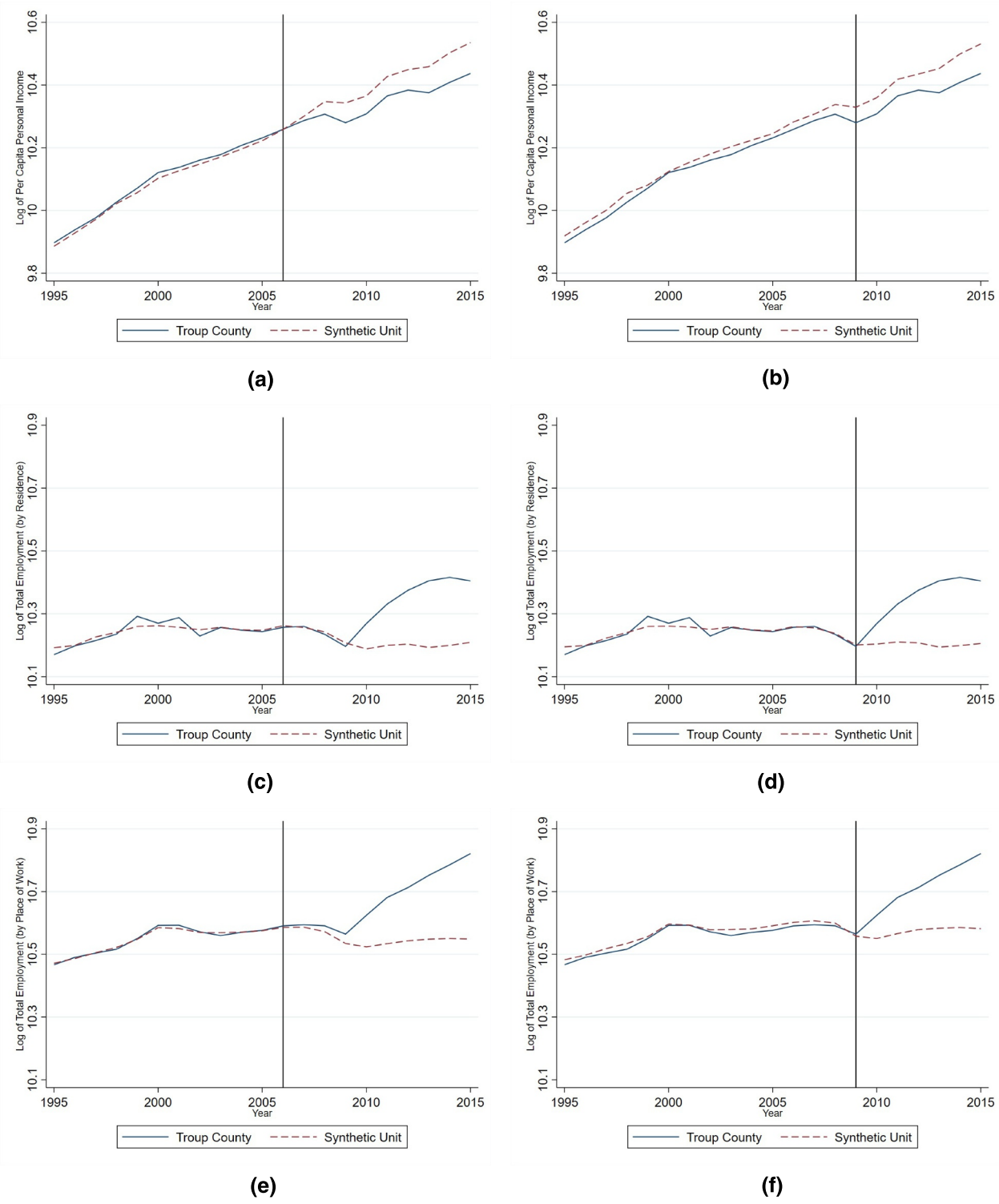


FIGURE 1 Synthetic control fit, 1995–2015. (a) PCPI, 2006 treatment year. (b) PCPI, 2009 treatment year. (c) Employment by residence, 2006 treatment year. (d) Employment by residence, 2009 treatment year. (e) Employment by POW, 2006 treatment year. (f) Employment by POW, 2009 treatment year. These figures graph the time series for Troup County and the synthetic unit for the following variables: log of PCPI, log of total county employment by residence, and log total county employment by POW. The treatment year for panels (a), (c), and (e) is 2006, and the treatment year for panels (b), (d), and (f) is 2009. PCPI and total county employment by POW data come from the Bureau of Economic Analysis. Data on total employment by residence comes from Bureau of Labor Statistics Local Area Unemployment Statistics. Years of analysis are 1995–2015. PCPI, per capita personal income; POW, place of work.

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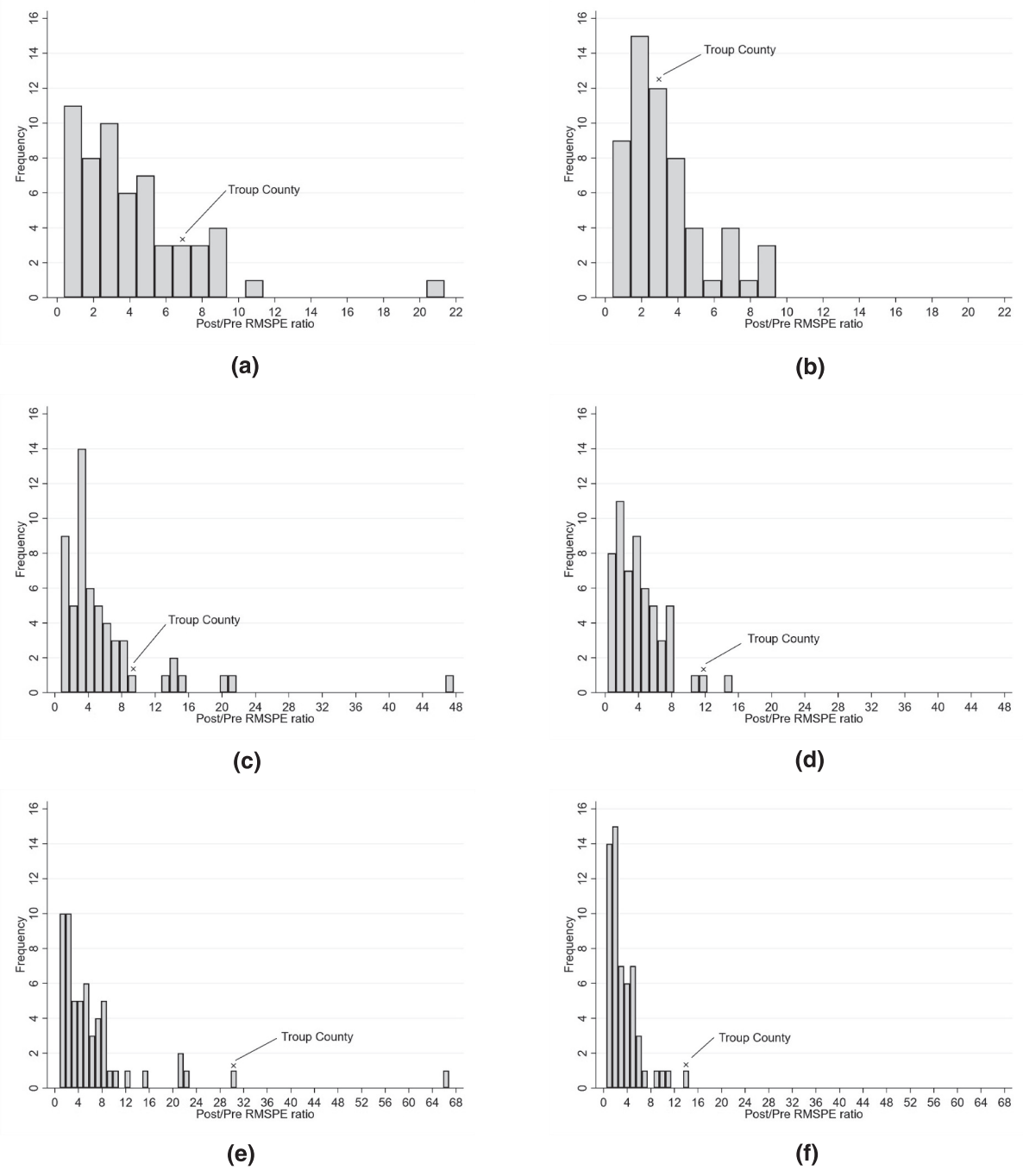


FIGURE 2 Histogram of RMSPE ratios for placebo tests. (a) PCPI, 2006 treatment year. (b) PCPI, 2009 treatment year. (c) Employment by residence, 2006 treatment year. (d) Employment by residence, 2009 treatment year. (e) Employment, 2006 treatment year. (f) Employment, 2009 treatment year. The histograms show the distribution of the ratio of the post- and pre-treatment RMPSE for Troup County and the 56 placebo counties for these variables: log of PCPI, log of total county employment by residence, and log total county employment by POW. I perform the synthetic control simulation for each donor county and calculate the pre- and post-treatment RMSPE. The treatment year for panels (a), (c), and (e) is 2006, and the treatment year for panels (b), (d), and (f) is 2009. The RMPSE ratio for Troup County is represented by the black X. PCPI and total county employment by POW data come from the Bureau of Economic Analysis. Data on total employment by residence comes from Bureau of Labor Statistics Local Area Unemployment Statistics. Years of analysis are 1995–2015. PCPI, per capita personal income; POW, place of work; RMSPE, root mean squared prediction error.

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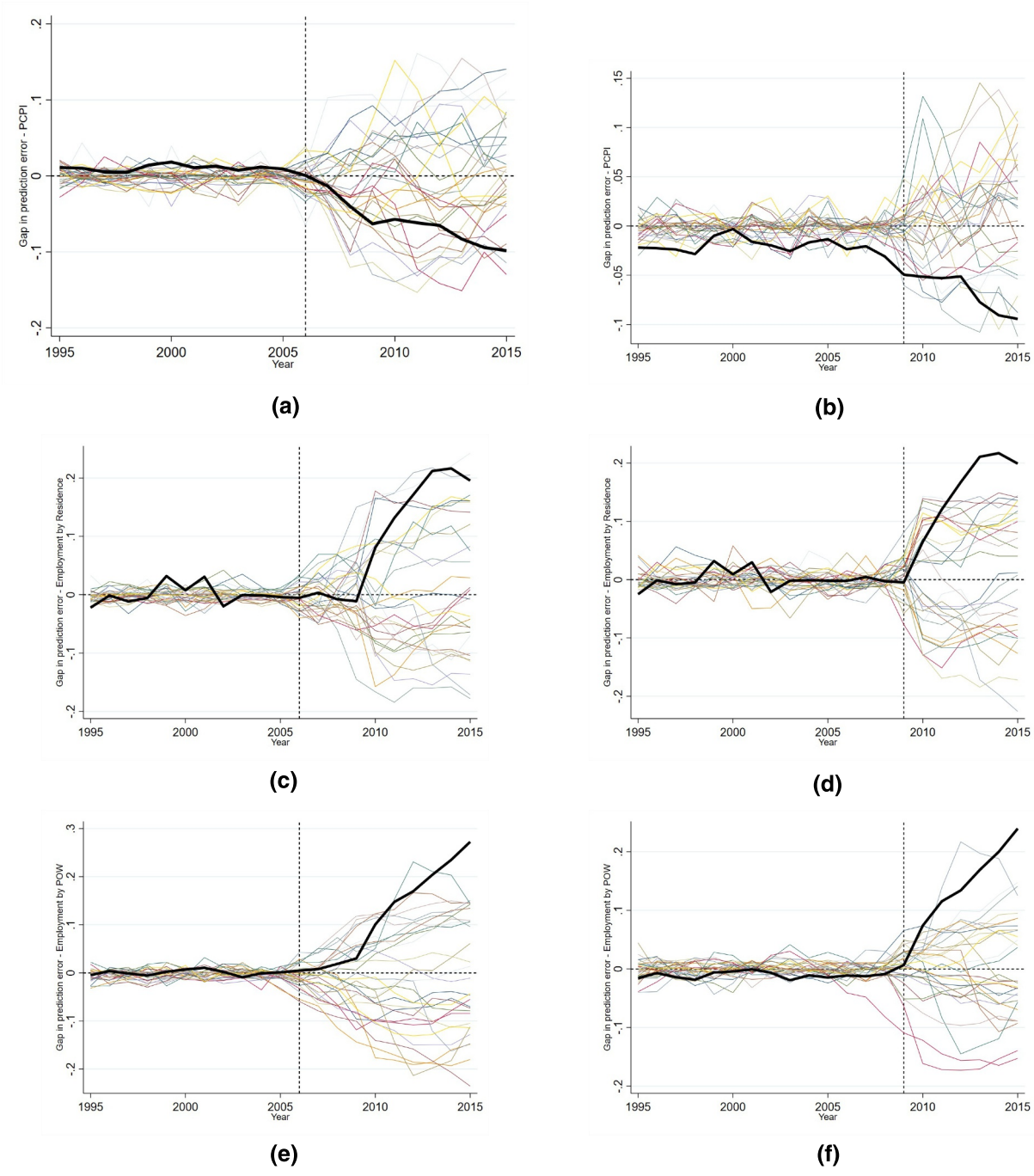


FIGURE 3 Gap in outcome variables for Troup County and placebo tests. (a) PCPI, 2006 treatment year. (b) PCPI, 2009 treatment year. (c) Employment by residence, 2006 treatment year. (d) Employment by residence, 2009 treatment year. (e) Employment by POW, 2006 treatment year. (f) Employment by POW, 2009 treatment year. This figure shows the prediction error between the treated unit and the synthetic unit in the pre-treatment period (1995–2005 or 1995–2008) and the post-treatment period (2006–2015 or 2009–2015). The prediction error is the difference between the outcome variables for the treated unit and the synthetic unit. The outcome variables are log of PCPI, log of total county employment by residence, and log of total county employment by POW. The thick black line represents Troup County. The other colored lines are the placebo tests where I simulated that the placebo county was treated. These figures include only the counties with the top 30 highest pre/post root mean squared prediction error ratio. The treatment year for panels (a), (c), and (e) is 2006, and the treatment year for panels (b), (d), and (f) is 2009. PCPI and total county employment by POW data come from the Bureau of Economic Analysis. Data on total employment by residence comes from Bureau of Labor Statistics Local Area Unemployment Statistics. Years of analysis are 1995–2015. PCPI, per capita personal income; POW, place of work.

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TABLE 7 Difference-in-differences regression estimates from synthetic control method.

Treatment year Dependent variable	2006 Log PCPI	2009 Log PCPI	2006 Log emp. by res.	2009 Log emp. by res.	2006 Log emp. by POW	2009 Log emp. by POW
Post*Troup	−0.0681 (0.0589)	−0.0471 (0.0600)	0.0982** (0.0301)	0.1389*** (0.0324)	0.1185** (0.0348)	0.1447*** (0.0373)
Percentage change	−6.58%	−4.60%	10.32%	14.90%	12.58%	15.57%
Dep. var. mean	10.22	10.23	10.25	10.25	10.58	10.59
RMSPE ratio	6.4719	3.0119	8.8409	11.7750	29.8577	14.0708
RMSPE rank	12	28	8	2	2	1
RMSPE *p*-value	0.2308	0.4912	0.1538	0.0385	0.0385	0.0175

Note: This table presents the regression results for the difference-in-difference specification comparing Troup County and the synthetic unit. Only the coefficient on Post*Troup, β_3, is reported. For columns (1) and (2), the dependent variable is the natural log of PCPI. The dependent variable for columns (3) and (4) is the log of total county employment by residency. The dependent variable for columns (5) and (6) is the log of county employment by POW. The treatment year is 2006 for columns (1), (3), and (5) and 2009 for columns (2), (4), and (6). Standard errors are in parentheses and clustered at the county level. The last row of the first panel is the percentage change, which is $\left(e^{\beta_3}-1\right)*100$. In the last panel, the RMSPE Ratio is the post-treatment RMPSE divided by the pre-treatment RMPSE. The RMSPE rank is the rank for Troup County's RMSPE ratio compared to the 56 placebo tests with the counties in the donor pool. The RMSPE Rank *p*-value is Troup County's RMSPE rank divided by 57, the total number of synthetic analysis (Troup County and the 56 placebo tests). Data come from the Bureau of Labor Statistics Local Area Unemployment Statistics and Bureau of Economic Analysis, years 1995–2015.

Abbreviations: PCPI, per capita personal income; POW, place of work; RMSPE, root mean squared prediction error.

*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Visually, the gap graphs support the findings from the placebo tests. The treatment effect measured via the synthetic control method on PCPI is not unique to Troup County. Other counties achieve a good fit in the pre-treatment and see a placebo effect in the post-treatment period. In the PCPI placebo tests, 11 counties in the first specification and 27 in the second have a more significant treatment effect than Troup County, although none of them received FDI in the same period. However, in the employment gap graphs, Troup County stands out with a notable treatment effect compared to the placebo tests, particularly for the 2009 treatment year.

3.4.3 | Synthetic control method: Main results

I contextualize the findings and their statistical significance with the DID regression results in Table 7. Neither of the coefficient estimates for PCPI are statistically significant at the conventional levels. These estimations support the findings from the placebo tests; the PCPI results are not distinguishable from randomness. The Kia Plant investment appears to have a null effect on PCPI.

Beyond aggregate growth outcomes, another strand of the literature examines how FDI affects local wages and finds mixed evidence. FDI leads to more foreign company presence in the host county, which should consequently increase labor demand and average wages. However, if the labor pool remains unchanged, FDI could have no impact on wages (Kato-Vidal, 2013). Foreign companies also may be incentivized to pay higher wages to reduce turnover and the possible sharing of company propriety information. Companies may also offer higher wages to employ more skilled workers or to meet host country regulations (Breau & Brown, 2011; Kucharčíková et al., 2015).

I estimate FDI increases the number of employed Troup County residents by 10% (5% significance level) on average from 2006 to 2015. Additionally, I estimate a 15% increase (1% significance level) in the employment of Troup residents each year on average after the Kia plant began production in 2009. The difference in the point estimates for 2006 and 2009 are consistent with the theory that the opening of the Kia plant in 2009 impacted residential county employment more than the investment in the building, infrastructure, and job up-skilling for the project between 2006 and 2009.

Columns (5) and (6) of Table 7 present the regression results for the analysis of total county employment by POW. For the 2006 specification, the Kia Motors FDI is associated with a 13% annual increase in employment in Troup County on average (5% significance level). Similarly, I estimate that county employment by POW increases by 16% per year on average from the Kia plant opening in 2009 (1% significance level). The low RMSPE rank and RMSPE *p*-value reported in the last panel of Table 7 also suggest a substantial change in county employment by POW due to the FDI.

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The effect of FDI on employment extends beyond direct employment within the Kia plant itself. Kia's incentive package was structured around a minimum commitment of 2500 new jobs. Yet between 2005 and 2015, the number of employed residents in Troup County increased by approximately 5,000, while total jobs located in the county, regardless of worker residence, rose by roughly 11,000. These magnitudes suggest that the employment impact of the Kia investment reached well beyond the plant's direct payroll.

Part of this expansion reflects the entry of supplier firms that were anticipated as part of the original investment negotiations. However, the employment growth likely also reflects complementarities between foreign and domestic investment. The presence of a large manufacturing anchor can deepen local supply chains, expand intermediate demand, and reduce fixed costs for related firms. In this way, the Kia investment may have crowded in both foreign and domestically sourced establishments, contributing to broader job creation in Troup County and surrounding areas. The observed employment gains should therefore be interpreted as the combined effect of direct FDI employment, supplier entry, and induced local investment.

3.4.4 | Robustness check: Alternate control group

In this section, I augment the main results by repeating the synthetic control method DID estimation with a different control group. Even with the synthetic control, underlying characteristics of counties that receive FDI could still be fundamentally different from the counties that do not. To address this concern, I change the control group to be rural counties with RUCC 4 and 5 that did not receive FDI between 2005 and 2015 but later received FDI in 2016–2019. I refer to this control group of 15 donor counties as later treated counties. Since these counties eventually receive FDI, their unobservable characteristics are more likely to match that of Troup County. I include the graphs comparing the treated and synthetic unit for this control group in Figure 4.

The results presented in Table 8 with the alternate control group echo the findings of the previous section. The Kia Plant FDI is associated with no change in PCPI and a 10%–14% increase in county employment by residency. I also estimate that the investment increased county employment by POW by 8%–15%. These estimates with the later treated control group are in line with the estimates from the never treated control group in Table 7. Furthermore, the 95% confidence intervals for the estimated coefficients overlap.

4 | EVIDENCE FROM FDI IN RURAL U.S. COUNTIES

The subsequent analysis builds on the case study results with more generalizable results for the average rural county. The case study provides insight into the economic effects of a very particular investment in a single rural Georgia county. The following empirical analysis includes rural counties across the United States. I combine PSM with a DID model to estimate the impact of FDI on personal income and employment growth.

4.1 | Methodology: Propensity score matching

I employ PSM to create the control group for the analysis of U.S. rural counties. The PSM addresses endogeneity concerns that treated counties have underlying, unobserved characteristics that explain why they receive FDI. The propensity score is the conditional probability that the rural county will receive FDI between 2004 and 2019, given the industry and demographic profile of the rural county in the year 2000.

I calculate each county's propensity score based on the following logit model, as in O'Keefe (2004):

$$\Pr(EverFDI = 1) = \left(e^{\beta' X}\right) / \left(1 + e^{\beta' X}\right) \tag{2}$$

EverFDI is a dummy variable, taking the value of 1 if the county received any foreign investment projects between 2004 and 2019 and zero otherwise. Matrix **X** is the county profile characteristics that determine the county's probability of receiving FDI. This county profile is based on a matrix of characteristics associated with FDI location decisions and is described in more detail in Section 4.2. I bootstrap the standard errors 50 times. Figure 5 shows that the propensity score distributions of treated and untreated counties overlap.

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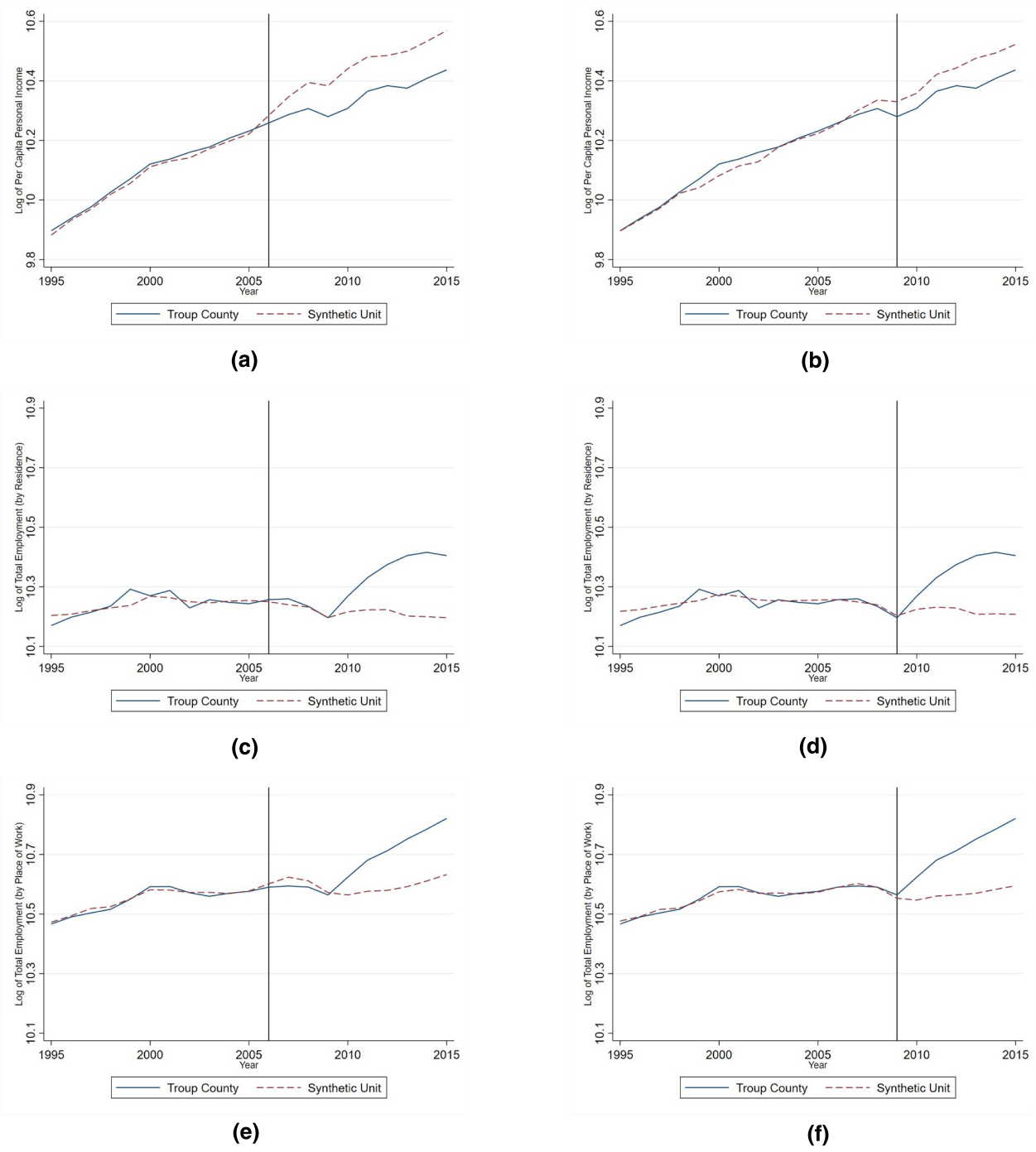


FIGURE 4 Alternate control group: synthetic control fit, 1995–2015. (a) PCPI, 2006 treatment year. (b) PCPI, 2009 treatment year. (c) Employment by residence, 2006 treatment year. (d) Employment by residence, 2009 treatment year. (e) Employment by POW, 2006 treatment year. (f) Employment by POW, 2009 treatment year. These figures graph the time series for Troup County and the synthetic unit for the following variables: log of PCPI, log of total county employment by residence, and log total county employment by POW. The treatment year for panels (a), (c), and (e) is 2006, and the treatment year for panels (b), (d), and (f) is 2009. PCPI and total county employment by POW data come from the Bureau of Economic Analysis. Data on total employment by residence comes from Bureau of Labor Statistics Local Area Unemployment Statistics. Years of analysis are 1995–2015. The control group includes rural counties with Rural-Urban Continuum Code 4 and 5 that did not receive FDI between 2003 and 2015 but later received FDI in 2016–2019. FDI, foreign direct investment; PCPI, per capita personal income; POW, place of work.

With the propensity scores, I then perform matching using the nearest neighbor method with replacement. I allow replacement in the matching algorithm, so a control county can serve as a match for one or more treated counties. On average, an untreated county serves as a control for two treated counties.

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TABLE 8 Difference-in-differences regression estimates from synthetic control method, alternate control group.

Treatment year Dependent variable	2006 Log PCPI	2009 Log PCPI	2006 Log emp. by res.	2009 Log emp. by res.	2006 Log emp. by POW	2009 Log emp. by POW
Post*Troup	−0.1107	−0.0777	0.0958**	0.1324***	0.0774*	0.1387***
	(0.0589)	(0.0607)	(0.0293)	(0.0323)	(0.0347)	(0.0373)
Percentage change	−10.48%	−7.48%	10.06%	14.16%	8.05%	14.88%
Dep. var. mean	10.23	10.22	10.25	10.26	10.59	10.58

Note: This table presents the regression results for the difference-in-difference specification comparing Troup County and the synthetic unit with the alternate control group. The alternate control group includes rural counties with Rural-Urban Continuum Code 4 and 5 that did not receive FDI between 2003 and 2015 but later received FDI in 2016–2019. Only the coefficient on Post*Troup, β_3, is reported. For columns (1) and (2), the dependent variable is the natural log of PCPI. The dependent variable for columns (3) and (4) is the log of total county employment by residency. The dependent variable for columns (5) and (6) is the log of county employment by POW. The treatment year is 2006 for columns (1), (3), and (5) and 2009 for columns (2), (4), and (6). Standard errors are in parentheses and clustered at the county level. The last row of the first panel is the percentage change, which is $\left(e^{\beta_3} - 1\right) * 100$. Data come from the Bureau of Labor Statistics Local Area Unemployment Statistics and Bureau of Economic Analysis, years 1995–2015.

Abbreviations: FDI, foreign direct investment; PCPI, per capita personal income; POW, place of work.

*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

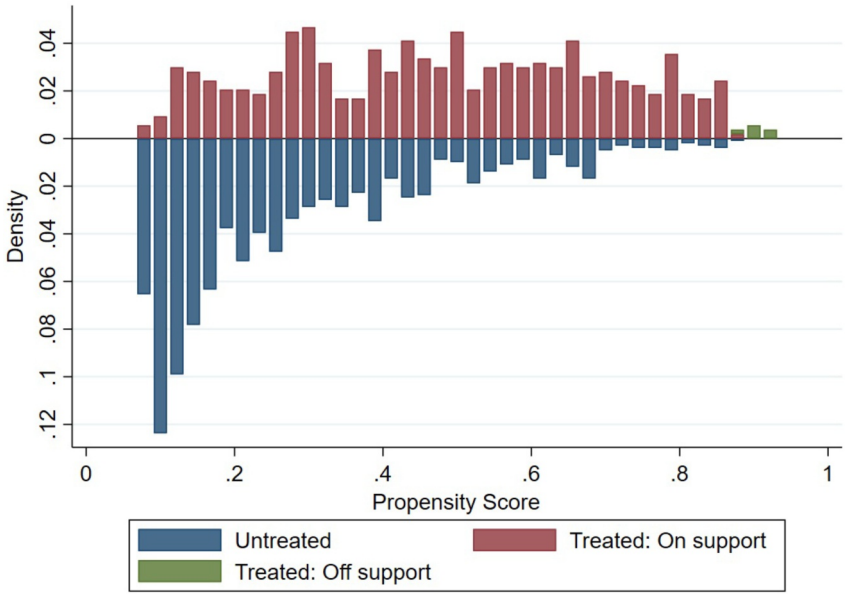


FIGURE 5 Visual overlap diagram for potential nearest neighbor matches. This figure shows the propensity score for treated and control counties on the following covariates: the natural log of population, the natural log of total establishments, the weekly wage location quotient (LQ) and the Manufacturing Employment LQ in the commuting zone, the number of railroad tracks in the county, the number of highway miles per square mile of county, and the unemployment rate. All covariates are from 2000 except for poverty (1999), highway miles (2025), and railroad tracks (2025). Covariate data come from the USDA, U.S. Census Bureau, U.S. Department of Transportation and Bureau of Labor Statistics. Treatment status comes from fDi Markets. FDI, foreign direct investment.

The choice between matching with or without replacement has little effect when there is sufficient overlap between the propensity scores for the treated and control groups or there are a large number of untreated units (Dehejia & Wahba, 2002). With substantial differences between the PSM distributions, matching with replacement is recommended (Caliendo & Kopeinig, 2008; Dehejia & Wahba, 2002). Additionally, matching with replacement improves the quality of matches, decreases bias, and places no importance on the order of matching unlike matching without replacement. However, these benefits come at the expense of a higher variance for treatment estimates (Caliendo & Kopeinig, 2008).

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TABLE 9 Control variables for propensity score matching logit models.

Logit specification variable	L1**	L2	L3	Relation to FDI location decisions	Data source
Ln population	X	X	X	Labor force size	USDA
Ln total establishment	X	X	X	Agglomeration, other industry	QCEW-BLS
CZ: Total weekly wage LQ	X		X	Labor costs in the commuting zone	QCEW-BLS
Manufacturing employment LQ		X	X	Specialization in manufacturing, agglomeration	QCEW-BLS
Poverty rate in 1999			X	Macroeconomic conditions	U.S. Census Bureau
Rail track count	X	X	X	Involvement/Readiness for supply chain	U.S. DOT
Highway miles per sq. mile	X	X	X	Access to major markets, supply chain readiness	U.S. DOT
Unemployment rate	X	X		Macroeconomic conditions	BLS-LAUS
Total weekly wage LQ		X		Labor costs, labor market characteristics	QCEW-BLS
CZ: Manufacturing employment LQ	X			Manufacturing specialization, agglomeration in CZ	QCEW-BLS

Note: This table describes matrix X, the county characteristics that determine a county's propensity score or likelihood to receive FDI, in the logit model. The symbols ** denote the preferred logit specification, which is used for the main tables and results. All data is from 2000 except highway miles per square mile, poverty rate, and rail track count. For a detailed description of the variables, see Data Supporting Information S2: Appendix.

Abbreviations: CZ, commuting zone; FDI, foreign direct investment; LQ, location quotient; QCEW-BLS, Quarterly Census of Employment and Wages—Bureau of Labor Statistics; USDA, United States Department of Agriculture.

4.2 | Data: Propensity score matching

As with the case study, I use the RUCCs and fDi Markets data to form the treated and control groups. First, I select the sample of rural counties, which includes 1981 rural counties with RUCC of four or higher in 1993 and 2003. Second, I determine the treatment status of the counties with the fDi Markets data. Counties with FDI projects between 2004 and 2019 are considered treated. Rural counties that did not receive any FDI between 2003 and 2019 are control counties.[9] This treatment definition, referred to as T1, will be used in the main results analysis.

In an alternate specification, I compare counties that received FDI early with counties that eventually receive FDI. For that specification, I define treated counties as having received FDI between 2004 and 2015. Then, I compare these with control counties that received FDI between 2016 and 2019. Unfortunately, this specification reduces the sample size to about one-fourth of the original size. I refer to this treatment definition as T2.

The final specification combines T1 and T2. Treated counties receive FDI in 2004–2015, and control counties either never receive FDI or receive FDI later from 2016 to 2019. I restrict years of analysis to 2003–2015 in this case. In the subsequent analysis, I note this treatment definition as T3. The propensity score overlap diagrams for the T2 and T3 specifications are included in Supporting Information S2: Appendix.

To construct propensity scores, I estimate a logit model that predicts the likelihood a rural county receives FDI between 2004 and 2019. County characteristics are selected based on established determinants of FDI and site selection from the prior literature. These include measures of agglomeration, infrastructure, labor market conditions, and the macroeconomic environment. Specifically, the model controls for the scale and diversity of local economic activity, manufacturing specialization, transportation connectivity, and labor market indicators at both the county and commuting zone levels. Full definitions, data sources, and robustness specifications (L1–L3) are described in the Data Supporting Information S2: Appendix. All variables and sources are listed in Table 9. As a robustness check, I employ 3 different logit specifications (L1, L2, L3), each including a different set of FDI location determinants.

4.3 | Results: Propensity score matching

In the following section, I present the results from the propensity score logit model and the nearest neighbor matching. First, I discuss the model's suitability for providing a counterfactual for rural counties that receive FDI. Then, I review the logit model results and provide evidence that the PSM balanced characteristics between treatment and control counties.

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TABLE 10 Full sample descriptive statistics, 2000.

Variable	Treatment mean (std. deviation)	Control mean (std. deviation)	$T - C$	p-value: (H0: diff = 0)
Ln population 2000	10.2049	9.3452	0.8597	0.0000
	(0.8368)	(0.9837)		
Ln total count	6.5287	5.7523	0.7764	0.0000
	(0.8218)	(0.9020)		
CZ: Manuf. emp. LQ (%)	134.2951	106.6078	27.6873	0.0000
	(70.4320)	(70.2298)		
CZ: Average wkly. wage (all)	498.4623	464.1858	34.2765	0.0000
	(75.6707)	(69.4058)		
County rail tracks count	25.8291	15.6749	10.1542	0.0000
	(26.0814)	(19.8460)		
Highway miles	0.0304	0.0117	0.0187	0.0211
	(0.2471)	(0.1124)		
Unemployment rate (%)	4.6858	4.6470	0.0388	0.6553
	(1.6761)	(1.8112)		
N	597	1384		

Note: This table presents the descriptive statistics for all U.S. rural counties in 2000. Data come from the U.S. Department of Agriculture, U.S. Census Bureau, and Bureau of Labor Statistics. Ln total count is the log of the total number of establishments in the county. Share manufacturing is the fraction of manufacturing establishments in the county compared to the total number of establishments. Vitality is the share of the county population between ages 25 and 49. Treated counties received FDI between 2003 and 2019. Untreated counties did not receive any FDI in the same time period. The fourth column reports the difference between the mean of the treated counties and the untreated counties. Column five tests for a statistically significant difference between the sample means. The reported statistics include 1981 rural counties; 597 are treated and 1384 are untreated.

Abbreviations: CZ, commuting zone; FDI, foreign direct investment; LQ, location quotient.

Table 10 depicts how treated and control rural counties compare for the characteristics that are considered in the analysis. The counties eventually experiencing FDI have higher populations and average wages. Additionally, these counties contain a higher count of establishments and are located in a CZ with a higher concentration of manufacturing establishments. These counties can access urban markets and the supply chain more easily as they have a more railroad tracks and highway miles. The unemployment rate in treated counties is slightly below that of untreated counties.

The last column of Table 10 shows that the differences in the means across these groups are statistically significant for most of the variables. Treatment counties appear inherently different from control counties and seem to already be on a different growth trajectory. This validates the need to use PSM. Without such a model, the estimates would be biased upwards.

Table 11 details the logit estimation results from the PSM model.[10] The model coefficients of log of population, average weekly wage in CZ, manufacturing employment LQ in CZ, and county railroad tracks are statistically significant at least at the 5% significance level. The estimated coeffients for concentration of highways in the county, total count of establishments, and unemployment rate are not statistically significant. However, all variables included in the model have the expected signs. A higher unemployment rate decreases the probability of FDI while more establishments increase the probability.

The logit model results support the findings of the FDI literature. Agglomeration economies, regional infrastructure, and connection to markets play a key role in FDI location decisions. Measures relating to population size, railroad tracks, and highway miles increase the probability of investment. The average weekly wage in the CZ also increases the likelihood of FDI.

Overall, rural counties that are economically linked to more urbanized areas are more likely to attract foreign investment projects. The higher wages signal a more skilled and productive workforce, although this comes with increased labor costs. Since most of the investment projects locating in rural areas are within manufacturing, the manufacturing LQ in an area also increases the probability of investment. Measures capturing industrial agglomeration, such as the total count of establishments, are also positive factors in the location decision.

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TABLE 11 Logit results, propensity score matching.

	(1) EverFDI
Ln population 2000	0.675***
	(0.196)
Ln total count	0.271
	(0.180)
CZ: Manuf. emp. LQ (%)	0.00263***
	(0.000858)
CZ: Average wkly. wage (all)	0.00277***
	(0.000806)
County rail tracks count	0.00544**
	(0.00246)
Highway miles	0.270
	(3.274)
Unemployment rate (%)	−0.00531
	(0.0333)
N	1981

Note: This table presents estimates from the logit estimation where the dependent variable EverFDI is an indicator variable equal to 1 if a rural county received any FDI projects from 2004 to 2019. This estimation is used to calculate the propensity score for each county, which is the predicted probability of receiving FDI. Data come from the U.S. Department of Agriculture, U.S. Census Bureau, and the Quarterly Census of Employment and Wages—Bureau of Labor Statistics for the year 2000, U.S. Department of Transportation, and fDi Markets. Ln total count is the natural log of the total number of establishments in the county. All county characteristics are for the year 2000 except number of rail tracks and highway miles, which comes from 2025 data.

Abbreviations: CZ, commuting zone; FDI, foreign direct investment; LQ, location quotient.

*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

I validate the suitability of the propensity score model with the propensity score overlap diagram in Figure 5 and matched pair descriptive statistics in Table 12. Figure 5 ensures the PSM model has sufficient overlap for treated counties to be matched with close neighbors and that there are no gaps at any point in the distributions. I remove treated counties that are not on common support from my sample.[11] Counties that are not used as controls are also taken out of the sample. Lastly, I remove treated counties that receive their first investment project in 2003 to ensure each county has at least one pre-treatment period in the DID regression.

Table 12 depicts how the remaining control and treated counties compare among the same group of characteristics. After the PSM model was employed, the treatment and control counties within propensity score blocks are balanced across the characteristics considered. The differences between the means of the treatment and control groups in the matched sample are no longer statistically significant for any of the variables. The final sample includes 965 rural counties, of which 591 are treated and 374 are untreated.

4.4 | Methodology and data: Difference-in-differences

With the treated and control groups established via the PSM model, I proceed to analyze of the impact of FDI on rural counties with a DID specification. The DID estimation equation takes the following form:

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TABLE 12 Matched pair descriptive statistics, 2000.

Variable	Treatment mean (std. deviation)	Control mean (std. deviation)	$T - C$	p-value (H0: diff! = 0)
Ln population 2000	10.1905	10.2007	−0.0102	0.8294
	(0.8272)	(0.8047)		
Ln total count	6.5132	6.5537	−0.0405	0.3876
	(0.8098)	(0.8035)		
CZ: Manuf. emp. LQ (%)	134.4462	129.1028	5.3434	0.1887
	(70.6643)	(69.0142)		
CZ: Average wkly. wage (all)	496.4049	494.1729	2.2320	0.5904
	(72.3040)	(70.2185)		
County rail tracks count	25.3773	26.0981	−0.7208	0.6563
	(24.9987)	(30.4060)		
Highway miles	0.0265	0.0198	0.0067	0.5705
	(0.2323)	(0.1708)		
Unemployment rate (%)	4.6998	4.6988	0.0010	0.9916
	(1.6730)	(1.6347)		
N	591	374		

Note: This table presents the descriptive statistics for U.S. rural counties included in the propensity score matching model. Data come from the U.S. Department of Agriculture, U.S. Census Bureau, the BLS, and the Quarterly Census of Employment and Wages—Bureau of Labor Statistics. Ln total count is the natural log of the total number of establishments in the county. Share manufacturing is the fraction of manufacturing establishments in the county compared to the total number of establishments. Vitality is the share of the county population between ages 25 and 49. Treated counties received FDI between 2003 and 2019. Untreated counties did not receive any FDI during the same time period. The fourth column reports the difference between the mean of the treated counties and the untreated counties. Column five tests for a statistically significant difference between the sample means. The table only includes the counties on common support. The untreated counties are weighted according to how many times they serve as a control county match.

Abbreviations: CZ, commuting zone; FDI, foreign direct investment; LQ, location quotient.

$$\ln\left(PCPI_{ijst}\right) = \alpha + \beta EverFDI_i + \delta Post_t \cdot EverFDI_i + \omega_i + \psi_j + \tau_t + \gamma_s + \epsilon_{ijst} \tag{3}$$

The dependent variable is the log of PCPI in county i of state s and matched pair j at time t. The variable $EverFDI_i$ is a dummy variable that is equal to 1 if a county receives FDI from 2004 to 2019 and zero otherwise. $Post_t \cdot EverFDI_i$ is the DID estimator and equals 1 for counties that have received FDI in time periods after the county's first FDI project. The coefficient δ on the variable $Post \cdot EverFDI_{it}$ is the coefficient of interest. It estimates the average annual percentage change in PCPI after the FDI. ψ_j are matched-pair fixed effects, and τ_t and γ_s are time and state fixed effects, respectively. I also include county fixed effects ω_i and cluster standard errors by county. Regression estimates are weighted according to how many times control counties serve as a match.[12]

With the DID specification, I also test the effect of FDI on employment. I replace the dependent variable in Equation (3) with the log of county employment by residency and by POW. As with the case study, data for the outcome variables comes from the BEA for PCPI and total county employment by POW, and the BLS-LAUS for total county employment by residence.

The DID approach addresses the issue of unobserved confounders as long as its identifying assumption is met. The assumption is that the treated and control group have parallel trends prior to treatment. In context, this assumption means that the outcomes of the counties with FDI and the counties without FDI trend similarly prior to the first FDI project. With the parallel trends assumption met, the DID estimator reduces endogeneity concerns by subtracting the initial mean differences in treatment and control groups.

Separately, PSM and DID present different innate weaknesses, yet together, compensate for those shortcomings. For instance, PSM results are sensitive to the omission of covariates that affect the probability of treatment. However, as long as those omitted explanatory characteristics are time-invariant factors, the DID estimation will remove the effect of those fixed factors on the outcome. Additionally, if any confounding factors impact the treated and control groups in the same way, those impacts will be subtracted out of the DID estimate. With the PSM model determining the control

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group, I decrease the likelihood that the outcomes of the treated and control counties would have evolved differently without treatment since matches are based on pre-treatment characteristics. Matching also ensures the treated counties are compared to control counties that are similar across various explanatory variables impacting the probability of receiving FDI (Pender & Reeder, 2011).

The event studies in Figure 6 show no pretrends in the period leading up to the FDI for PCPI or employment. This indicates that the PSM was successful in satisfying the parallel trends assumption for DID. After a county's first FDI project, the event studies show no change in PCPI and an increase in total county employment. Total employment has a small increase at first, which then grows over time, reaching its peak 10 years after the FDI. The regression results in the next section quantify this effect and its statistical significance.

4.5 | Results: Difference-in-differences

The regression results in Table 13 show that FDI in rural areas has no impact on PCPI and increases overall employment. I estimate that FDI projects increase total county employment by residency and POW by 2.8% and 2.4%, respectively, every year on average. The employment gains are about one-sixth of what Troup County, GA experienced from the Kia Plant opening. Similar to the case study, there is no impact on PCPI.

For counties which receive FDI, the amount of FDI investment has no impact on PCPI or employment, as shown in Table 14. The units of the capital investment is 100 millions. I demean the capital investment in each county, so the results are relative to the mean capital investment, which is approximately 65 million. The estimated coefficients on the variable *Post*Relative FDI Dosage* are not statistically significant for employment or PCPI. Receiving any type of FDI project has a more substantial impact on employment than the dollar amount. Additionally, increased capital investment still does not move the needle on PCPI.

4.5.1 | Heterogeneous treatment effects by rural code

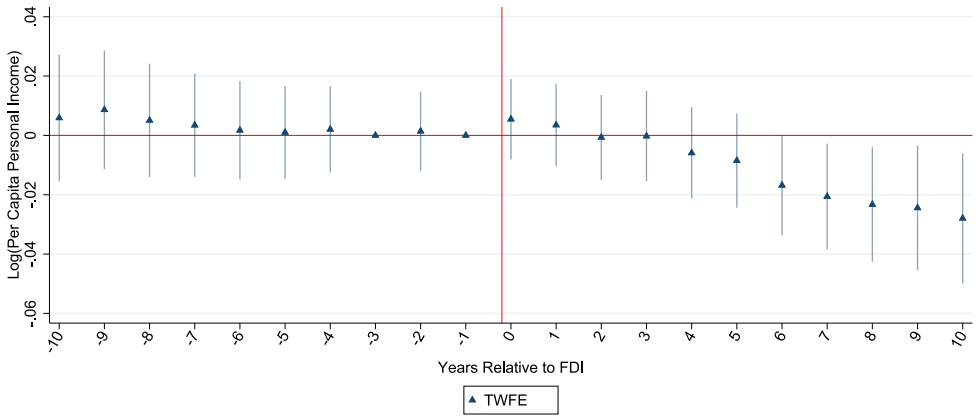
The aggregate results in the preceding section impose a common treatment effect across all rural counties. However, the Alonso–Muth–Mills model of monocentric city structure suggests this homogeneity is unlikely to hold (Alonso, 1964; Mills, 1972; Muth, 1969). In that framework, land-intensive production activities locate at the urban fringe, where land rents are lower but access to metropolitan labor markets remains feasible through commuting. Counties classified as rural under administrative definitions may therefore function as part of a broader metropolitan labor market if they are adjacent to, or economically integrated with, a major urban center.

This has direct implications for FDI: edge-of-metro rural counties benefit from agglomeration spillovers and labor market depth, while more remote counties face tighter labor supply constraints and weaker multiplier effects. The location of the Kia plant in Troup County, which is southwest of Atlanta and within commuting distance of the Atlanta metro area, illustrates this dynamic precisely. Treating all rural counties as equivalent would therefore obscure the conditions under which FDI generates employment and income gains. Therefore, I re-estimate the baseline DID specification separately for counties with RUCCs 4–5, 6–7, and 8–9, with results reported in Tables 15 and 16.

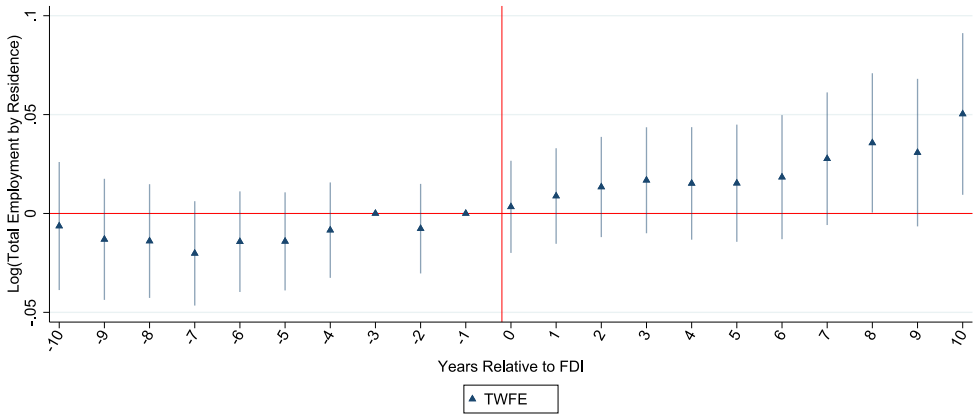
The results reveal that the employment response to FDI is concentrated in moderately rural counties. FDI projects in RUCC 6–7 counties are associated with approximately a 3.6% increase in residential employment and a 4.9% increase in total county employment per year on average. By contrast, FDI has no statistically significant effect on employment in counties with RUCC 4–5 or RUCC 8–9. The pattern for PCPI is consistent with the aggregate results: FDI is associated with no change in per capita income across all RUCC groups, with one exception. Counties with RUCC 8–9 experience a 2.3% increase in PCPI for each additional $100 million in FDI capital investment. The presence of FDI alone produces no detectable income effect in any subgroup.

These differences across RUCC categories are consistent with the absorptive capacity framework. RUCC 4–5 counties are relatively close to metro areas and tend to have higher wages, more establishments, and better infrastructure than their more rural counterparts. As a result, they are better positioned to attract FDI in the first place, but the marginal impact of an additional investment project on an already developed local economy is smaller. RUCC 6–7 counties occupy a middle ground: they have lower wages and fewer establishments than RUCC 4–5 counties, giving new FDI more room to generate employment gains, while still possessing enough existing manufacturing activity—as reflected in their higher manufacturing location quotients—to support the supply chain linkages and workforce skill

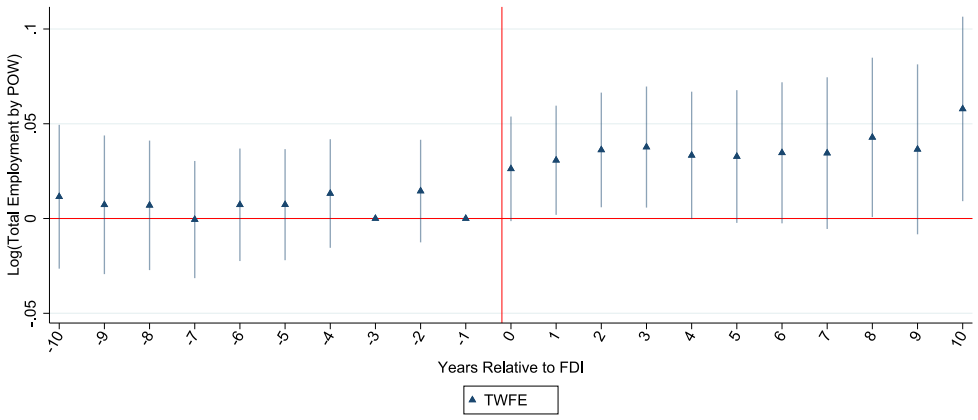
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(a)



(b)



(c)

FIGURE 6 Event studies for difference-in-differences estimator. (a) PCPI. (b) Total employment by residence. (c) Total employment by POW. These event studies show the coefficient estimates and confidence intervals for the years prior to and after an FDI announcement in a rural county. The outcome variables are PCPI in panel (a), total county employment by residency in panel (b), and total county employment by POW in panel (c). The untreated counties are weighted according to how many times they served as a control county match. Data on FDI project dates come from fDi Markets. Data for the dependent variables are from the Bureau of Economic Analysis and Bureau of Labor Statistics Local Area Unemployment Statistics. Years of analysis are 2003–2019. FDI, foreign direct investment; PCPI, per capita personal income; POW, place of work.

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TABLE 13 Results from DiD analysis with propensity score matching.

	(1) Log of PCPI	(2) Log of employment by residence	(3) Log of employment by POW
EverFDI	0.00253	−0.00836	0.0145
	(0.00806)	(0.0135)	(0.0162)
Post*EverFDI	−0.00936	0.0275**	0.0239*
	(0.00666)	(0.0111)	(0.0130)
Dep. var. mean	10.40	9.386	9.525
N	20,094	20,094	20,094

Note: This table presents the results from a DiD estimation, controlling for state, time, county, and matched pair fixed effects. The dependent variables are log of PCPI, log of total county employment by residency, and log of county employment by POW. *EverFDI* is an indicator variable equal to 1 if the county received FDI between 2004 and 2019. *Post*EverFDI* is an indicator variable equal to 1 if the county had an FDI announcement in that year and every year after until 2019. Standard errors are clustered at the county level. Only the counties on common support are included. The final sample has 965 rural counties, of which 591 are treated and 374 are untreated. The untreated counties are weighted according to how many times they serve as a control county match. Data come from the U.S. Department of Agriculture, U.S. Census Bureau, BLS-LAUS, the Bureau of Economic Analysis, and fDi Markets. Years of analysis are 2003–2019.

Abbreviations: BLS-LAUS, Bureau of Labor Statistics Local Area Unemployment Statistics; DiD, difference-in-differences; FDI, foreign direct investment; PCPI, per capita personal income; POW, place of work.

*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

TABLE 14 Results from DiD analysis with propensity score matching, by FDI capital investment.

	(1) Log of PCPI	(2) Log of employment by residence	(3) Log of employment by POW
EverFDI	−0.00231	0.00611	0.0267*
	(0.00731)	(0.0132)	(0.0158)
Post*Relative FDI dosage	0.000806	−0.00350	−0.00109
	(0.00221)	(0.00407)	(0.00503)
Dep. var. mean	10.40	9.386	9.525
N	20,094	20,094	20,094

Note: This table presents the results from a DiD estimation, controlling for state, time, county, and matched pair fixed effects. The dependent variables are log of PCPI, log of total county employment by residency, and log of county employment by POW. *EverFDI* is an indicator variable equal to 1 if the county received FDI between 2004 and 2019. *Relative FDI Dosage* is the county's cumulative FDI capital investment in $100 millions, relative to the mean capital investment of 65 million. Standard errors are clustered at the county level. Only the counties on common support are included. The final sample has 965 rural counties, of which 591 are treated and 374 are untreated. The untreated counties are weighted according to how many times they serve as a control county match. Data come from the U.S. Department of Agriculture, U.S. Census Bureau, Bureau of Labor Statistics Local Area Unemployment Statistics, the Bureau of Economic Analysis, and fDi Markets. Years of analysis are 2003–2019.

Abbreviations: DiD, difference-in-differences; FDI, foreign direct investment; PCPI, per capita personal income; POW, place of work.

*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

demands that FDI requires. This combination of latent absorptive capacity and room for growth explains why employment effects are strongest in this group.

Remote counties with RUCC 8–9 present a different constraint. Despite having the lowest wages and fewest establishments—conditions that in principle create substantial room for growth—they lack the infrastructure, labor force skills, and market connectivity needed to sustain the complementary activity that converts FDI into lasting employment gains (Bermejo Carbonell & Werner, 2018; Carkovic & Levine, 2005; Hermes & Lensink, 2003). The one exception is the positive PCPI response to large capital investments in RUCC 8–9 counties. This result likely reflects wage pressure rather than broad-based income growth: with a thin local labor market, a sufficiently large investment may bid up wages for those employed, even as the overall employment response remains limited. This interpretation is supported by the population data—the mean annualized population growth rate for RUCC 8–9 treated counties is −0.26%, compared with −0.15% for untreated RUCC 8–9 counties between 2005 and 2019. Rising PCPI alongside

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TABLE 15 DiD estimates by rural codes, FDI project.

	PCPI			Employment by res.			Employment by POW		
RUCC	**4–5**	**6–7**	**8–9**	**4–5**	**6–7**	**8–9**	**4–5**	**6–7**	**8–9**
EverFDI	0.0425***	0.00197	0.104***	0.0775**	0.00134	−0.345***	0.0618	0.0124	−0.266***
	(0.0153)	(0.00882)	(0.0307)	(0.0368)	(0.0149)	(0.0810)	(0.0415)	(0.0199)	(0.0809)
Post*EverFDI	−0.0128	0.00721	−0.00810	−0.0110	0.0366***	0.0613	−0.0230	0.0493***	0.0541
	(0.0116)	(0.00757)	(0.0238)	(0.0249)	(0.0124)	(0.0476)	(0.0275)	(0.0150)	(0.0386)
Dep. var. mean	10.46	10.38	10.41	10.27	9.264	7.953	10.49	9.395	8.047
N	3842	11,560	2006	3842	11,560	2006	3842	11,560	2006

Note: This table presents DiD estimation results by RUCC category, controlling for state, time, county, and matched pair fixed effects. Data come from the U.S. Department of Agriculture, U.S. Census Bureau, Bureau of Labor Statistics Local Area Unemployment Statistics, Bureau of Economic Analysis, and fDi Markets. Years of analysis are 2003–2019.

Abbreviations: DiD, difference-in-differences; FDI, foreign direct investment; PCPI, per capita personal income; POW, place of work; RUCC, Rural-Urban Continuum Code.

*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

TABLE 16 DiD estimates by rural codes, FDI capital investment.

	PCPI			Employment by res.			Employment by POW		
RUCC	**4–5**	**6–7**	**8–9**	**4–5**	**6–7**	**8–9**	**4–5**	**6–7**	**8–9**
EverFDI	0.0368***	0.00448	0.0919***	0.0727**	0.0188	−0.323***	0.0514	0.0344*	−0.255***
	(0.0135)	(0.00780)	(0.0321)	(0.0340)	(0.0138)	(0.0877)	(0.0372)	(0.0185)	(0.0820)
Post*Relative FDI dosage	−0.00284	0.00551	0.0226***	−0.00270	0.00184	−0.000345	−0.00489	0.0110	0.0200
	(0.00335)	(0.00539)	(0.00833)	(0.00666)	(0.00514)	(0.0173)	(0.00836)	(0.00686)	(0.0171)
Dep. var. mean	10.46	10.38	10.41	10.27	9.264	7.953	10.49	9.395	8.047
N	3842	11,560	2006	3842	11,560	2006	3842	11,560	2006

Note: This table presents DiD estimation results by RUCC category for the FDI capital investment dosage specification, controlling for state, time, county, and matched pair fixed effects. Data come from the U.S. Department of Agriculture, U.S. Census Bureau, Bureau of Labor Statistics Local Area Unemployment Statistics, Bureau of Economic Analysis, and fDi Markets. Years of analysis are 2003–2019.

Abbreviations: DiD, difference-in-differences; FDI, foreign direct investment; PCPI, per capita personal income; POW, place of work; RUCC, Rural-Urban Continuum Code.

*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

population decline points to wage gains among a shrinking resident base rather than an expansion of the local workforce.

4.5.2 | Robustness

Different Matching Characteristics and Treatment Definitions: As a robustness check, I repeat the DID analysis with different logit specifications (L1, L2, L3) and treatment and control groups (T1, T2, T3). The Supporting Information S2: Appendix reports the estimated coefficients for PCPI, employment by residence, and employment by POW for each logit and treatment combination.

Overall, the results are stable among the different specifications. Consistent with the main results, FDI is not associated with any change in county PCPI. Estimates for FDI's impact on employment range from 2% to 7% and hover around 3%. The highest statistically significant estimate for county employment is approximately 6%–7% for L3 and T3. Under this specification, counties with FDI in 2004–2015 are compared to counties that either received FDI later from 2016 to 2019 or didn't receive any FDI from 2003 to 2019. The years of analysis under T3 are 2003–2015, so the sample period is smaller. However, the 95% confidence interval for the higher estimates include 3%.

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Counties with large FDI projects only: As a further robustness check, I re-estimate the main DID specifications restricting the sample to counties that received large FDI shocks, defined as those at or above the 75th and 90th percentiles of cumulative capital investment among treated counties. This addresses whether very large investments—more comparable in scale to the Kia Motors project—replicate the employment effects observed in the case study. [13] Results are reported in Supporting Information S2: Appendix.

The large-shock subsamples produce consistently larger employment effects than the full sample. Among counties above the 75th percentile, residential employment increases by approximately 3%–8% per year on average, compared with 2%–6% in the full sample. Employment by place of work follows a similar pattern, with estimates ranging from 3% to 9% for large-shock counties. While these results are consistent with the case study—where a very large investment generated above-average employment gains—they do not fully replicate the magnitude of the Kia effect. Even among the largest FDI shocks in the national sample, estimated employment gains remain well below the 13%–16% annual increases observed in Troup County. This gap likely reflects the exceptional scale of the Kia investment relative to even the top decile of rural FDI projects, as well as the coordinated package of supplier entry, government incentives, and workforce training that accompanied it.

Restricting the sample to large FDI shocks does not alter the PCPI result: per capita income remains statistically indistinguishable from zero across most specifications and thresholds. The null income effect therefore holds even for the largest rural FDI projects, reinforcing the paper's central finding that employment and income effects are decoupled regardless of investment scale.

Spatial dependence and spillovers: Labor markets, housing markets, and commuting patterns do not respect county boundaries, which may generate spatial correlation in county-level residuals if unobserved shocks are correlated across neighboring counties. I formally test for and address spatial dependence by applying Moran's I to the baseline DiD residuals using a 100-mile spatial threshold (Moran, 1950).[14] The test finds statistically significant spatial autocorrelation in the preferred T1–L1 specification (Moran's $I = 0.071$, $p < 0.001$) and in the T3–L1 specification (Moran's $I = 0.036$, $p = 0.048$), while the smaller T2–L1 subsample shows no significant spatial autocorrelation (Moran's $I = 0.026$, $p = 0.41$). The Moran's I statistics are positive but small in magnitude, indicating that spatial correlation is present but modest after conditioning on the fixed effects structure. I therefore re-estimate the main specifications using Conley (1999) spatially robust standard errors with bandwidths of 100 and 200 miles.

The corrected standard errors are modestly larger than county-clustered estimates, as expected, but the core results are unchanged. Employment effects remain positive and statistically significant across most specifications, while per capita income remains statistically indistinguishable from zero. One exception is employment by place of work, where statistical significance is sensitive to the spatial correction, although the point estimates remain stable. Overall, these results indicate that spatial correlation affects inference at the margin but does not alter the main conclusions. The Conley-corrected estimates for per capita income, employment by residence, and employment by place of work are reported in Supporting Information S2: Appendix.

Next, I examine spatial spillovers using a neighbor-based difference-in-differences specification, with results reported in Supporting Information S2: Appendix. While treated counties experience employment gains without corresponding increases in per capita income, neighboring counties show modest spillovers: per capita income increases by approximately 1.0%–1.2%, and employment by place of work rises slightly, while employment by residence remains largely unchanged. This pattern is consistent with labor market integration across counties through commuting and inter-county linkages. While spillover effects are present, their magnitude is smaller than the direct treatment effects, indicating that spatial spillovers do not meaningfully alter the magnitude or interpretation of the estimated county-level coefficients.

Last, I re-estimate the model at the commuting zone level to capture broader regional adjustment. The commuting zone results are reported in Supporting Information S2: Appendix. Aggregating the data attenuates the estimated effects, suggesting that part of the adjustment occurs across local labor markets rather than strictly within county boundaries. In summary, these additional results imply that FDI generates both local and regional employment responses. Evaluating FDI at the county level therefore remains policy-relevant for creating incentive packages at the county level, but the resulting impacts should be interpreted within the broader context of integrated regional labor markets.

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5 | DISCUSSION

I estimate that the Kia Motors' FDI in Troup County led to 10%–15% more residents employed and an increase of 13%–16% more jobs in the county on average per year. The placebo tests also confirm the significance of FDI's positive effect on total county employment for Troup County. I would expect to see the greatest impact on the outcome variables from the start of plant production, 2009 and onward. Therefore, it's not surprising the estimate of FDI's impact on total employment with 2009 as the treatment year is greater than that of the 2006 specification. The coefficient estimates for the effect of FDI on PCPI are not statistically different from zero. Furthermore, when performing the placebo tests for the untreated donor counties, the PCPI results for Troup County were not statistically significant in the distribution of placebo effects. Together, these findings indicate that the Kia Motors' FDI had a null effect on PCPI.

The absence of a statistically significant effect on PCPI is consistent with a simple two-sector labor reallocation framework. In this framework, a local economy is composed of a manufacturing sector and a non-manufacturing sector. A large FDI project represents a positive demand shock to manufacturing labor. In the presence of labor mobility across sectors, workers shift from the non-manufacturing sector into manufacturing until wages equalize across sectors in equilibrium. Under this adjustment process, total employment in the expanding sector increases, and aggregate employment may rise if previously non-employed workers enter the labor force. However, average wages need not increase proportionally.

If the shock primarily induces reallocation rather than sustained wage pressure, the equilibrium wage may remain largely unchanged. This labor mobility mechanism is particularly relevant in rural counties that are economically integrated with nearby labor markets. Commuting flows and cross-sector mobility allow labor supply to respond elastically to localized demand shocks, dampening wage effects while amplifying employment responses. Other empirical studies provide evidence that the effect of FDI on wages depends on industry composition (Bogliaccini & Egan, 2017), the skills of the existing labor force (Saucedo et al., 2020), labor bargaining power, and institutional factors (Onaran, 2012). These factors explain why FDI can produce markedly different outcomes across regions, shaping both the magnitude of its benefits and their distribution within local economies.

A lack of PCPI growth can further be explained by changes in personal income (the numerator) and population (the denominator). For no growth, these variables must be constant or moving in the same direction. Figure 7a,b depict Troup County's population and PCPI growth from 5 years before the Kia investment in 2006 to 10 years after. Population grows at a constant rate before and after the Kia Plant FDI. Therefore, for PCPI to be constant, county personal income must also be increasing at a constant rate, which the data and figures support. To explore further why FDI didn't change the population growth or personal income growth rates of Troup County, I turn to the broader analysis of U.S. rural counties.

In the second part of the paper, I estimate that FDI impacts employment on average by 2%–3% annually and has no statistically significant effect on PCPI. Positive employment growth is concentrated in counties with RUCC codes 6–7. Similar to the case study, employment by POW starts at a greater level pre-treatment than employment by residence. Both employment variables grow at between 2.39% and 2.75%, with their 95% confidence intervals overlapping. Population levels remain flat during the pre and post periods of the FDI for the treated rural counties. Likewise, county personal income grows at a steady rate over the same time period.

I group the treated counties into quartiles based on PCPI growth during the first 5 years after the FDI investment and analyze them separately. Grouping counties by post-FDI PCPI growth reveals the housing, wage, and business environment conditions that distinguish high-income-growth counties from those with stagnant incomes. To address population and moving behavior, I examine the number of new housing permits and new households in the county before and after the FDI. For personal income, I analyze the changes in average weekly wages and total establishments. The time series of these variables are depicted in Figure 7c–f.

While population remains steady, the number of households increases for counties in the top quartile of PCPI growth. Additionally, the top quartile counties experience higher growth in housing permits, particularly 5 years after the FDI compared to the bottom quartile. Together, these facts suggest that high PCPI growth counties have above-average growth in new households and new housing.

Differences in weekly wages and total establishment counts indicate stronger labor market competition among counties in the top quartile of PCPI growth. From the FDI until 10 years later, top quartile counties have weekly wages that increased 10 percentage points more than bottom quartile counties over the same period. The increase in weekly wages could be due to a shortage of workers, competition among firms for laborers, or both. The resulting increase in wages explains the increase in PCPI while population levels stay the same.

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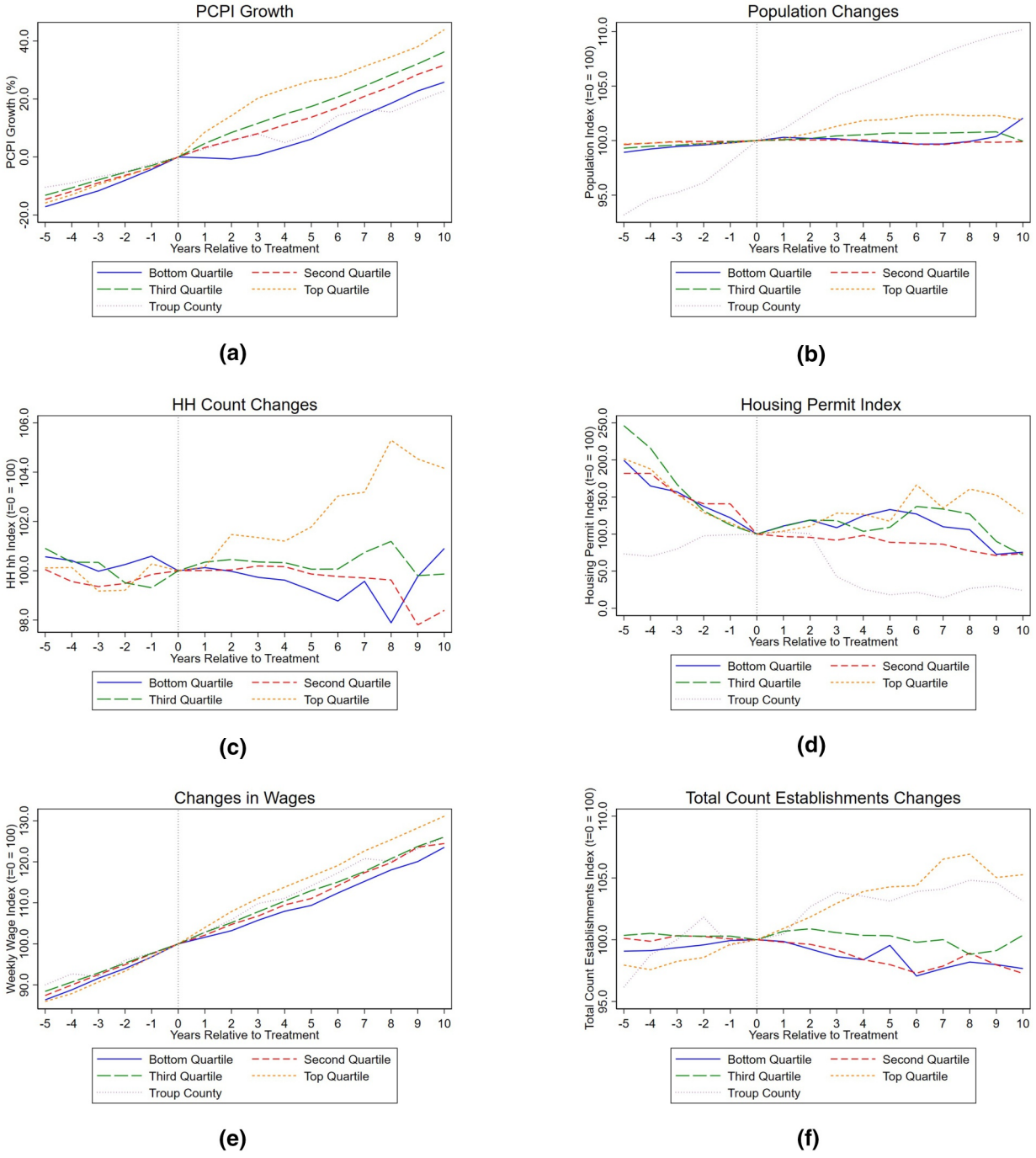


FIGURE 7 PCPI growth quartile graphs. (a) PCPI growth. (b) Population growth. (c) Growth in number of households. (d) Growth in housing permits. (e) Growth in weekly wages. (f) Growth in total establishments. These figures graph the time series for treated rural counties that received FDI in year $t = 0$. The treated counties are separated into quartiles based on the amount of PCPI growth the county experience in the first 5 years after the FDI. Only 5 years prior to the FDI and 10 years after the FDI are included. Years of analysis are 1995–2019. FDI, foreign direct investment; PCPI, per capita personal income.

Although the aforementioned discussion does not provide causal evidence of the mechanism behind the null effect of FDI on PCPI, the data present a set of stylized facts that support a likely mechanism. Treated counties that experience more of an economic boom from the FDI have more total establishments, higher wages, more new housing, more migrating households, and a higher PCPI. Meanwhile, the counties with the lowest PCPI see less wage growth, fewer new housing units, and fewer total establishments.

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Troup County is in the middle of the pack when it comes to PCPI growth. The steadily increasing population appears to be limited by the lack of new housing permits. Furthermore, because county jobs grow more than residential employment, the data suggest that some of the new wage earners are not residing in the county. Thus, the growth rate of personal income remains steady although wages for county jobs increase. In a separate report for Troup County, Bleakly Advisory Group (2019) cited similar findings. The report described how the lack of available housing options and new affordable residential construction stunted household and population growth. Consequently, a large share of Troup County workers commute into the county while living in other nearby counties. Bleakly Advisory Group (2019) notes that Troup ranks in the 83rd percentile for wages paid in Georgia, but only the 60th percentile for household income. This gap suggests that while high-paying jobs are located in Troup, many of the earners likely reside elsewhere. The report attributes this to a lack of suitable housing options, which limits Troup's ability to convert job growth into local income gains.

These results speak to a key policy trade-off. If the goal is simply to create jobs, FDI attraction strategies can succeed in a wide range of rural contexts. But if the goal is to raise local incomes, then FDI must be paired with investments that expand housing supply, facilitate worker in-migration, and encourage firm competition for labor. Without these complementary measures, the returns to public spending on FDI incentives will be uneven. In many cases, the income benefits will diffuse to surrounding counties and diminish local gains.

6 | CONCLUSION

This paper provides the first county-level, national analysis of FDI's long-term impact on rural U.S. economies. It combines a large panel dataset with synthetic control, propensity score matching, and difference-in-differences methods. Across both the Troup County case study and the national sample, the evidence shows that rural FDI reliably increases employment but does not, on average, raise PCPI in the host county. Where incomes do rise, they are associated with higher wage growth, expanded housing supply, and more business establishments.

In light of these findings, how policymakers and economic development organizations use their time and resources to pursue future FDI projects depends on their goals. This paper shows that FDI in rural counties can increase the total number of jobs in a county, as demonstrated by the employment by POW estimates. The employment by residency estimates also show how FDI increases residential employment. When it comes to the effectiveness of FDI as a catalyst for increasing PCPI in a host rural county, the evidence is lacking.

Even with large FDI investments, PCPI appears to be a hard measure to move for rural counties. Counties with the highest PCPI have the highest sustained growth in new housing permits, new households, and weekly wages. Together, these attributes set up the high-PCPI-growth counties to convert the high-skilled workers to residents and increase the county's personal income overall. Therefore, regional governments wanting to increase PCPI alongside employment should prepare the residential housing markets for new workers and look to enact policies that encourage more firm competition and upskilling workers, pushing up weekly wages.

Among FDI projects in rural counties, Kia Motors and Troup County are an anomaly and one worth replicating for employment growth. The case study demonstrates the potential for a large-scale FDI project to greatly increase county employment with additional supplier and regional government investment. The greater employment gains in Troup County may reflect not only the scale of Kia's investment, but also targeted support from local and state governments, such as a training program for potential employees through the QuickStart program. Future research with data on government incentives can further explore this potential mechanism to explain the differing employment effects.

With the federal government's push to spark economic development and reindustrialize the U.S., FDI in rural counties is likely to become more common. However, regional development in rural regions through FDI should be reframed. State and local governments seeking to raise incomes alongside employment should view FDI as one component of a broader development strategy. This strategy should include proactive residential planning to expand housing supply, measures to foster competitive labor markets, and support for complementary business formation. Without this groundwork, a significant share of the benefits from FDI may diffuse to surrounding areas through commuting and migration patterns. Consequently, FDI's promise of economic growth will remain only partially fulfilled.

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DATA AVAILABILITY STATEMENT

The publicly available data that support the findings of this study are available in the ICPSR data repository at https://doi.org/10.3886/E249865V1 (Jones, 2026). These data were derived from the following resources available in the public domain: Bureau of Economic Analysis Regional Data (PCPI and county-level GDP, CAGDP1 series), available at https://www.bea.gov/data/income-saving/personal-income-county-metro-and-other-areas; Bureau of Labor Statistics Local Area Unemployment Statistics (county employment by residence), available at https://www.bls.gov/lau/; U.S. Department of Agriculture Rural-Urban Continuum Codes, available at https://www.ers.usda.gov/data-products/rural-urban-continuum-codes/; U.S. Census Bureau population estimates, available at https://www.census.gov/programs-surveys/popest.html; U.S. Department of Transportation highway and railroad data, available at https://geodata.bts.gov/; and Bureau of Labor Statistics Quarterly Census of Employment and Wages, available at https://www.bls.gov/cew/. The foreign direct investment data that support the findings of this study are available from fDi Markets (Financial Times Ltd, 2024). Restrictions apply to the availability of these data, which were used under license for this study. Data are available from fDi Markets.

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ENDNOTES

[1] A potential concern is that fDi Markets may capture only job-creating FDI projects, which would mean the treatment group excludes foreign investments that do not involve new employment. To assess the extent of this in the rural sample, I document whether each project reported expected job creation at the time of announcement. All projects in the rural PSM sample reported positive expected employment at announcement, confirming that the dataset captures announced job-creating investments rather than all forms of foreign capital inflows. The estimates should therefore be interpreted as the effects of employment-generating FDI announcements. Because project entry into the dataset is based on announced investment activity rather than realized employment outcomes, selection on ex-post performance is not a concern.

[2] If an FDI project is announced without an estimated investment or job creation number, fDi Markets predicts these values based on their proprietary algorithm.

[3] See BEA's note for more information about the PCPI and population estimates: https://www.bea.gov/note-capita-personal-income-and-population.

[4] I estimate the synthetic control with a treatment year of 2006 and 2009 in case the construction of the Kia Plant from 2006 to 2009 temporarily inflated the employment or PCPI. Additionally, since my treatment period spans 6–9 years, I expect any temporary inflation or deflation of the income to be averaged out across the several years where I focus my analysis.

[5] See Data Supporting Information S2: Appendix for additional discussion of PCPI measurement in Troup County.

[6] See Data Supporting Information S2: Appendix for additional information on differences between BEA and BLS employment estimates.

[7] Kia Motors did engage in talks with Mississippi about locating their plant in the state before eventually choosing Troup County, Georgia (Bruns, 2006).

[8] A low ratio means that after the treatment occurred, the treated unit does not deviate from its pre-treatment trend nor that of the synthetic unit. On the other hand, a high ratio suggests that the synthetic unit closely matches the treated unit in the pre-treatment period and that the treated unit sees a substantial deviation from the trend after treatment (Cunningham, 2021).

[9] The fDi Markets data begins in 2003. I do not include rural counties that were treated in 2003 since they would have no pretreatment years for the DID specification. This applies to four counties.

[10] See Data Supporting Information S2: Appendix for results from alternate logit specifications (L2, L3) and treatment definitions.

[11] For these counties, their propensity score is too high to find an appropriate control county match.

[12] Because FDI treatment in rural counties is staggered, I report in Supporting Information S2: Appendix an alternative difference-in-differences estimator following the approach of Callaway and Sant'Anna (2021).

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[13] For the preferred specification, the 75th percentile of cumulative FDI capital investment among treated rural counties corresponds to approximately $189.70 million, and the 90th percentile to approximately $336.50 million. These thresholds are computed from the treated counties in the matched sample.

[14] Moran's I is a measure of spatial autocorrelation that tests whether nearby geographic units tend to have more similar values than would be expected under spatial randomness. The statistic ranges from −1 (perfect dispersion) to +1 (perfect clustering), with values near zero indicating spatial randomness. A positive and significant Moran's I on DiD residuals indicates that the model's errors are not independently distributed across space, which can lead to understated standard errors and overconfident inference. I test for spatial autocorrelation in the residuals rather than in the outcome variables themselves to test whether unmodeled spatial dependence remains after conditioning on the fixed effects structure of the estimator.

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